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(19) **United States**(12) **Patent Application Publication**
MEISSNER et al.(10) **Pub. No.: US 2025/0280829 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **USE OF****(5S)-3-[3-(3-CHLORO-2-FLUOROPHENOXY)-
6-METHYLPYRIDAZIN-4-YL]-5-(2-CHLORO-
4-METHYLBENZYL)-5,6-DIHYDRO-4H-1,2,4-
OXADIAZINE FOR CONTROLLING
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Sylvain TAFFOREAU, Paris (FR)(21) Appl. No.: **18/862,344**(22) PCT Filed: **Apr. 26, 2023**(86) PCT No.: **PCT/EP2023/060917**

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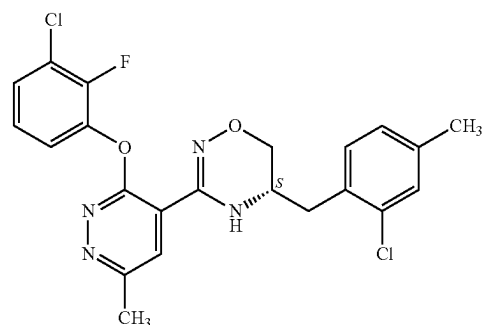
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(57)

ABSTRACT

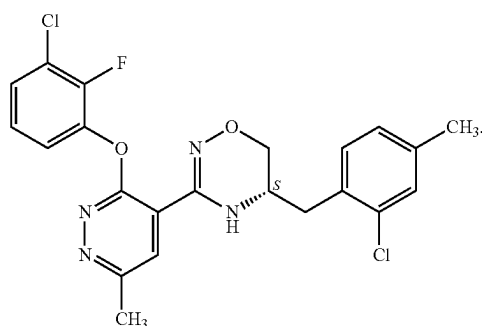
The present invention relates to the use of (5S)-3-[3-(3-chloro-2-fluorophenoxy)-6-methylpyridazin-4-yl]-5-(2-chloro-4-methylbenzyl)-5,6-dihydro-4H-1,2,4-oxadiazine according to formula (I) for controlling unwanted microorganisms causing a variety of different plant diseases,

(I)



**USE OF
(5S)-3-[3-(3-CHLORO-2-FLUOROPHENOXY)-6-METHYLPYRIDAZIN-4-YL]-5-(2-CHLORO-4-METHYLBENZYL)-5,6-DIHYDRO-4H-1,2,4-OXADIAZINE FOR CONTROLLING UNWANTED MICROORGANISMS**

[0001] The present invention relates to the use of (5S)-3-[3-(3-chloro-2-fluorophenoxy)-6-methylpyridazin-4-yl]-5-(2-chloro-4-methylbenzyl)-5,6-dihydro-4H-1,2,4-oxadiazine according to formula (I) for controlling unwanted microorganisms causing a variety of different plant diseases,



[0002] (5S)-3-[3-(3-chloro-2-fluorophenoxy)-6-methylpyridazin-4-yl]-5-(2-chloro-4-methylbenzyl)-5,6-dihydro-4H-1,2,4-oxadiazine and a process for its preparation is known from WO 2020/127780.

[0003] Numerous crop protection agents to combat or prevent microorganisms' infestations have been developed until now. However, the need remains for the development of new compounds as such, in order to provide compounds being effective against a broad spectrum of phytopathogenic microorganisms, such as fungi, having low toxicity, high selectivity or that can be used at low application rate whilst still allowing effective pest control. It may also be desired to have new compounds to prevent the emergence of resistances.

[0004] It has now been surprisingly found that (5S)-3-[3-(3-chloro-2-fluorophenoxy)-6-methylpyridazin-4-yl]-5-(2-chloro-4-methylbenzyl)-5,6-dihydro-4H-1,2,4-oxadiazine is very active against a variety of different unwanted microorganisms while exhibiting an overall beneficial profile with respect to human and environmental safety, toxicity, low application rate and resistance profile.

[0005] Accordingly, subject of this invention is a method of controlling unwanted microorganisms, in particular phytopathogenic fungi and phytopathogenic viruses, causing a variety of different plant diseases, comprising application of (5S)-3-[3-(3-chloro-2-fluorophenoxy)-6-methylpyridazin-4-yl]-5-(2-chloro-4-methylbenzyl)-5,6-dihydro-4H-1,2,4-oxadiazine of formula (I) to the plant or a part thereof.

[0006] In the context of the present invention, "controlling unwanted microorganisms" means a reduction in infestation by the respective microorganisms, compared with the untreated plant measured as fungicidal efficacy, preferably a reduction by 25-50%, compared with the untreated plant (100%), more preferably a reduction by 40-79%, compared with the untreated plant (100%). Even more preferably, the infection by the microorganism is suppressed by 80-100%.

Controlling comprises curative control, i.e. treatment of already infected plants in order to reduce or eliminate infection, and protective/preventive control, i.e. treatment of plants which have not yet been infected in order to prevent infection.

[0007] In the method according to the invention plants or parts thereof are treated with a compound of formula (I).

[0008] Plant parts as understood herein are all above ground parts and organs of plants such as shoot, leaf and blossom, whereby for example leaves, stems, blossoms, fruiting bodies, fruits and seed are listed.

Uses

[0009] The compound and the composition of the invention have potent microbicidal activity and/or plant defense modulating potential. They can be used for controlling unwanted microorganisms, such as unwanted fungi, on plants. They can be particularly useful in crop protection (they control microorganisms that cause plants diseases) or for protecting materials (e.g. industrial materials, timber, storage goods) as described in more details herein below. More specifically, the compound and the composition of the invention can be used to protect seeds, germinating seeds, emerged seedlings, plants, plant parts, fruits, harvest goods and/or the soil in which the plants grow from unwanted microorganisms.

[0010] Control or controlling as used herein encompasses protective, curative and eradication treatment of unwanted microorganisms. Unwanted microorganisms may be pathogenic virus, or pathogenic fungi, more specifically phytopathogenic virus, or phytopathogenic fungi. As detailed herein below, these phytopathogenic microorganisms are the causal agents of a broad spectrum of plants diseases.

[0011] More specifically, the compound and the composition of the invention can be used as fungicides. For the purpose of the specification, the term "fungicide" refers to a compound or composition that can be used in crop protection for the control of unwanted fungi, such as Plasmodiophoromycetes, Chytridiomycetes, Zygomycetes, Ascomycetes, Basidiomycetes and Deuteromycetes.

[0012] The compound and the composition of the invention may also be used as antiviral agent in crop protection. For example the compound and the composition of the invention may have effects on diseases from plant viruses, such as the tobacco mosaic virus (TMV), tobacco rattle virus, tobacco stunt virus (TStuV), tobacco leaf curl virus (VLCV), tobacco nervilia mosaic virus (TVBMV), tobacco necrotic dwarf virus (TNDV), tobacco streak virus (TSV), potato virus X (PVX), potato viruses Y, S, M, and A, potato acuba mosaic virus (PAMV), potato mop-top virus (PMTV), potato leaf-roll virus (PLRV), alfalfa mosaic virus (AMV), cucumber mosaic virus (CMV), cucumber green mottle-mosaic virus (CGMMV), cucumber yellows virus (CuYV), watermelon mosaic virus (WMV), tomato spotted wilt virus (TSWV), tomato ringspot virus (TomRSV), sugarcane mosaic virus (SCMV), rice drawf virus, rice stripe virus, rice black-streaked drawf virus, strawberry mottle virus (SMoV), strawberry vein banding virus (SVBV), strawberry mild yellow edge virus (SMYEV), strawberry crinkle virus (SCrV), broad beanwilt virus (BBWV), and melon necrotic spot virus (MNSV).

[0013] The present invention also relates to a method for controlling unwanted microorganisms, such as unwanted fungi, on plants comprising the step of applying the compound of the invention or at least one composition of the

invention to the microorganisms and/or their habitat (to the plants, plant parts, seeds, fruits or to the soil in which the plants grow).

[0014] Typically, when the compound and the composition of the invention are used in curative or protective methods for controlling phytopathogenic fungi, an effective and plant-compatible amount thereof is applied to the plants, plant parts, fruits, seeds or to the soil or substrates in which the plants grow. Suitable substrates that may be used for cultivating plants include inorganic based substrates, such as mineral wool, in particular stone wool, perlite, sand or gravel; organic substrates, such as peat, pine bark or sawdust; and petroleum based substrates such as polymeric foams or plastic beads. Effective and plant-compatible amount means an amount that is sufficient to control or destroy the fungi present or liable to appear on the cropland and that does not entail any appreciable symptom of phytotoxicity for said crops. Such an amount can vary within a wide range depending on the fungus to be controlled, the type of crop, the crop growth stage, the climatic conditions and the respective compound or composition of the invention used. This amount can be determined by systematic field trials that are within the capabilities of a person skilled in the art.

Plants and Plant Parts

[0015] The compound and the composition of the invention may be applied to any plants or plant parts.

[0016] Plants mean all plants and plant populations, such as desired and undesired wild plants or crop plants (including naturally occurring crop plants). Crop plants may be plants which can be obtained by conventional breeding and optimization methods or by biotechnological and genetic engineering methods or combinations of these methods, including the genetically modified plants (GMO or transgenic plants) and the plant cultivars which are protectable and non-protectable by plant breeders' rights.

[0017] Plant cultivars are understood to mean plants which have new properties ("traits") and have been obtained by conventional breeding, by mutagenesis or by recombinant DNA techniques. They can be cultivars, varieties, bio- or genotypes.

[0018] Plant parts are understood to mean all parts and organs of plants above and below the ground, such as shoots, leaves, needles, stalks, stems, flowers, fruit bodies, fruits, seeds, roots, tubers and rhizomes. The plant parts also include harvested material and vegetative and generative propagation material, for example cuttings, tubers, rhizomes, slips and seeds.

[0019] Plants which may be treated in accordance with the methods of the invention include the following: cotton, flax, grapevine, fruit, vegetables, such as *Rosaceae* sp. (for example pome fruits such as apples and pears, but also stone fruits such as apricots, cherries, almonds and peaches, and soft fruits such as strawberries), *Ribesioideae* sp., *Juglandaceae* sp., *Betulaceae* sp., *Anacardiaceae* sp., *Fagaceae* sp., *Moraceae* sp., *Oleaceae* sp., *Actinidaceae* sp., *Lauraceae* sp., *Musaceae* sp. (for example banana trees and plantations), *Rubiaceae* sp. (for example coffee), *Theaceae* sp., *Sterculiaceae* sp., *Rutaceae* sp. (for example lemons, oranges and grapefruit), *Solanaceae* sp. (for example potatoes and tomatoes), *Liliaceae* sp., *Asteraceae* sp. (for example lettuce), *Umbelliferae* sp., *Cruciferae* sp., *Chenopodiaceae* sp., *Cucurbitaceae* sp. (for example cucum-

ber), *Alliaceae* sp. (for example leek, onion), *Papilionaceae* sp. (for example peas); major crop plants, such as *Gramineae* sp. (for example maize, turf, cereals such as wheat, rye, rice, barley, oats, millet and triticale), *Asteraceae* sp. (for example sunflower), *Brassicaceae* sp. (for example white cabbage, red cabbage, broccoli, cauliflower, Brussels sprouts, pak choi, kohlrabi, radishes, oilseed rape, mustard, horseradish and cress), *Fabaceae* sp. (for example bean, peanuts), *Papilionaceae* sp. (for example soya bean), *Chenopodiaceae* sp. (for example sugar beet, fodder beet, swiss chard, beetroot); useful plants and ornamental plants for gardens and wooded areas; and genetically modified varieties of each of these plants.

[0020] Preferably, plants which may be treated in accordance with the methods of the invention include the following: grapevine, fruit, vegetables, such as *Rosaceae* sp. (for example pome fruits such as apples and pears, but also stone fruits such as apricots, cherries, almonds and peaches, and soft fruits such as strawberries), *Oleaceae* sp., *Actinidaceae* sp., *Musaceae* sp. (for example banana and plantations), *Rubiaceae* sp. (for example coffee), *Theaceae* sp., *Sterculiaceae* sp., *Rutaceae* sp. (for example lemons, oranges and grapefruit), *Solanaceae* sp. (for example potatoes and tomatoes), *Liliaceae* sp., *Asteraceae* sp. (for example lettuce), *Umbelliferae* sp., *Cruciferae* sp., *Chenopodiaceae* sp., *Cucurbitaceae* sp. (for example cucumber), *Alliaceae* sp. (for example leek, onion), *Papilionaceae* sp. (for example peas); major crop plants, such as *Gramineae* sp. (for example maize, turf, cereals such as wheat, rye, rice, barley, oats, millet and triticale), *Asteraceae* sp. (for example sunflower), *Brassicaceae* sp. (for example white cabbage, red cabbage, broccoli, cauliflower, Brussels sprouts, pak choi, kohlrabi, radishes, oilseed rape, mustard, horseradish and cress), *Fabaceae* sp. (for example bean, peanuts), *Papilionaceae* sp. (for example soya bean), *Chenopodiaceae* sp. (for example sugar beet, fodder beet, swiss chard, beetroot) and genetically modified varieties of each of these plants.

[0021] Plants and plant cultivars which may be treated by the above disclosed methods include plants and plant cultivars which are resistant against one or more biotic stresses, i.e. said plants show a better defense against animal and microbial pests, such as against nematodes, insects, mites, phytopathogenic fungi, bacteria, viruses and/or viroids.

[0022] Plants and plant cultivars which may be treated by the above disclosed methods include those plants which are resistant to one or more abiotic stresses. Abiotic stress conditions may include, for example, drought, cold temperature exposure, heat exposure, osmotic stress, flooding, increased soil salinity, increased mineral exposure, ozone exposure, high light exposure, limited availability of nitrogen nutrients, limited availability of phosphorus nutrients, shade avoidance.

[0023] Plants and plant cultivars which may be treated by the above disclosed methods include those plants characterized by enhanced yield characteristics. Increased yield in said plants may be the result of, for example, improved plant physiology, growth and development, such as water use efficiency, water retention efficiency, improved nitrogen use, enhanced carbon assimilation, improved photosynthesis, increased germination efficiency and accelerated maturation. Yield may furthermore be affected by improved plant architecture (under stress and non-stress conditions), including but not limited to, early flowering, flowering control for hybrid seed production, seedling vigor, plant size, internode

number and distance, root growth, seed size, fruit size, pod size, pod or ear number, seed number per pod or ear, seed mass, enhanced seed filling, reduced seed dispersal, reduced pod dehiscence and lodging resistance. Further yield traits include seed composition, such as carbohydrate content and composition for example cotton or starch, protein content, oil content and composition, nutritional value, reduction in anti-nutritional compounds, improved processability and better storage stability.

[0024] Plants and plant cultivars which may be treated by the above disclosed methods include plants and plant cultivars which are hybrid plants that already express the characteristic of heterosis or hybrid vigor which results in generally higher yield, vigor, health and resistance towards biotic and abiotic stresses.

Transgenic Plants, Seed Treatment and Integration Events

[0025] The compound according to the invention can be advantageously used to treat transgenic plants, plant cultivars or plant parts that received genetic material which imparts advantageous and/or useful properties (traits) to these plants, plant cultivars or plant parts. Therefore, it is contemplated that the present invention may be combined with one or more recombinant traits or transgenic event(s) or a combination thereof. For the purposes of this application, a transgenic event is created by the insertion of a specific recombinant DNA molecule into a specific position (locus) within the chromosome of the plant genome. The insertion creates a novel DNA sequence referred to as an "event" and is characterized by the inserted recombinant DNA molecule and some amount of genomic DNA immediately adjacent to/flanking both ends of the inserted DNA. Such trait(s) or transgenic event(s) include, but are not limited to, pest resistance, water use efficiency, yield performance, drought tolerance, seed quality, improved nutritional quality, hybrid seed production, and herbicide tolerance, in which the trait is measured with respect to a plant lacking such trait or transgenic event. Concrete examples of such advantageous and/or useful properties (traits) are better plant growth, vigor, stress tolerance, standability, lodging resistance, nutrient uptake, plant nutrition, and/or yield, in particular improved growth, increased tolerance to high or low temperatures, increased tolerance to drought or to levels of water or soil salinity, enhanced flowering performance, easier harvesting, accelerated ripening, higher yields, higher quality and/or a higher nutritional value of the harvested products, better storage life and/or processability of the harvested products, and increased resistance against animal and microbial pests, such as against insects, arachnids, nematodes, mites, slugs and snails.

[0026] Among DNA sequences encoding proteins which confer properties of tolerance to such animal and microbial pests, in particular insects, mention will particularly be made of the genetic material from *Bacillus thuringiensis* encoding the Bt proteins widely described in the literature and well known to those skilled in the art. Mention will also be made of proteins extracted from bacteria such as *Photorhabdus* (WO97/17432 and WO98/08932). In particular, mention will be made of the Bt Cry or VIP proteins which include the CryIA, CryIAb, CryIAc, CryIIA, CryIIIA, CryIIIB2, Cry9c Cry2Ab, Cry3Bb and CryIF proteins or toxic fragments thereof and also hybrids or combinations thereof, especially the CryIF protein or hybrids derived from a CryIF protein (e.g. hybrid CryIA-CryIF proteins or toxic fragments

thereof), the CryIA-type proteins or toxic fragments thereof, preferably the CryIAc protein or hybrids derived from the CryIAc protein (e.g. hybrid CryIAb-CryIAc proteins) or the CryIAb or Bt2 protein or toxic fragments thereof, the Cry2Ae, Cry2Af or Cry2Ag proteins or toxic fragments thereof, the CryIA. 105 protein or a toxic fragment thereof, the VIP3Aa19 protein, the VIP3Aa20 protein, the VIP3A proteins produced in the COT202 or COT203 cotton events, the VIP3Aa protein or a toxic fragment thereof as described in Estruch et al. (1996), Proc Natl Acad Sci US A. 28; 93 (11): 5389-94, the Cry proteins as described in WO2001/47952, the insecticidal proteins from *Xenorhabdus* (as described in WO98/50427), *Serratia* (particularly from *S. entomophila*) or *Photorhabdus* species strains, such as Tc-proteins from *Photorhabdus* as described in WO98/08932. Also any variants or mutants of any one of these proteins differing in some amino acids (1-10, preferably 1-5) from any of the above named sequences, particularly the sequence of their toxic fragment, or which are fused to a transit peptide, such as a plastid transit peptide, or another protein or peptide, is included herein.

[0027] Another and particularly emphasized example of such properties is conferred tolerance to one or more herbicides, for example imidazolinones, sulphonylureas, glyphosate or phosphinothricin. Among DNA sequences encoding proteins which confer properties of tolerance to certain herbicides on the transformed plant cells and plants, mention will be particularly be made to the bar or PAT gene or the *Streptomyces coelicolor* gene described in WO2009/152359 which confers tolerance to glufosinate herbicides, a gene encoding a suitable EPSPS (5-Enolpyruvylshikimate-3-phosphat-synthase) which confers tolerance to herbicides having EPSPS as a target, especially herbicides such as glyphosate and its salts, a gene encoding glyphosate-n-acetyltransferase, or a gene encoding glyphosate oxidoreductase. Further suitable herbicide tolerance traits include at least one ALS (acetolactate synthase) inhibitor (e.g. WO2007/024782), a mutated Arabidopsis ALS/AHAS gene (e.g. U.S. Pat. No. 6,855,533), genes encoding 2,4-D-monooxygenases conferring tolerance to 2,4-D (2,4-dichlorophenoxyacetic acid) and genes encoding Dicamba monooxygenases conferring tolerance to dicamba (3,6-dichloro-2-methoxybenzoic acid).

[0028] Yet another example of such properties is resistance to one or more phytopathogenic fungi, for example Asian Soybean Rust. Among DNA sequences encoding proteins which confer properties of resistance to such diseases, mention will particularly be made of the genetic material from *glycine tomentella*, for example from any one of publically available accession lines PI441001, PI483224, PI583970, PI446958, PI499939, PI505220, PI499933, PI441008, PI505256 or PI446961 as described in WO2019/103918.

[0029] Further and particularly emphasized examples of such properties are increased resistance against bacteria and/or viruses owing, for example, to systemic acquired resistance (SAR), systemin, phytoalexins, elicitors and also resistance genes and correspondingly expressed proteins and toxins.

[0030] Particularly useful transgenic events in transgenic plants or plant cultivars which can be treated with preference in accordance with the invention include Event 531/PV-GHBK04 (cotton, insect control, described in WO2002/040677), Event 1143-14A (cotton, insect control, not deposited, described in WO2006/128569); Event 1143-51B

(cotton, insect control, not deposited, described in WO2006/128570); Event 1445 (cotton, herbicide tolerance, not deposited, described in US-A 2002-120964 or WO2002/034946); Event 17053 (rice, herbicide tolerance, deposited as PTA-9843, described in WO2010/117737); Event 17314 (rice, herbicide tolerance, deposited as PTA-9844, described in WO2010/117735); Event 281-24-236 (cotton, insect control-herbicide tolerance, deposited as PTA-6233, described in WO2005/103266 or US-A 2005-216969); Event 3006-210-23 (cotton, insect control-herbicide tolerance, deposited as PTA-6233, described in US-A 2007-143876 or WO2005/103266); Event 3272 (corn, quality trait, deposited as PTA-9972, described in WO2006/098952 or US-A 2006-230473); Event 33391 (wheat, herbicide tolerance, deposited as PTA-2347, described in WO2002/027004); Event 40416 (corn, insect control-herbicide tolerance, deposited as ATCC PTA-11508, described in WO 11/075593); Event 43A47 (corn, insect control-herbicide tolerance, deposited as ATCC PTA-11509, described in WO2011/075595); Event 5307 (corn, insect control, deposited as ATCC PTA-9561, described in WO2010/077816); Event ASR-368 (bent grass, herbicide tolerance, deposited as ATCC PTA-4816, described in US-A 2006-162007 or WO2004/053062); Event B16 (corn, herbicide tolerance, not deposited, described in US-A 2003-126634); Event BPS-CV127-9 (soybean, herbicide tolerance, deposited as NCIMB No. 41603, described in WO2010/080829); Event BLRI (oilseed rape, restoration of male sterility, deposited as NCIMB 41193, described in WO2005/074671); Event CE43-67B (cotton, insect control, deposited as DSM ACC2724, described in US-A 2009-217423 or WO2006/128573); Event CE44-69D (cotton, insect control, not deposited, described in US-A 2010-0024077); Event CE44-69D (cotton, insect control, not deposited, described in WO2006/128571); Event CE46-02A (cotton, insect control, not deposited, described in WO2006/128572); Event COT102 (cotton, insect control, not deposited, described in US-A 2006-130175 or WO2004/039986); Event COT202 (cotton, insect control, not deposited, described in US-A 2007-067868 or WO2005/054479); Event COT203 (cotton, insect control, not deposited, described in WO2005/054480); Event DAS21606-3/1606 (soybean, herbicide tolerance, deposited as PTA-11028, described in WO2012/033794); Event DAS40278 (corn, herbicide tolerance, deposited as ATCC PTA-10244, described in WO2011/022469); Event DAS-44406-6/pDAB8264.44.06.1 (soybean, herbicide tolerance, deposited as PTA-11336, described in WO2012/075426); Event DAS-14536-7/pDAB8291.45.36.2 (soybean, herbicide tolerance, deposited described in WO2012/075429); Event DAS-59122-7 (corn, insect control-herbicide tolerance, deposited as ATCC PTA 11384, described in US-A 2006-070139); Event DAS-59132 (corn, insect control-herbicide tolerance, not deposited, described in WO2009/100188); Event DAS68416 (soybean, herbicide tolerance, deposited as ATCC PTA-10442, described in WO2011/066384 or WO2011/066360); Event DP-098140-6 (corn, herbicide tolerance, deposited as ATCC PTA-8296, described in US-A 2009-137395 or WO 08/112019); Event DP-305423-1 (soybean, quality trait, not deposited, described in US-A 2008-312082 or WO2008/054747); Event DP-32138-1 (corn, hybridization system, deposited as ATCC PTA-9158, described in US-A 2009-0210970 or WO2009/103049); Event DP-356043-5 (soybean, herbicide tolerance, deposited as ATCC PTA-8287,

described in US-A 2010-0184079 or WO2008/002872); Event EE-I (brinjal, insect control, not deposited, described in WO 07/091277); Event Fil 17 (corn, herbicide tolerance, deposited as ATCC 209031, described in US-A 2006-059581 or WO 98/044140); Event FG72 (soybean, herbicide tolerance, deposited as PTA-11041, described in WO2011/063413); Event GA21 (corn, herbicide tolerance, deposited as ATCC 209033, described in US-A 2005-086719 or WO 98/044140); Event GG25 (corn, herbicide tolerance, deposited as ATCC 209032, described in US-A 2005-188434 or WO98/044140); Event GHB119 (cotton, insect control-herbicide tolerance, deposited as ATCC PTA-8398, described in WO2008/151780); Event GHB614 (cotton, herbicide tolerance, deposited as ATCC PTA-6878, described in US-A 2010-050282 or WO2007/017186); Event GJ11 (corn, herbicide tolerance, deposited as ATCC 209030, described in US-A 2005-188434 or WO98/044140); Event GM RZ13 (sugar beet, virus resistance, deposited as NCIMB-41601, described in WO2010/076212); Event H7-1 (sugar beet, herbicide tolerance, deposited as NCIMB 41158 or NCIMB 41159, described in US-A 2004-172669 or WO 2004/074492); Event JOPLINI (wheat, disease tolerance, not deposited, described in US-A 2008-064032); Event LL27 (soybean, herbicide tolerance, deposited as NCIMB41658, described in WO2006/108674 or US-A 2008-320616); Event LL55 (soybean, herbicide tolerance, deposited as NCIMB 41660, described in WO 2006/108675 or US-A 2008-196127); Event LLCotton25 (cotton, herbicide tolerance, deposited as ATCC PTA-3343, described in WO2003/013224 or US-A 2003-097687); Event LLRICE06 (rice, herbicide tolerance, deposited as ATCC 203353, described in U.S. Pat. No. 6,468,747 or WO2000/026345); Event LLRice62 (rice, herbicide tolerance, deposited as ATCC 203352, described in WO2000/026345); Event LLRICE601 (rice, herbicide tolerance, deposited as ATCC PTA-2600, described in US-A 2008-2289060 or WO2000/026356); Event LY038 (corn, quality trait, deposited as ATCC PTA-5623, described in US-A 2007-028322 or WO2005/061720); Event MIR162 (corn, insect control, deposited as PTA-8166, described in US-A 2009-300784 or WO2007/142840); Event MIR604 (corn, insect control, not deposited, described in US-A 2008-167456 or WO2005/103301); Event MON15985 (cotton, insect control, deposited as ATCC PTA-2516, described in US-A 2004-250317 or WO2002/100163); Event MON810 (corn, insect control, not deposited, described in US-A 2002-102582); Event MON863 (corn, insect control, deposited as ATCC PTA-2605, described in WO2004/011601 or US-A 2006-095986); Event MON87427 (corn, pollination control, deposited as ATCC PTA-7899, described in WO2011/062904); Event MON87460 (corn, stress tolerance, deposited as ATCC PTA-8910, described in WO2009/111263 or US-A 2011-0138504); Event MON87701 (soybean, insect control, deposited as ATCC PTA-8194, described in US-A 2009-130071 or WO2009/064652); Event MON87705 (soybean, quality trait-herbicide tolerance, deposited as ATCC PTA-9241, described in US-A 2010-0080887 or WO2010/037016); Event MON87708 (soybean, herbicide tolerance, deposited as ATCC PTA-9670, described in WO2011/034704); Event MON87712 (soybean, yield, deposited as PTA-10296, described in WO2012/051199); Event MON87754 (soybean, quality trait, deposited as ATCC PTA-9385, described in WO2010/024976); Event MON87769 (soybean, quality trait, depos-

ited as ATCC PTA-8911, described in US-A 2011-0067141 or WO2009/102873); Event MON88017 (corn, insect control-herbicide tolerance, deposited as ATCC PTA-5582, described in US-A 2008-028482 or WO2005/059103); Event MON88913 (cotton, herbicide tolerance, deposited as ATCC PTA-4854, described in WO2004/072235 or US-A 2006-059590); Event MON88302 (oilseed rape, herbicide tolerance, deposited as PTA-10955, described in WO2011/153186), Event MON88701 (cotton, herbicide tolerance, deposited as PTA-11754, described in WO2012/134808), Event MON89034 (corn, insect control, deposited as ATCC PTA-7455, described in WO 07/140256 or US-A 2008-260932); Event MON89788 (soybean, herbicide tolerance, deposited as ATCC PTA-6708, described in US-A 2006-282915 or WO2006/130436); Event MSI 1 (oilseed rape, pollination control-herbicide tolerance, deposited as ATCC PTA-850 or PTA-2485, described in WO2001/031042); Event MS8 (oilseed rape, pollination control-herbicide tolerance, deposited as ATCC PTA-730, described in WO2001/041558 or US-A 2003-188347); Event NK603 (corn, herbicide tolerance, deposited as ATCC PTA-2478, described in US-A 2007-292854); Event PE-7 (rice, insect control, not deposited, described in WO2008/114282); Event RF3 (oilseed rape, pollination control herbicide tolerance, deposited as ATCC PTA-730, described in WO2001/041558 or US-A 2003-188347); Event RT73 (oilseed rape, herbicide tolerance, not deposited, described in WO2002/036831 or US-A 2008-070260); Event SYHTOH2/SYN-000H2-5 (soybean, herbicide tolerance, deposited as PTA-11226, described in WO2012/082548), Event T227-1 (sugar beet, herbicide tolerance, not deposited, described in WO2002/44407 or US-A 2009-265817); Event T25 (corn, herbicide tolerance, not deposited, described in US-A 2001-029014 or WO2001/051654); Event T304-40 (cotton, insect control-herbicide tolerance, deposited as ATCC PTA-8171, described in US-A 2010-077501 or WO2008/122406); Event T342-142 (cotton, insect control, not deposited, described in WO2006/128568); Event TC1507 (corn, insect control-herbicide tolerance, not deposited, described in US-A 2005-039226 or WO2004/099447); Event VIP1034 (corn, insect control-herbicide tolerance, deposited as ATCC PTA-3925, described in WO2003/052073), Event 32316 (corn, insect control-herbicide tolerance, deposited as PTA-11507, described in WO2011/084632), Event 4114 (corn, insect control-herbicide tolerance, deposited as PTA-11506, described in WO2011/084621), event EE-GM3/FG72 (soybean, herbicide tolerance, ATCC Accession No PTA-11041) optionally stacked with event EE-GM1/LL27 or event EE-GM2/LL55 (WO2011/063413A2), event DAS-68416-4 (soybean, herbicide tolerance, ATCC Accession No PTA-10442, WO2011/066360A1), event DAS-68416-4 (soybean, herbicide tolerance, ATCC Accession No PTA-10442, WO2011/066384A1), event DP-040416-8 (corn, insect control, ATCC Accession No PTA-11508, WO2011/075593A1), event DP-043A47-3 (corn, insect control, ATCC Accession No PTA-11509, WO2011/075595A1), event DP-004114-3 (corn, insect control, ATCC Accession No PTA-11506, WO2011/084621A1), event DP-032316-8 (corn, insect control, ATCC Accession No PTA-11507, WO2011/084632A1), event MON-88302-9 (oilseed rape, herbicide tolerance, ATCC Accession No PTA-10955, WO2011/153186A1), event DAS-21606-3 (soybean, herbicide tolerance, ATCC Accession No. PTA-11028, WO2012/033794A2), event MON-87712-4 (soybean, quality trait, ATCC Accession No.

PTA-10296, WO2012/051199A2), event DAS-44406-6 (soybean, stacked herbicide tolerance, ATCC Accession No. PTA-11336, WO2012/075426A1), event DAS-14536-7 (soybean, stacked herbicide tolerance, ATCC Accession No. PTA-11335, WO2012/075429A1), event SYN-000H2-5 (soybean, herbicide tolerance, ATCC Accession No. PTA-11226, WO2012/082548A2), event DP-061061-7 (oilseed rape, herbicide tolerance, no deposit No available, WO2012071039A1), event DP-073496-4 (oilseed rape, herbicide tolerance, no deposit No available, US2012131692), event 8264.44.06.1 (soybean, stacked herbicide tolerance, Accession No PTA-11336, WO2012075426A2), event 8291.45.36.2 (soybean, stacked herbicide tolerance, Accession No. PTA-11335, WO2012075429A2), event SYHTOH2 (soybean, ATCC Accession No. PTA-11226, WO2012/082548A2), event MON88701 (cotton, ATCC Accession No PTA-11754, WO2012/134808A1), event KK179-2 (alfalfa, ATCC Accession No PTA-11833, WO2013/003558A1), event pDAB8264.42.32.1 (soybean, stacked herbicide tolerance, ATCC Accession No PTA-11993, WO2013/010094A1), event MZDT09Y (corn, ATCC Accession No PTA-13025, WO2013/012775A1).

[0031] Further, a list of such transgenic event(s) is provided by the United States Department of Agriculture's (USDA) Animal and Plant Health Inspection Service (APHIS) and can be found on their website on the world wide web at aphis.usda.gov. For this application, the status of such list as it is/was on the filing date of this application, is relevant.

[0032] The genes/events which impart the desired traits in question may also be present in combinations with one another in the transgenic plants. Examples of transgenic plants which may be mentioned are the important crop plants, such as cereals (wheat, rice, triticale, barley, rye, oats), maize, soya beans, potatoes, sugar beet, sugar cane, tomatoes, peas and other types of vegetable, cotton, tobacco, oilseed rape and also fruit plants (with the fruits apples, pears, citrus fruits and grapes), with particular emphasis being given to maize, soya beans, wheat, rice, potatoes, cotton, sugar cane, tobacco and oilseed rape. Traits which are particularly emphasized are the increased resistance of the plants to insects, arachnids, nematodes and slugs and snails, as well as the increased resistance of the plants to one or more herbicides.

[0033] Commercially available examples of such plants, plant parts or plant seeds that may be treated with preference in accordance with the invention include commercial products, such as plant seeds, sold or distributed under the GENUITY®, DROUGHTGARD®, SMARTSTAX®, RIB COMPLETE®, ROUNDUP READY®, VT DOUBLE PRO®, VT TRIPLE PRO®, BOLLGARD II®, ROUNDUP READY 2 YIELD®, YIELDGARD®, ROUNDUP READY® 2 XTEND™, INTACTA RR2 PRO®, VISTIVE GOLD®, and/or XTENDFLEX™ trade names.

[0034] The compound of the formula (I) can preferably be used as fungicide.

Pathogens

[0035] Non-limiting examples of pathogens of fungal diseases which may be treated in accordance with the invention include:

[0036] diseases caused by powdery mildew pathogens, for example *Blumeria* species, for example *Blumeria graminis*; *Leveillula* species, for example *Leveillula*

Taurica; *Podosphaera* species, for example *Podosphaera leucotricha* or *Podosphaera xanthii*; *Sphaerotheca* species, for example *Sphaerotheca fuliginea*; *Erysiphe* species, for example *Erysiphe necator*, *Erysiphe betae*;

[0037] diseases caused by rust disease pathogens, for example *Gymnosporangium* species, for example *Gymnosporangium sabinae*; *Hemileia* species, for example *Hemileia vastatrix*; *Phakopsora* species, for example *Phakopsora pachyrhizi* or *Phakopsora meibomiae*; *Puccinia* species, for example *Puccinia recondita*, *Puccinia graminis* oder *Puccinia striiformis*; *Uromyces* species, for example *Uromyces appendiculatus*;

[0038] leaf blotch diseases and leaf wilt diseases caused, for example, by *Alternaria* species, for example *Alternaria solani* or *Alternaria mali* or *Alternaria alternata*; *Cercospora* species, for example *Cercospora beticola*; *Cladosporium* species, for example *Cladosporium cucumerinum*; *Cochliobolus* species, for example *Cochliobolus sativus* (conidial form: *Drechslera*, syn: *Helminthosporium*) or *Cochliobolus miyabeanus*; *Colletotrichum* species, for example *Colletotrichum lindemuthianum* or *Colletotrichum capsica* or *Colletotrichum acutatum*; *Corynespora* species, for example *Corynespora cassiicola*; *Cyloconium* species, for example *Cyloconium oleaginum*; *Diaporthe* species, for example *Diaporthe citri*; *Elsinoe* species, for example *Elsinoe fawcettii*; *Gloeosporium* species, for example *Gloeosporium laeticolor*; *Glomerella* species, for example *Glomerella cingulata*; *Guignardia* species, for example *Guignardia bidwelli*; *Leptosphaeria* species, for example *Leptosphaeria maculans*; *Magnaporthe* species, for example *Magnaporthe grisea*; *Marsonina* species, for example *Marsonina mali*; *Microdochium* species, for example *Microdochium nivale*; *Monilinia* species, for example *Monilinia laxa*; *Mycosphaerella* species, for example *Mycosphaerella graminicola*, *Mycosphaerella arachidicola* or *Mycosphaerella fijiensis*; *Phaeosphaeria* species, for example *Phaeosphaeria nodorum*; *Phyllachora* species, for example *Phyllachora maydis*, *Pyrenophora* species, for example *Pyrenophora teres* or *Pyrenophora tritici repentis*; *Ramularia* species, for example *Ramularia collo-cygni* or *Ramularia areola*; *Rhynchosporium* species, for example *Rhynchosporium secalis*; *Septoria* species, for example *Septoria apii* or *Septoria lycopersici*; *Setosphaeria* species, for example *Setosphaeria turcica*; *Sclerotinia* species, for example *Sclerotinia sclerotiorum*; *Stagonospora* species, for example *Stagonospora nodorum*; *Stemphylium* species, for example *Stemphylium vesicarium*; *Typhula* species, for example *Typhula incarnata*; *Venturia* species, for example *Venturia inaequalis*;

[0039] root and stem diseases caused, for example, by *Corticium* species, for example *Corticium graminearum*; *Fusarium* species, for example *Fusarium oxysporum*; *Gaeumannomyces* species, for example *Gaeumannomyces graminis*; *Marsonina* species, for example *Marsonina mali*; *Rhizoctonia* species, for example *Rhizoctonia solani*; *Sarocladium* species, for example *Sarocladium oryzae*; *Sclerotium* species, for example *Sclerotium oryzae*; *Tapesia* species, for example *Tapesia acufiformis*; *Thielaviopsis* species, for example *Thielaviopsis basicola*;

[0040] ear and panicle diseases (including corn cobs) caused, for example, by *Alternaria* species, for example *Alternaria* spp.; *Aspergillus* species, for example *Aspergillus flavus*; *Cladosporium* species, for example *Cladosporium cladosporioides*; *Claviceps* species, for example *Claviceps purpurea*; *Fusarium* species, for example *Fusarium culmorum*; *Gibberella* species, for example *Gibberella zeae*; *Monographella* species, for example *Monographella nivalis*; *Stagonospora* species, for example *Stagonospora nodorum*;

[0041] diseases caused by smut fungi, for example *Sphacelotheca* species, for example *Sphacelotheca reiliana*; *Tilletia* species, for example *Tilletia caries* or *Tilletia controversa*; *Urocystis* species, for example *Urocystis occulta*; *Ustilago* species, for example *Ustilago nuda*;

[0042] fruit rot caused, for example, by *Aspergillus* species, for example *Aspergillus flavus*; *Botrytis* species, for example *Botrytis cinerea*; *Monilinia* species, for example *Monilinia laxa*; *Penicillium* species, for example *Penicillium expansum* or *Penicillium purpogenum*; *Phomopsis* species, for example *Phomopsis viticola*; *Rhizopus* species, for example *Rhizopus stolonifer*; *Sclerotinia* species, for example *Sclerotinia sclerotiorum*; *Verticillium* species, for example *Verticillium albo-atrum*;

[0043] seed- and soil-borne rot and wilt diseases, and also diseases of seedlings, caused, for example, by *Alternaria* species, for example *Alternaria brassicicola*; *Aphanomyces* species, for example *Aphanomyces euteiches*; *Ascochyta* species, for example *Ascochyta lentis*; *Aspergillus* species, for example *Aspergillus flavus*; *Cladosporium* species, for example *Cladosporium herbarum*; *Cochliobolus* species, for example *Cochliobolus sativus* (conidial form: *Drechslera*, *Bipolaris* Syn: *Helminthosporium*); *Colletotrichum* species, for example *Colletotrichum coccodes*; *Fusarium* species, for example *Fusarium culmorum*; *Gibberella* species, for example *Gibberella zeae*; *Macrophomina* species, for example *Macrophomina phaseolina*; *Microdochium* species, for example *Microdochium nivale*; *Monographella* species, for example *Monographella nivalis*; *Penicillium* species, for example *Penicillium expansum*; *Phoma* species, for example *Phoma lingam*; *Phomopsis* species, for example *Phomopsis sojae*; *Pyrenophora* species, for example *Pyrenophora graminea*; *Pyricularia* species, for example *Pyricularia oryzae*; *Rhizoctonia* species, for example *Rhizoctonia solani*; *Rhizopus* species, for example *Rhizopus oryzae*; *Sclerotium* species, for example *Sclerotium rolfsii*; *Septoria* species, for example *Septoria nodorum*; *Typhula* species, for example *Typhula incarnata*; *Verticillium* species, for example *Verticillium dahliae*;

[0044] cancers, galls and witches' broom caused, for example, by *Nectria* species, for example *Nectria galligena*;

[0045] wilt diseases caused, for example, by *Verticillium* species, for example *Verticillium longisporum*; *Fusarium* species, for example *Fusarium oxysporum*;

[0046] deformations of leaves, flowers and fruits caused, for example, by *Exobasidium* species, for example *Exobasidium vexans*; *Taphrina* species, for example *Taphrina deformans*;

- [0047] degenerative diseases in woody plants, caused, for example, by *Esca* species, for example *Phaeomonella chlamydospora*, *Phaeoacremonium aleophilum* or *Fomitiporia mediterranea*; *Ganoderma* species, for example *Ganoderma boninense*;
- [0048] diseases of plant tubers caused, for example, by *Rhizoctonia* species, for example *Rhizoctonia solani*; *Helminthosporium* species, for example *Helminthosporium solani*;
- [0049] diseases of soya beans:
- [0050] fungal diseases on leaves, stems, pods and seeds caused, for example, by *Alternaria* leaf spot (*Alternaria* spec. *atrans tenuissima*), *Anthrachnose* (*Colletotrichum gloeosporoides dematium* var. *truncatum*), brown spot (*Septoria glycines*), *cercospora* leaf spot and blight (*Cercospora kikuchii*), choanephora leaf blight (*Choanephora infundibulifera trispora* (Syn.)), *cordana* leaf spot on bananas (*Cordana musae*), *dactuliophora* leaf spot (*Dactuliophora glycines*), downy mildew (*Peronospora manshurica*), *drechslera* blight (*Drechslera glycini*), frog-eye leaf spot (*Cercospora sojae*), *leptosphaerulina* leaf spot (*Leptosphaerulina trifolii*), *Leveillula* powdery mildew (*Leveillula Taurica*), *phyllosticta* leaf spot (*Phyllosticta sojaecola*), pod and stem blight (*Phomopsis sojae*), powdery mildew (*Microsphaera diffusa*), *pyrenochaeta* leaf spot (*Pyrenochaeta glycines*), *rhizoctonia* aerial, foliage, and web blight (*Rhizoctonia solani*), rust (*Phakopsora pachyrhizi*, *Phakopsora meibomia*), scab (*Sphaceloma glycines*), stemphylium leaf blight (*Stemphylium botryosum*), sudden death syndrome (*Fusarium virguliforme*), target spot (*Corynespora cassiicola*);
- [0051] fungal diseases on roots and the stem base caused, for example, by black root rot (*Calonectria crotalariae*), charcoal rot (*Macrophomina phaseolina*), fusarium blight or wilt, root rot, and pod and collar rot (*Fusarium oxysporum*, *Fusarium orthoceras*, *Fusarium semitectum*, *Fusarium equiseti*), mycoleptodiscus root rot (*Mycoleptodiscus terrestris*), neocosmospora (*Neocosmospora vasinfecta*), pod and stem blight (*Diaporthe phaseolorum*), stem canker (*Diaporthe phaseolorum* var. *caulivora*), brown stem rot (*Phialophora gregata*), *rhizoctonia* root rot, stem decay, and damping-off (*Rhizoctonia solani*), sclerotinia stem decay (*Sclerotinia sclerotiorum*), sclerotinia southern blight (*Sclerotinia rolfsii*), *thielaviopsis* root rot (*Thielaviopsis basicola*).
- [0052] Preferably, the present invention relates to the use of compound of formula (I) for controlling diseases caused by powdery mildew pathogens, for example *Blumeria* species, for example *Blumeria graminis*; *Leveillula* species, for example *Leveillula Taurica*; *Podosphaera* species, for example *Podosphaera leucotricha* or *Podosphaera xanthii*; *Sphaerotheca* species, for example *Sphaerotheca fuliginea*; *Erysiphe* species, for example *Erysiphe necator*, *Erysiphe betae*;
- [0053] leaf blotch diseases and leaf wilt diseases caused, for example, by *Alternaria* species, for example *Alternaria solani* or *Alternaria mali* or *Alternaria alternata*; *Cercospora* species, for example *Cercospora beticola*; *Cladosporium* species, for example *Cladosporium cucumerinum*; *Cochliobolus* species, for example *Cochliobolus sativus* (conidial form: *Drechslera*, syn: *Helminthosporium*) or *Cochliobolus miyabeanus*; *Colletotrichum* species, for example *Colletotrichum lindemuthianum* or *Colletotrichum capsica* or *Colletotrichum acutatum*; *Corynespora* species, for example *Corynespora cassiicola*; *Cycloconium* species, for example *Cycloconium oleaginum*; *Diaporthe* species, for example *Diaporthe citri*; *Elsinoe* species, for example *Elsinoe fawcettii*; *Gloeosporium* species, for example *Gloeosporium laeticolor*; *Glomerella* species, for example *Glomerella cingulata*; *Guignardia* species, for example *Guignardia bidwellii*; *Leptosphaeria* species, for example *Leptosphaeria maculans*; *Magnaporthe* species, for example *Magnaporthe grisea*; *Marsonina* species, for example *Marsonina mali*; *Microdochium* species, for example *Microdochium nivale*; *Monilinia* species, for example *Monilinia laxa*; *Mycosphaerella* species, for example *Mycosphaerella graminicola*, *Mycosphaerella arachidicola* or *Mycosphaerella fijiensis*; *Phaeosphaeria* species, for example *Phaeosphaeria nodorum*; *Phyllachora* species, for example *Phyllachora maydis*, *Pyrenophora* species, for example *Pyrenophora teres* or *Pyrenophora tritici repentis*; *Ramularia* species, for example *Ramularia collo-cygni* or *Ramularia areola*; *Rhynchosporium* species, for example *Rhynchosporium secalis*; *Septoria* species, for example *Septoria apii* or *Septoria lycopersici*; *Setosphaeria* species, for example *Setosphaeria turcica*; *Sclerotinia* species, for example *Sclerotinia sclerotiorum*; *Stagonospora* species, for example *Stagonospora nodorum*; *Stemphylium* species, for example *Stemphylium vesicarium*; *Typhula* species, for example *Typhula incarnata*; *Venturia* species, for example *Venturia inaequalis*;
- [0054] root and stem diseases caused, for example, by *Corticium* species, for example *Corticium graminearum*; *Fusarium* species, for example *Fusarium oxysporum*; *Gaeumannomyces* species, for example *Gaeumannomyces graminis*; *Marsonina* species, for example *Marsonina mali*; *Rhizoctonia* species, for example *Rhizoctonia solani*; *Sarocladium* species, for example *Sarocladium oryzae*; *Sclerotium* species, for example *Sclerotium oryzae*; *Tapesia* species, for example *Tapesia acuformis*; *Thielaviopsis* species, for example *Thielaviopsis basicola*;
- [0055] ear and panicle diseases (including corn cobs) caused, for example, by *Alternaria* species, for example *Alternaria* spp.; *Aspergillus* species, for example *Aspergillus flavus*; *Cladosporium* species, for example *Cladosporium cladosporioides*; *Claviceps* species, for example *Claviceps purpurea*; *Fusarium* species, for example *Fusarium culmorum*; *Gibberella* species, for example *Gibberella zeae*; *Monographella* species, for example *Monographella nivalis*; *Stagnospora* species, for example *Stagnospora nodorum*;
- [0056] fruit rot caused, for example, by *Aspergillus* species, for example *Aspergillus flavus*; *Botrytis* species, for example *Botrytis cinerea*; *Monilinia* species, for example *Monilinia laxa*; *Penicillium* species, for example *Penicillium expansum* or *Penicillium purpogenum*; *Phomopsis* species, for example *Phomopsis viticola*; *Rhizopus* species, for example *Rhizopus stolonifer*; *Sclerotinia* species, for example *Sclerotinia sclerotiorum*; *Verticillium* species, for example *Verticillium albo-atrum*;

- [0057] seed- and soil-borne rot and wilt diseases, and also diseases of seedlings, caused, for example, by *Alternaria* species, for example *Alternaria brassicicola*; *Aphanomyces* species, for example *Aphanomyces euteiches*; *Ascochyta* species, for example *Ascochyta lentis*; *Aspergillus* species, for example *Aspergillus flavus*; *Cladosporium* species, for example *Cladosporium herbarum*; *Cochliobolus* species, for example *Cochliobolus sativus* (conidial form: *Drechslera*, *Bipolaris* Syn: *Helminthosporium*); *Colletotrichum* species, for example *Colletotrichum coccodes*; *Fusarium* species, for example *Fusarium culmorum*; *Gibberella* species, for example *Gibberella zeae*; *Macrophomina* species, for example *Macrophomina phaseolina*; *Microdochium* species, for example *Microdochium nivale*; *Monographella* species, for example *Monographella nivalis*; *Penicillium* species, for example *Penicillium expansum*; *Phoma* species, for example *Phoma lingam*; *Phomopsis* species, for example *Phomopsis sojae*; *Pyrenophora* species, for example *Pyrenophora graminea*; *Pyricularia* species, for example *Pyricularia oryzae*; *Rhizoctonia* species, for example *Rhizoctonia solani*; *Rhizopus* species, for example *Rhizopus oryzae*; *Sclerotium* species, for example *Sclerotium rolfsii*; *Septoria* species, for example *Septoria nodorum*; *Typhula* species, for example *Typhula incarnata*; *Verticillium* species, for example *Verticillium dahliae*;
- [0058] cancers, galls and witches' broom caused, for example, by *Nectria* species, for example *Nectria galligena*;
- [0059] wilt diseases caused, for example, by *Fusarium* species, for example *Fusarium oxysporum*;
- [0060] diseases of soya beans:
- [0061] fungal diseases on leaves, stems, pods and seeds caused, for example, by *Alternaria* leaf spot (*Alternaria spec. atrans tenuissima*), *Anthraco*se (*Colletotrichum gloeosporoides dematium* var. *truncatum*), brown spot (*Septoria glycines*), *cercospora* leaf spot and blight (*Cercospora kikuchii*), choanephora leaf blight (*Choanephora infundibulifera trispora* (Syn.)), *dactulio*phora leaf spot (*Dactuliothra glycines*), downy mildew (*Peronospora manshurica*), *drechslera* blight (*Drechslera glycini*), frog-eye leaf spot (*Cercospora sojae*), *leptosphaerulina* leaf spot (*Leptosphaerulina trifolii*), *Leveillula* powdery mildew (*Leveillula taurica*), *phyllosticta* leaf spot (*Phyllosticta sojaecola*), pod and stem blight (*Phomopsis sojae*), powdery mildew (*Microspora diffusa*), *pyrenochaeta* leaf spot (*Pyrenochaeta glycines*), *rhizoctonia* aerial, foliage, and web blight (*Rhizoctonia solani*), scab (*Sphaceloma glycines*), *stemphyllium* leaf blight (*Stemphyllium botryosum*), sudden death syndrome (*Fusarium virguliforme*), target spot (*Corynespora cassiicola*);
- [0062] fungal diseases on roots and the stem base caused, for example, by black root rot (*Calonectria crotalariae*), charcoal rot (*Macrophomina phaseolina*), *fusarium* blight or wilt, root rot, and pod and collar rot (*Fusarium oxysporum*, *Fusarium orthoceras*, *Fusarium semitectum*, *Fusarium equiseti*), *mycolepto*-discus root rot (*Mycoleptodiscus terrestris*), *neocos*-mospora (*Neocosmopora vasinfecta*), pod and stem blight (*Diaporthe phaseolorum*), stem canker (*Diaporthe phaseolorum* var. *caulivora*), brown stem rot (*Phialophora gregata*), *rhizoctonia* root rot, stem

decay, and damping-off (*Rhizoctonia solani*), *sclero*-tinia stem decay (*Sclerotinia sclerotiorum*), *sclerotinia* southern blight (*Sclerotinia rolfsii*), *thielaviopsis* root rot (*Thielaviopsis basicola*).

[0063] Most preferably, the present invention relates to the use of compound of formula (I) for controlling diseases caused by powdery mildew pathogens, *Sphaerotheca* species, for example *Sphaerotheca fuliginea*;

[0064] leaf blotch diseases and leaf wilt diseases caused, for example, by *Alternaria* species, for example *Alternaria solani* or *Alternaria mali* or *Alternaria alternata*, *Cercospora* species, for example *Cercospora beticola*, *Colletotrichum* species, for example *Colletotrichum lindemuthianum* or *Colletotrichum capsica* or *Colletotrichum acutatum*; *Diaporthe* species, for example *Diaporthe citri*; *Phaeosphaeria* species, for example *Phaeosphaeria nodorum*; *Pyrenophora* species, for example *Pyrenophora teres* or *Pyrenophora tritici repentis*; *Septoria* species, for example *Septoria apii* or *Septoria lycopersici*;

[0065] root and stem diseases caused, for example, *Fusarium* species, for example *Fusarium oxysporum*;

[0066] ear and panicle diseases (including corn cobs) caused, for example, by *Alternaria* species, for example *Alternaria* spp.; *Fusarium* species, for example *Fusarium culmorum*;

[0067] fruit rot caused, for example by *Botrytis* species, for example *Botrytis cinerea*;

[0068] seed- and soil-borne rot and wilt diseases, and also diseases of seedlings, caused, for example, by *Alternaria* species, for example *Alternaria brassicicola*; *Colletotrichum* species, for example *Colletotrichum coccodes*; *Fusarium* species, for example *Fusarium culmorum*; *Pyrenophora* species, for example *Pyrenophora graminea*; *Pyricularia* species, for example *Pyricularia oryzae*;

[0069] wilt diseases caused, for example, by *Fusarium* species, for example *Fusarium oxysporum*.

Mycotoxins

[0070] In addition, the compound of formula (I) and composition comprising thereof may reduce the mycotoxin content in the harvested material and the foods and feeds prepared therefrom. Mycotoxins include particularly, but not exclusively, the following: deoxynivalenol (DON), nivalenol, 15-Ac-DON, 3-Ac-DON, T2- and HT2-toxin, fumonisins, zearalenon, moniliformin, fusarin, diacetoxyscirpenol (DAS), beauvericin, enniatin, fusaroproliferin, fusarenol, ochratoxins, patulin, ergot alkaloids and aflatoxins which can be produced, for example, by the following fungi: *Fusarium* spec., such as *F. acuminatum*, *F. asiaticum*, *F. avenaceum*, *F. crookwellense*, *F. culmorum*, *F. graminearum* (*Gibberella zeae*), *F. equiseti*, *F. fujikuroi*, *F. musarum*, *F. oxysporum*, *F. proliferatum*, *F. poae*, *F. pseudo-graminearum*, *F. sambucinum*, *F. scirpi*, *F. semitectum*, *F. solani*, *F. sporotrichoides*, *F. langsethiae*, *F. subglutinans*, *F. tricinctum*, *F. verticillioides* etc., and also by *Aspergillus* spec., such as *A. flavus*, *A. parasiticus*, *A. nomius*, *A. ochraceus*, *A. clavatus*, *A. terreus*, *A. versicolor*, *Penicillium* spec., such as *P. verrucosum*, *P. viridicatum*, *P. citrinum*, *P. expansum*, *P. claviforme*, *P. roqueforti*, *Claviceps* spec., such as *C. purpurea*, *C. fusiformis*, *C. paspali*, *C. africana*, *Stachybotrys* spec. and others.

Material Protection

[0071] The compound of formula (I) and composition comprising thereof may also be used in the protection of materials, especially for the protection of industrial materials against attack and destruction by phytopathogenic fungi.

[0072] In addition, the compound of formula (I) and composition comprising thereof may be used as antifouling compositions, alone or in combinations with other active ingredients.

[0073] Industrial materials in the present context are understood to mean inanimate materials which have been prepared for use in industry. For example, industrial materials which are to be protected from microbial alteration or destruction may be adhesives, glues, paper, wallpaper and board/cardboard, textiles, carpets, leather, wood, fibers and tissues, paints and plastic articles, cooling lubricants and other materials which can be infected with or destroyed by microorganisms. Parts of production plants and buildings, for example cooling-water circuits, cooling and heating systems and ventilation and air-conditioning units, which may be impaired by the proliferation of microorganisms may also be mentioned within the scope of the materials to be protected. Industrial materials within the scope of the present invention preferably include adhesives, sizes, paper and card, leather, wood, paints, cooling lubricants and heat transfer fluids, more preferably wood.

[0074] The compound of formula (I) and composition comprising thereof may prevent adverse effects, such as rotting, decay, discoloration, decoloration or formation of mould.

[0075] In the case of treatment of wood the compound of formula (I) and composition comprising thereof may also be used against fungal diseases liable to grow on or inside timber.

[0076] Timber means all types of species of wood, and all types of working of this wood intended for construction, for example solid wood, high-density wood, laminated wood, and plywood. In addition, the compound of formula (I) and composition comprising thereof may be used to protect objects which come into contact with saltwater or brackish water, especially hulls, screens, nets, buildings, moorings and signalling systems, from fouling.

[0077] The compound of formula (I) and composition comprising thereof may also be employed for protecting storage goods. Storage goods are understood to mean natural substances of vegetable or animal origin or processed products thereof which are of natural origin, and for which long-term protection is desired. Storage goods of vegetable origin, for example plants or plant parts, such as stems, leaves, tubers, seeds, fruits, grains, may be protected freshly harvested or after processing by (pre) drying, moistening, comminuting, grinding, pressing or roasting. Storage goods also include timber, both unprocessed, such as construction timber, electricity poles and barriers, or in the form of finished products, such as furniture. Storage goods of animal origin are, for example, hides, leather, furs and hairs. The compound of formula (I) and composition comprising thereof may prevent adverse effects, such as rotting, decay, discoloration, decoloration or formation of mould.

[0078] Microorganisms capable of degrading or altering industrial materials include, for example, bacteria, fungi, yeasts, algae and slime organisms. The compound of formula (I) and composition comprising thereof preferably act against fungi, especially moulds, wood-discoloring and

wood-destroying fungi (Ascomycetes, Basidiomycetes, Deuteromycetes and Zygomycetes), and against slime organisms and algae. Examples include microorganisms of the following genera: *Alternaria*, such as *Alternaria tenuis*; *Aspergillus*, such as *Aspergillus niger*; *Chaetomium*, such as *Chaetomium globosum*; *Coniophora*, such as *Coniophora puetana*; *Lentinus*, such as *Lentinus tigrinus*; *Penicillium*, such as *Penicillium glaucum*; *Polyporus*, such as *Polyporus versicolor*; *Aureobasidium*, such as *Aureobasidium pullulans*; *Sclerophoma*, such as *Sclerophoma pityophila*; *Trichoderma*, such as *Trichoderma viride*; *Ophiostoma* spp., *Ceratocystis* spp., *Humicola* spp., *Petriella* spp., *Trichurus* spp., *Coriolus* spp., *Gloeophyllum* spp., *Pleurotus* spp., *Poria* spp., *Serpula* spp. and *Tyromyces* spp., *Cladosporium* spp., *Paecilomyces* spp. *Mucor* spp., *Escherichia*, such as *Escherichia coli*; *Pseudomonas*, such as *Pseudomonas aeruginosa*; *Staphylococcus*, such as *Staphylococcus aureus*, *Candida* spp. and *Saccharomyces* spp., such as *Saccharomyces cerevisiae*.

[0079] The plant protection agent may additionally comprise one or more further active ingredients like fungicides, bactericides, acaricides, nematocides, insecticides, biological control agents or herbicides. Mixtures with fertilizers, growth regulators, safeners, nitrification inhibitors, semiochemicals and/or other agriculturally beneficial agents are also possible. This may allow to broaden the activity spectrum or to prevent development of resistance.

[0080] The active compounds identified here by their common names are known and are described, for example, in the pesticide handbook ("The Pesticide Manual" 17th Ed., British Crop Protection Council 2015) or can be found on the Internet (e.g. <http://www.alanwood.net/pesticides>).

[0081] Examples of fungicides which could be mixed with the compound and the composition of the invention are:

[0082] 1) Inhibitors of the ergosterol biosynthesis, for example (1.001) cyproconazole, (1.002) difenoconazole, (1.003) epoxiconazole, (1.004) fenbuconazole, (1.005) fenhexamid, (1.006) fenpropidin, (1.007) fenpropimorph, (1.008) fenpyrazamine, (1.009) Fluoxystroconazole, (1.010) fluquinconazole, (1.011) flutriafol, (1.012) hexaconazole, (1.013) imazalil, (1.014) imazalil sulfate, (1.015) ipconazole, (1.016) ipfentrifluconazole, (1.017) mefentrifluconazole, (1.018) metconazole, (1.019) myclobutanil, (1.020) paclobutrazol, (1.021) penconazole, (1.022) prochloraz, (1.023) propiconazole, (1.024) prothioconazole, (1.025) pyrisoxazole, (1.026) spiroxamine, (1.027) tebuconazole, (1.028) tetraconazole, (1.029) triadimenol, (1.030) tridemorph, (1.031) triticonazole, (1.032) (1R,2S,5S)-5-(4-chlorobenzyl)-2-(chloromethyl)-2-methyl-1-(1H-1,2,4-triazol-1-ylmethyl) cyclopentanol, (1.033) (1S,2R,5R)-5-(4-chlorobenzyl)-2-(chloromethyl)-2-methyl-1-(1H-1,2,4-triazol-1-ylmethyl)cyclopentanol, (1.034) (2R)-2-(1-chlorocyclopropyl)-4-[(1R)-2,2-dichlorocyclopropyl]-1-(1H-1,2,4-triazol-1-yl) butan-2-ol, (1.035) (2R)-2-(1-chlorocyclopropyl)-4-[(1S)-2,2-dichlorocyclopropyl]-1-(1H-1,2,4-triazol-1-yl) butan-2-ol, (1.036) (2R)-2-[4-(4-chlorophenoxy)-2-(trifluoromethyl)phenyl]-1-(1H-1,2,4-triazol-1-yl) propan-2-ol, (1.037) (2S)-2-(1-chlorocyclopropyl)-4-[(1R)-2,2-dichlorocyclopropyl]-1-(1H-1,2,4-triazol-1-yl) butan-2-ol, (1.038) (2S)-2-(1-chlorocyclopropyl)-4-[(1S)-2,2-dichlorocyclopropyl]-1-(1H-1,2,4-triazol-1-yl) butan-2-ol, (1.039) (2S)-2-[4-(4-chlorophenoxy)-2-(trifluoromethyl)phenyl]-1-(1H-1,2,4-triazol-1-yl) propan-2-ol, (1.040) (R)-[3-(4-chloro-2-

fluorophenyl)-5-(2,4-difluorophenyl)-1,2-oxazol-4-yl](pyridin-3-yl) methanol, (1.041) (S)-[3-(4-chloro-2-fluorophenyl)-5-(2,4-difluorophenyl)-1,2-oxazol-4-yl](pyridin-3-yl) methanol, (1.042) [3-(4-chloro-2-fluorophenyl)-5-(2,4-difluorophenyl)-1,2-oxazol-4-yl](pyridin-3-yl) methanol, (1.043) 1-({(2R,4S)-2-[2-chloro-4-(4-chlorophenoxy)phenyl]-4-methyl-1,3-dioxolan-2-yl)methyl}-1H-1,2,4-triazole, (1.044) 1-({(2S,4S)-2-[2-chloro-4-(4-chlorophenoxy)phenyl]-4-methyl-1,3-dioxolan-2-yl)methyl}-1H-1,2,4-triazole, (1.045) 1-{{[3-(2-chlorophenyl)-2-(2,4-difluorophenyl) oxiran-2-yl]methyl}-1H-1,2,4-triazol-5-yl thiocyanate, (1.046) 1-{{[rel (2R,3R)-3-(2-chlorophenyl)-2-(2,4-difluorophenyl) oxiran-2-yl]methyl}-1H-1,2,4-triazol-5-yl thiocyanate, (1.047) 1-{{[rel (2R,3S)-3-(2-chlorophenyl)-2-(2,4-difluorophenyl) oxiran-2-yl]methyl}-1H-1,2,4-triazol-5-yl thiocyanate, (1.048) 2-[(2R,4R,5R)-1-(2,4-dichlorophenyl)-5-hydroxy-2,6,6-trimethylheptan-4-yl]-2,4-dihydro-3H-1,2,4-triazole-3-thione, (1.049) 2-[(2R,4R,5S)-1-(2,4-dichlorophenyl)-5-hydroxy-2,6,6-trimethylheptan-4-yl]-2,4-dihydro-3H-1,2,4-triazole-3-thione, (1.050) 2-[(2R,4S,5R)-1-(2,4-dichlorophenyl)-5-hydroxy-2,6,6-trimethylheptan-4-yl]-2,4-dihydro-3H-1,2,4-triazole-3-thione, (1.051) 2-[(2R,4S,5S)-1-(2,4-dichlorophenyl)-5-hydroxy-2,6,6-trimethylheptan-4-yl]-2,4-dihydro-3H-1,2,4-triazole-3-thione, (1.052) 2-[(2S,4R,5R)-1-(2,4-dichlorophenyl)-5-hydroxy-2,6,6-trimethylheptan-4-yl]-2,4-dihydro-3H-1,2,4-triazole-3-thione, (1.053) 2-[(2S,4R,5S)-1-(2,4-dichlorophenyl)-5-hydroxy-2,6,6-trimethylheptan-4-yl]-2,4-dihydro-3H-1,2,4-triazole-3-thione, (1.054) 2-[(2S,4S,5R)-1-(2,4-dichlorophenyl)-5-hydroxy-2,6,6-trimethylheptan-4-yl]-2,4-dihydro-3H-1,2,4-triazole-3-thione, (1.055) 2-[(2S,4S,5S)-1-(2,4-dichlorophenyl)-5-hydroxy-2,6,6-trimethylheptan-4-yl]-2,4-dihydro-3H-1,2,4-triazole-3-thione, (1.056) 2-[1-(2,4-dichlorophenyl)-5-hydroxy-2,6,6-trimethylheptan-4-yl]-2,4-dihydro-3H-1,2,4-triazole-3-thione, (1.057) 2-[6-(4-bromophenoxy)-2-(trifluoromethyl)-3-pyridyl]-1-(1,2,4-triazol-1-yl) propan-2-ol, (1.058) 2-[6-(4-chlorophenoxy)-2-(trifluoromethyl)-3-pyridyl]-1-(1,2,4-triazol-1-yl) propan-2-ol, (1.059) 2-{{[3-(2-chlorophenyl)-2-(2,4-difluorophenyl) oxiran-2-yl]methyl}-2,4-dihydro-3H-1,2,4-triazole-3-thione, (1.060) 2-{{[rel (2R,3R)-3-(2-chlorophenyl)-2-(2,4-difluorophenyl) oxiran-2-yl]methyl}-2,4-dihydro-3H-1,2,4-triazole-3-thione, (1.061) 2-{{[rel (2R,3S)-3-(2-chlorophenyl)-2-(2,4-difluorophenyl) oxiran-2-yl]methyl}-2,4-dihydro-3H-1,2,4-triazole-3-thione, (1.062) 3-[2-(1-chlorocyclopropyl)-3-(3-chloro-2-fluorophenyl)-2-hydroxy-propyl]imidazole-4-carbonitrile, (1.063) 5-(4-chlorobenzyl)-2-(chloromethyl)-2-methyl-1-(1H-1,2,4-triazol-1-ylmethyl)cyclopentanol, (1.064) 5-(allylsulfanyl)-1-{{[3-(2-chlorophenyl)-2-(2,4-difluorophenyl) oxiran-2-yl]methyl}-1H-1,2,4-triazole, (1.065) 5-(allylsulfanyl)-1-{{[rel (2R,3R)-3-(2-chlorophenyl)-2-(2,4-difluorophenyl) oxiran-2-yl]methyl}-1H-1,2,4-triazole, (1.066) 5-(allylsulfanyl)-1-{{[rel (2R,3S)-3-(2-chlorophenyl)-2-(2,4-difluorophenyl) oxiran-2-yl]methyl}-1H-1,2,4-triazole, (1.067) methyl 2-[2-chloro-4-(4-chlorophenoxy)phenyl]-2-hydroxy-3-(1H-1,2,4-triazol-1-yl) propanoate, (1.068) N'-(2,5-dimethyl-4-(2-methylbenzyl)phenyl)-N-ethyl-N-methylformimidamide, (1.069) N'-(2-chloro-4-(4-cyanobenzyl)-5-methylphenyl)-N-ethyl-N-methylformimidamide, (1.070) N'-(2-chloro-4-(4-methoxybenzyl)-5-methylphenyl)-N-ethyl-N-methylformimidamide, (1.071) N'-(2-chloro-5-methyl-4-

phenoxyphenyl)-N-ethyl-N-methylimidoformamide, (1.072) N'-(4-benzyl-2-chloro-5-methylphenyl)-N-ethyl-N-methylformimidamide, (1.073) N'-(2-chloro-4-(2-fluorophenoxy)-5-methylphenyl)-N-ethyl-N-methylimidoformamide, (1.074) N'-[5-bromo-6-(2,3-dihydro-1H-inden-2-yloxy)-2-methylpyridin-3-yl]-N-ethyl-N-methylimidoformamide, (1.075) N'-(4-[(4,5-dichloro-1,3-thiazol-2-yl)oxy]-2,5-dimethylphenyl)-N-ethyl-N-methylimidoformamide, (1.076) N'-(5-bromo-2-methyl-6-[(1-propoxypropan-2-yl)oxy]pyridin-3-yl)-N-ethyl-N-methylimidoformamide, (1.077) N'-(5-bromo-6-[(1R)-1-(3,5-difluorophenyl) ethoxy]-2-methylpyridin-3-yl)-N-ethyl-N-methylimidoformamide, (1.078) N'-(5-bromo-6-[(1S)-1-(3,5-difluorophenyl) ethoxy]-2-methylpyridin-3-yl)-N-ethyl-N-methylimidoformamide, (1.079) N'-(5-bromo-6-[(cis-4-isopropylcyclohexyl)oxy]-2-methylpyridin-3-yl)-N-ethyl-N-methylimidoformamide, (1.080) N'-(5-bromo-6-[(trans-4-isopropylcyclohexyl)oxy]-2-methylpyridin-3-yl)-N-ethyl-N-methylimidoformamide, (1.081) N'-(5-bromo-6-[1-(3,5-difluorophenyl) ethoxy]-2-methylpyridin-3-yl)-N-ethyl-N-methylimidoformamide, (1.082) N-isopropyl-N'-[5-methoxy-2-methyl-4-(2,2,2-trifluoro-1-hydroxy-1-phenylethyl)phenyl]-N-methylimidoformamide, (1.083) p-tolylmethyl 4-[(E)-[ethyl(methyl)amino]methyleneamino]-2,5-dimethylbenzoate.

[0083] 2) Inhibitors of the respiratory chain at complex I or II, for example (2.001) benzovindiflupyr, (2.002) bixafen, (2.003) boscalid, (2.004) carboxin, (2.005) cyclobutrifluram, (2.006) flubeneteram, (2.007) fluindapyr, (2.008) fluopyram, (2.009) flutolanil, (2.010) fluxapyroxad, (2.011) furametpyr, (2.012) inpyrfluxam, (2.013) isofetamid, (2.014) isoflucypram, (2.015) isopyrazam, (2.016) penflufen, (2.017) pen thiopyrad, (2.018) pydiflumetofen, (2.019) pyrapropoyne, (2.020) pyraziflumid, (2.021) sedaxane, (2.022) Thifluzamide (aka trifluzamide), (2.023) 5,8-difluoro-N-[2-(2-fluoro-4-{{[4-(trifluoromethyl)pyridin-2-yl]oxy}phenyl}ethyl)quinazolin-4-amine, (2.024) 5-chloro-N-[2-[1-(4-chlorophenyl) pyrazol-3-yl]oxyethyl]-6-ethyl-pyrimidin-4-amine, (2.025) N-[2-[1-(4-chlorophenyl) pyrazol-3-yl]oxyethyl]quinazolin-4-amine, (2.026) 1,3-dimethyl-N-(1,1,3-trimethyl-2,3-dihydro-1H-inden-4-yl)-1H-pyrazole-4-carboxamide, (2.027) 1,3-dimethyl-N-[(3R)-1,1,3-trimethyl-2,3-dihydro-1H-inden-4-yl]-1H-pyrazole-4-carboxamide, (2.028) 1,3-dimethyl-N-[(3S)-1,1,3-trimethyl-2,3-dihydro-1H-inden-4-yl]-1H-pyrazole-4-carboxamide, (2.029) 1-methyl-3-(trifluoromethyl)-N-[2'-(trifluoromethyl) biphenyl-2-yl]-1H-pyrazole-4-carboxamide, (2.030) 2-fluoro-6-(trifluoromethyl)-N-(1,1,3-trimethyl-2,3-dihydro-1H-inden-4-yl)benzamide, (2.031) 3-(difluoromethyl)-1-methyl-N-(1,1,3-trimethyl-2,3-dihydro-1H-inden-4-yl)-1H-pyrazole-4-carboxamide, (2.032) 3-(difluoromethyl)-1-methyl-N-[(3S)-1,1,3-trimethyl-2,3-dihydro-1H-inden-4-yl]-1H-pyrazole-4-carboxamide, (2.033) 3-(difluoromethyl)-N-[(3R)-7-fluoro-1,1,3-trimethyl-2,3-dihydro-1H-inden-4-yl]-1-methyl-1H-pyrazole-4-carboxamide, (2.034) 3-(difluoromethyl)-N-[(3S)-7-fluoro-1,1,3-trimethyl-2,3-dihydro-1H-inden-4-yl]-1-methyl-1H-pyrazole-4-carboxamide, (2.035) N-[(1R,4S)-9-(dichloromethylene)-1,2,3,4-tetrahydro-1,4-methanonaphthalen-5-yl]-3-(difluoromethyl)-1-methyl-1H-pyrazole-4-carboxamide, (2.036) N-[(1S,4R)-9-(dichloromethylene)-1,2,3,4-tetrahydro-1,4-methanonaphthalen-5-yl]-3-(difluoromethyl)-1-methyl-1H-pyrazole-4-carboxamide, (2.037) N-[1-(2,4-

dichlorophenyl)-1-methoxypropan-2-yl]-3-(difluoromethyl)-1-methyl-1H-pyrazole-4-carboxamide, (2.038) N-[rac-(1S,2S)-2-(2,4-dichlorophenyl)cyclobutyl]-2-(trifluoromethyl) nicotinamide.

[0084] 3) Inhibitors of the respiratory chain at complex III, for example (3.001) ametoctradin, (3.002) amisulbrom, (3.003) azoxystrobin, (3.004) coumethoxystrobin, (3.005) coumoxystrobin, (3.006) cyazofamid, (3.007) dimoxystrobin, (3.008) enoxastrobin, (3.009) famoxadone, (3.010) fenamidone, (3.011) fenpicoxamid, (3.012) florylpicoxamid, (3.013) flufenoxystrobin, (3.014) fluoxastrobin, (3.015) kresoxim-methyl, (3.016) mandestrobin, (3.017) metarylpicoxamid, (3.018) metominostrobin, (3.019) metyltetraprole, (3.020) orysastrobin, (3.021) picoxystrobin, (3.022) pyraclostrobin, (3.023) pyrametostrobin, (3.024) pyraoxystrobin, (3.025) trifloxystrobin, (3.026) (2E)-2-[2-[[[(1E)-1-(3-{[(E)-1-fluoro-2-phenylvinyl]oxy}phenyl)ethylidene]amino]oxy]methyl]phenyl]-2-(methoxyimino)-N-methylacetamide, (3.027) (2E,3Z)-5-{[1-(4-chloro-2-fluorophenyl)-1H-pyrazol-3-yl]oxy}-2-(methoxyimino)-N,3-dimethylpent-3-enamide, (3.028) (2E,3Z)-5-{[1-(4-chlorophenyl)-1H-pyrazol-3-yl]oxy}-2-(methoxyimino)-N,3-dimethylpent-3-enamide, (3.029) (2R)-2-[2-[(2,5-dimethylphenoxy)methyl]phenyl]-2-methoxy-N-methylacetamide, (3.030) (2S)-2-[2-[(2,5-dimethylphenoxy)methyl]phenyl]-2-methoxy-N-methylacetamide, (3.031) (Z,2E)-5-[1-(2,4-dichlorophenyl)pyrazol-3-yl]oxy-2-methoxyimino-N,3-dimethylpent-3-enamide, (3.032) methyl (Z)-2-(5-cyclohexyl-2-methylphenoxy)-3-methoxyprop-2-enoate, (3.033) methyl (Z)-2-(5-cyclopentyl-2-methylphenoxy)-3-methoxyprop-2-enoate, (3.034) methyl (Z)-3-methoxy-2-[2-methyl-5-(3-propylpyrazol-1-yl)phenoxy]prop-2-enoate, (3.035) methyl (Z)-3-methoxy-2-[2-methyl-5-[3-(trifluoromethyl)pyrazol-1-yl]phenoxy]prop-2-enoate, (3.036) methyl {5-[3-(2,4-dimethylphenyl)-1H-pyrazol-1-yl]-2-methylbenzyl}carbamate, (3.037) [rac-2-(4-bromo-7-fluoro-indol-1-yl)-1-methyl-propyl](2S)-2-[(3-hydroxy-4-methoxy-pyridine-2-carbonyl)amino]propanoate, (3.038) [rac-2-(7-bromo-4-fluoro-indol-1-yl)-1-methyl-propyl](2S)-2-[(3-acetoxy-4-methoxy-pyridine-2-carbonyl)amino]propanoate, (3.039) [rac-2-(7-bromoindol-1-yl)-1-methyl-propyl](2S)-2-[(3-hydroxy-4-methoxy-pyridine-2-carbonyl)amino]propanoate, (3.040) [rac-2-(3,5-dichloro-2-pyridyl)-1-methyl-propyl](2S)-2-[(3-hydroxy-4-methoxy-pyridine-2-carbonyl)amino]propanoate, (3.041) [(1S)-1-[1-(1-naphthyl)cyclopropyl]ethyl](2S)-2-[(3-acetoxy-4-methoxy-pyridine-2-carbonyl)amino]propanoate, (3.042) [(1S)-1-[1-(1-naphthyl)cyclopropyl]ethyl](2S)-2-[(3-hydroxy-4-methoxy-pyridine-2-carbonyl)amino]propanoate, (3.043) [(1S)-1-[1-(1-naphthyl)cyclopropyl]ethyl](2S)-2-[(3-acetoxymethoxy)-4-methoxy-pyridine-2-carbonyl]amino]propanoate, (3.044) [2-[[[(1S)-2-[(1RS,2SR)-2-(3,5-dichloro-2-pyridyl)-1-methyl-propoxy]-1-methyl-2-oxo-ethyl]carbamoyl]-4-methoxy-3-pyridyl]oxymethyl 2-methylpropanoate, (3.045) N-(3-ethyl-3,5,5-trimethylcyclohexyl)-3-formamido-2-hydroxybenzamide.

[0085] 4) Inhibitors of the mitosis and cell division, for example (4.001) carbendazim, (4.002) diethofencarb, (4.003) ethaboxam, (4.004) fluopicolide, (4.005) flupimomide, (4.006) metrafenone, (4.007) pencycuron, (4.008) pyridachlometyl, (4.009) pyriofenone (chlazafenone), (4.010) thiabendazole, (4.011) thiophanate-methyl, (4.012) zoxamide, (4.013) 3-chloro-5-(4-chlorophenyl)-4-(2,6-difluorophenyl)-6-methylpyridazine, (4.014) 3-chloro-5-(6-chloropyridin-3-yl)-6-methyl-4-(2,4,6-trifluorophenyl)pyridazine, (4.015) 4-(2-bromo-4-fluorophenyl)-N-(2,6-difluorophenyl)-1,3-dimethyl-1H-pyrazol-5-amine, (4.016) 4-(2-bromo-4-fluorophenyl)-N-(2-bromo-6-fluorophenyl)-1,3-dimethyl-1H-pyrazol-5-amine, (4.017) 4-(2-bromo-4-fluorophenyl)-N-(2-bromophenyl)-1,3-dimethyl-1H-pyrazol-5-amine, (4.018) 4-(2-bromo-4-fluorophenyl)-N-(2-chloro-6-fluorophenyl)-1,3-dimethyl-1H-pyrazol-5-amine, (4.019) 4-(2-bromo-4-fluorophenyl)-N-(2-chlorophenyl)-1,3-dimethyl-1H-pyrazol-5-amine, (4.020) 4-(2-bromo-4-fluorophenyl)-N-(2-fluorophenyl)-1,3-dimethyl-1H-pyrazol-5-amine, (4.021) 4-(2-chloro-4-fluorophenyl)-N-(2,6-difluorophenyl)-1,3-dimethyl-1H-pyrazol-5-amine, (4.022) 4-(2-chloro-4-fluorophenyl)-N-(2-chloro-6-fluorophenyl)-1,3-dimethyl-1H-pyrazol-5-amine, (4.023) 4-(2-chloro-4-fluorophenyl)-N-(2-chlorophenyl)-1,3-dimethyl-1H-pyrazol-5-amine, (4.024) 4-(2-chloro-4-fluorophenyl)-N-(2-fluorophenyl)-1,3-dimethyl-1H-pyrazol-5-amine, (4.025) 4-(4-chlorophenyl)-5-(2,6-difluorophenyl)-3,6-dimethylpyridazine, (4.026) N-(2-bromo-6-fluorophenyl)-4-(2-chloro-4-fluorophenyl)-1,3-dimethyl-1H-pyrazol-5-amine, (4.027) N-(2-bromophenyl)-4-(2-chloro-4-fluorophenyl)-1,3-dimethyl-1H-pyrazol-5-amine, (4.028) N-(4-chloro-2,6-difluorophenyl)-4-(2-chloro-4-fluorophenyl)-1,3-dimethyl-1H-pyrazol-5-amine.

[0086] 5) Compounds capable to have a multisite action, for example (5.001) bordeaux mixture, (5.002) captafol, (5.003) captan, (5.004) chlorothalonil, (5.005) copper hydroxide, (5.006) copper naphthenate, (5.007) copper oxide, (5.008) copper oxychloride, (5.009) copper (2+) sulfate, (5.010) dithianon, (5.011) dodine, (5.012) folpet, (5.013) mancozeb, (5.014) maneb, (5.015) metiram, (5.016) metiram zinc, (5.017) oxine-copper, (5.018) propineb, (5.019) sulfur and sulfur preparations including calcium polysulfide, (5.020) thiram, (5.021) zineb, (5.022) ziram, (5.023) 6-ethyl-5,7-dioxo-6,7-dihydro-5H-pyrrolo[3',4':5,6][1,4]dithiino[2,3-c][1,2]thiazole-3-carbonitrile.

[0087] 6) Compounds capable to induce a host defence, for example (6.001) acibenzolar-S-methyl, (6.002) fosetyl-aluminium, (6.003) fosetyl-calcium, (6.004) fosetyl-sodium, (6.005) isotianil, (6.006) phosphorous acid and its salts, (6.007) probenazole, (6.008) tiadinil.

[0088] 7) Inhibitors of the amino acid and/or protein biosynthesis, for example (7.001) cyprodinil, (7.002) kasugamycin, (7.003) kasugamycin hydrochloride hydrate, (7.004) oxytetracycline, (7.005) pyrimethanil

[0089] 8) Inhibitors of the ATP production, for example (8.001) silthiofam.

[0090] 9) Inhibitors of the cell wall synthesis, for example (9.001) benthiavalicarb, (9.002) dimethomorph, (9.003) flumorph, (9.004) iprovalicarb, (9.005) mandipropamid, (9.006) pyrimorph, (9.007) valifenalate, (9.008) (2E)-3-(4-tert-butylphenyl)-3-(2-chloropyridin-4-yl)-1-(morpholin-4-yl) prop-2-en-1-one, (9.009) (2Z)-3-(4-tert-butylphenyl)-3-(2-chloropyridin-4-yl)-1-(morpholin-4-yl) prop-2-en-1-one.

[0091] 10) Inhibitors of the lipid synthesis or transport, or membrane synthesis, for example (10.001) fluoxapirolin, (10.002) natamycin, (10.003) oxathiapirolin, (10.004) propamocarb, (10.005) propamocarb hydrochloride, (10.006) propamocarb-fosetyl, (10.007) tolclifos-methyl, (10.008) 1-(4-{4-[(5R)-5-(2,6-difluorophenyl)-4,5-dihydro-1,2-oxazol-3-yl]-1,3-thiazol-2-yl}piperidin-1-yl)-2-[5-methyl-3-(trifluoromethyl)-1H-pyrazol-1-yl]ethanone, (10.009)

1-(4-{4-[(5S)-5-(2,6-difluorophenyl)-4,5-dihydro-1,2-oxazol-3-yl]-1,3-thiazol-2-yl}piperidin-1-yl)-2-[5-methyl-3-(trifluoromethyl)-1H-pyrazol-1-yl]ethanone, (10.010) 2-[3,5-bis(difluoro-methyl)-1H-pyrazol-1-yl]-1-[4-(4-{5-[2-(prop-2-yn-1-yloxy)phenyl]-4,5-dihydro-1,2-oxazol-3-yl]-1,3-thiazol-2-yl}piperidin-1-yl)ethanone, (10.011) 2-[3,5-bis(difluoromethyl)-1H-pyrazol-1-yl]-1-[4-(4-{5-[2-chloro-6-(prop-2-yn-1-yloxy)phenyl]-4,5-dihydro-1,2-oxazol-3-yl]-1,3-thiazol-2-yl}piperidin-1-yl)ethanone, (10.012) 2-[3,5-bis(difluoromethyl)-1H-pyrazol-1-yl]-1-[4-(4-{5-[2-fluoro-6-(prop-2-yn-1-yloxy)phenyl]-4,5-dihydro-1,2-oxazol-3-yl]-1,3-thiazol-2-yl}piperidin-1-yl)ethanone, (10.013) 2-{(5R)-3-[2-(1-{[3,5-bis(difluoromethyl)-1H-pyrazol-1-yl]acetyl}piperidin-4-yl)-1,3-thiazol-4-yl]-4,5-dihydro-1,2-oxazol-5-yl]-3-chlorophenyl methanesulfonate, (10.014) 2-{(5S)-3-[2-(1-{[3,5-bis(difluoromethyl)-1H-pyrazol-1-yl]acetyl}piperidin-4-yl)-1,3-thiazol-4-yl]-4,5-dihydro-1,2-oxazol-5-yl]-3-chlorophenyl methanesulfonate, (10.015) 2-[3-[2-(1-{[3,5-bis(difluoromethyl)-1H-pyrazol-1-yl]acetyl}piperidin-4-yl)-1,3-thiazol-4-yl]-4,5-dihydro-1,2-oxazol-5-yl]phenyl methanesulfonate, (10.016) 3-[2-(1-{[5-methyl-3-(trifluoromethyl)-1H-pyrazol-1-yl]acetyl}piperidin-4-yl)-1,3-thiazol-4-yl]-1,5-dihydro-2,4-benzodioxepin-6-yl methanesulfonate, (10.017) 9-fluoro-3-[2-(1-{[5-methyl-3-(trifluoromethyl)-1H-pyrazol-1-yl]acetyl}piperidin-4-yl)-1,3-thiazol-4-yl]-1,5-dihydro-2,4-benzodioxepin-6-yl methanesulfonate, (10.018) 3-[2-(1-{[3,5-bis(difluoromethyl)-1H-pyrazol-1-yl]acetyl}piperidin-4-yl)-1,3-thiazol-4-yl]-1,5-dihydro-2,4-benzodioxepin-6-yl methanesulfonate, (10.019) 3-[2-(1-{[3,5-bis(difluoromethyl)-1H-pyrazol-1-yl]acetyl}piperidin-4-yl)-1,3-thiazol-4-yl]-9-fluoro-1,5-dihydro-2,4-benzodioxepin-6-yl methanesulfonate.

[0092] 11) Inhibitors of the melanin biosynthesis, for example (11.001) tolprocarb, (11.002) tricyclazole.

[0093] 12) Inhibitors of the nucleic acid synthesis, for example (12.001) benalaxyl, (12.002) benalaxyl-M (kiralaxyl), (12.003) metalaxyl, (12.004) metalaxyl-M (mefenoxam).

[0094] 13) Inhibitors of the signal transduction, for example (13.001) fludioxonil, (13.002) iprodione, (13.003) procymidone, (13.004) proquinazid, (13.005) quinoxifen, (13.006) vinclozolin.

[0095] 14) Compounds capable to act as an uncoupler, for example (14.001) fluazinam, (14.002) meptyldinocap.

[0096] 15) Further compounds, for example (15.001) abscisic acid, (15.002) aminopyrifin, (15.003) benthiazole, (15.004) bethoxazin, (15.005) capsimycin, (15.006) carvone, (15.007) chinomethionat, (15.008) chloroinconazide, (15.009) cufraneb, (15.010) cyflufenamid, (15.011) cymoxanil, (15.012) cyprosulfamide, (15.013) dipymetitrone, (15.014) D-tagatose, (15.015) flufenoxadiazam, (15.016) flumetylsulforim, (15.017) flutianil, (15.018) ipflufenquin, (15.019) methyl isothiocyanate, (15.020) mildiomycin, (15.021) nickel dimethyldithiocarbamate, (15.022) nitrothal-isopropyl, (15.023) oxyfenthin, (15.024) pentachlorophenol and salts, (15.025) picarbutrazox, (15.026) quinofumelin, (15.027) tebufloquin, (15.028) tecloftalam, (15.029) tolnifanide, (15.030) 2-(6-benzylpyridin-2-yl) quinazoline, (15.031) 2-[6-(3-fluoro-4-methoxyphenyl)-5-methylpyridin-2-yl] quinazoline, (15.032) 2-phenylphenol and salts, (15.033) 4-amino-5-fluoropyrimidin-2-ol (tautomeric form: 4-amino-5-fluoropyrimidin-2 (1H)-one), (15.034) 4-oxo-4-[(2-phenylethyl)amino]butanoic acid, (15.035) 5-amino-1,3,4-thiadiazole-2-thiol, (15.036) 5-chloro-N'-phenyl-N'-(prop-2-yn-1-yl)thiophene-2-sulfonohydrazide, (15.037) 5-fluoro-2-[(4-fluorobenzyl)oxy]pyrimidin-4-amine, (15.038) 5-fluoro-2-[(4-methylbenzyl)oxy]pyrimidin-4-amine, (15.039) but-3-yn-1-yl {6-[[[(Z)-(1-methyl-1H-tetrazol-5-yl) (phenyl) methylene]amino]oxy)methyl]pyridin-2-yl} carbamate, (15.040) ethyl (2Z)-3-amino-2-cyano-3-phenylacrylate, (15.041) methyl 2-[acetyl-[2-ethylsulfonyl-4-(trifluoro-methyl) benzoyl]amino]-5-(trifluoromethoxy)benzoate, (15.042) N-acetyl-N-[2-bromo-4-(trifluoro-methoxy)phenyl]-2-ethylsulfonyl-4-(trifluoromethyl)benzamide, (15.043) phenazine-1-carboxylic acid, (15.044) propyl 3,4,5-trihydroxybenzoate, (15.045) quinolin-8-ol, (15.046) quinolin-8-ol sulfate (2:1), (15.047) (2R)-2-benzyl-N-(8-fluoro-2-methyl-3-quinolyl)-2,4-dimethyl-pentanamide, (15.048) (2S)-2-benzyl-N-(8-fluoro-2-methyl-3-quinolyl)-2,4-dimethyl-pentanamide, (15.049) 1-(4,5-dimethyl-1H-benzimidazol-1-yl)-4,4-difluoro-3,3-dimethyl-3,4-dihydroisoquinoline, (15.050) 1-(4,5-dimethylbenzimidazol-1-yl)-4,4,5-trifluoro-3,3-dimethyl-isoquinoline, (15.051) 1-(5-(fluoromethyl)-6-methyl-pyridin-3-yl)-4,4-difluoro-3,3-dimethyl-3,4-dihydroisoquinoline, (15.052) 1-(5,6-dimethylpyridin-3-yl)-4,4-difluoro-3,3-dimethyl-3,4-dihydroisoquinoline, (15.053) 1-(6-(difluoromethyl)-5-methoxy-pyridin-3-yl)-4,4-difluoro-3,3-dimethyl-3,4-dihydroisoquinoline, (15.054) 1-(6-(difluoromethyl)-5-methyl-pyridin-3-yl)-4,4-difluoro-3,3-dimethyl-3,4-dihydroisoquinoline, (15.055) 1-(6,7-dimethylpyrazolo[1,5-a]pyridin-3-yl)-4,4,5-trifluoro-3,3-dimethyl-isoquinoline, (15.056) 1-(6,7-dimethylpyrazolo[1,5-a]pyridin-3-yl)-4,4-difluoro-3,3-dimethyl-3,4-dihydroisoquinoline, (15.057) 2-{2-fluoro-6-[(8-fluoro-2-methylquinolin-3-yl)oxy]phenyl}propan-2-ol, (15.058) 3-(4,4,5-trifluoro-3,3-dimethyl-3,4-dihydroisoquinolin-1-yl) quinoline, (15.059) 3-(4,4-difluoro-3,3-dimethyl-3,4-dihydroisoquinolin-1-yl)-8-fluoroquinoline, (15.060) 3-(4,4-difluoro-5,5-dimethyl-4,5-dihydrothieno[2,3-c]pyridin-7-yl) quinoline, (15.061) 3-(5-fluoro-3,3,4,4-tetramethyl-3,4-dihydroisoquinolin-1-yl) quinoline, (15.062) 4,4-difluoro-3,3-dimethyl-1-(4-methylbenzimidazol-1-yl) isoquinoline, (15.063) 4,4-difluoro-3,3-dimethyl-1-(6-methylpyrazolo[1,5-a]pyridin-3-yl) isoquinoline, (15.064) 5-bromo-1-(5,6-dimethylpyridin-3-yl)-3,3-dimethyl-3,4-dihydroisoquinoline, (15.065) 7,8-difluoro-N-[rac-1-benzyl-1,3-dimethyl-butyl]quinoline-3-carboxamide, (15.066) 8-fluoro-3-(5-fluoro-3,3,4,4-tetramethyl-3,4-dihydroisoquinolin-1-yl)-quinoline, (15.067) 8-fluoro-3-(5-fluoro-3,3-dimethyl-3,4-dihydroisoquinolin-1-yl)-quinoline, (15.068) 8-fluoro-N-(4,4,4-trifluoro-2-methyl-1-phenylbutan-2-yl) quinoline-3-carboxamide, (15.069) 8-fluoro-N-[(1R)-1-[(3-fluorophenyl)methyl]-1,3-dimethyl-butyl]quinoline-3-carboxamide, (15.070) 8-fluoro-N-[(1S)-1-[(3-fluorophenyl)methyl]-1,3-dimethyl-butyl]quinoline-3-carboxamide, (15.071) 8-fluoro-N-[(2S)-4,4,4-trifluoro-2-methyl-1-phenylbutan-2-yl]quinoline-3-carboxamide, (15.072) 8-fluoro-N-[rac-1-[(3-fluorophenyl)methyl]-1,3-dimethyl-butyl]quinoline-3-carboxamide, (15.073) 9-fluoro-2,2-dimethyl-5-(quinolin-3-yl)-2,3-dihydro-1,4-benzoxazepine, (15.074) N-(2,4-dimethyl-1-phenylpentan-2-yl)-8-fluoroquinoline-3-carboxamide, (15.075) N-[(1R)-1-benzyl-1,3-dimethyl-butyl]-7,8-difluoro-quinoline-3-carboxamide, (15.076) N-[(1S)-1-benzyl-1,3-dimethyl-butyl]-7,8-difluoro-quinoline-3-carboxamide, (15.077) N-[(2R)-2,4-dimethyl-1-phenylpentan-2-yl]-8-fluoroquino-

line-3-carboxamide, (15.078) rac-2-benzyl-N-(8-fluoro-2-methyl-3-quinolyl)-2,4-dimethyl-pentanamide, (15.079) 1,1-diethyl-3-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]-3-dimethoxy-1-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]urea, (15.080) yl]phenyl methyl]urea, (15.081) 1-[[3-fluoro-4-(5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl)phenyl]methyl]azepan-2-one, (15.082) 1-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]piperidin-2-one, (15.083) 1-methoxy-1-methyl-3-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]urea, (15.084) 1-methoxy-3-methyl-1-[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]urea, (15.085) 1-methoxy-3-methyl-1-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]urea, (15.086) 2-(difluoromethyl)-5-[2-[1-(2,6-difluorophenyl)cyclopropoxy]pyrimidin-5-yl]-1,3,4-oxadiazole, (15.087) 2,2-difluoro-N-methyl-2-[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]acetamide, (15.088) 3,3-dimethyl-1-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]piperidin-2-one, (15.089) 3-ethyl-1-methoxy-1-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]urea, (15.090) 4,4-dimethyl-1-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]pyrrolidin-2-one, (15.091) 4,4-dimethyl-2-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]isoxazolidin-3-one, (15.092) 4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl dimethyl-carbamate, (15.093) 5,5-dimethyl-2-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]isoxazolidin-3-one, (15.094) 5-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-N-[(1R)-1-(2,6-difluorophenyl)ethyl]pyrimidin-2-amine, (15.095) 5-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-N-[(1R)-1-(2,6-difluorophenyl) propyl]pyrimidin-2-amine, (15.096) 5-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-N-[(1R)-1-(2-fluorophenyl)ethyl]pyrimidin-2-amine, (15.097) 5-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-N-[(1R)-1-(2-fluorophenyl)ethyl]pyrimidin-2-amine, (15.098) 5-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-N-[(1R)-1-(3,5-difluorophenyl)ethyl]pyrimidin-2-amine, (15.099) 5-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-N-[(1R)-1-phenylethyl]pyrimidin-2-amine, (15.100) 5-[5-(difluoromethyl)-1,3,4-oxadiazol-2-yl]-N-[1-(2-fluorophenyl)cyclopropyl]pyrimidin-2-amine, (15.101) 5-methyl-1-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]pyrrolidin-2-one, (15.102) ethyl 1-{4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzyl}-1H-pyrazole-4-carboxylate, (15.103) methyl {4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl}carbamate, (15.104) N-(1-methylcyclopropyl)-4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzamide, (15.105) N-(2,4-difluorophenyl)-4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzamide, (15.106) N,2-dimethoxy-N-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]propanamide, (15.107) N,N-dimethyl-1-{4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzyl}-1H-1,2,4-triazol-3-amine, (15.108) N-[(E)-methoxyiminomethyl]-4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzamide, (15.109) N-[(E)-N-methoxy-C-methyl-carbonimidoyl]-4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzamide, (15.110) N-[(Z)-methoxyiminomethyl]-4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzamide, (15.111) N-[(Z)-N-methoxy-C-methyl-carbonimidoyl]-4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzamide, (15.112) N-[[2,3-difluoro-4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]-3,3,3-trifluoro-propanamide, (15.113) N-[[4-[5-

(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]propanamide, (15.114) N-[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]cyclopropanecarboxamide, (15.115) N-{2,3-difluoro-4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzyl}butanamide, (15.116) N-{4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzyl}cyclopropanecarboxamide, (15.117) N-{4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl}propan-amide, (15.118) N-allyl-N-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]acetamide, (15.119) N-allyl-N-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]propanamide, (15.120) N-ethyl-2-methyl-N-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]propanamide, (15.121) N-methoxy-N-[[4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]phenyl]methyl]cyclopropanecarboxamide, (15.122) N-methyl-4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzamide, (15.123) N-methyl-4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzenecarbothioamide, (15.124) N-methyl-N-phenyl-4-[5-(trifluoromethyl)-1,2,4-oxadiazol-3-yl]benzamide.

[0097] All named mixing partners of the classes (1) to (15) as described here above can be present in the form of the free compound or, if their functional groups enable this, an agrochemically active salt thereof.

[0098] The compound and the composition of the invention may also be combined with one or more biological control agents.

[0099] As used herein, the term “biological control” is defined as control of harmful organisms such as a phytopathogenic fungi and/or insects and/or acarids and/or nematodes by the use or employment of a biological control agent.

[0100] As used herein, the term “biological control agent” is defined as an organism other than the harmful organisms and/or proteins or secondary metabolites produced by such an organism for the purpose of biological control. Mutants of the second organism shall be included within the definition of the biological control agent. The term “mutant” refers to a variant of the parental strain as well as methods for obtaining a mutant or variant in which the pesticidal activity is greater than that expressed by the parental strain. The “parent strain” is defined herein as the original strain before mutagenesis. To obtain such mutants the parental strain may be treated with a chemical such as N-methyl-N'-nitro-N-nitrosoguanidine, ethylmethanesulfone, or by irradiation using gamma, x-ray, or UV-irradiation, or by other means well known to those skilled in the art. Known mechanisms of biological control agents comprise enteric bacteria that control root rot by out-competing fungi for space on the surface of the root. Bacterial toxins, such as antibiotics, have been used to control pathogens. The toxin can be isolated and applied directly to the plant or the bacterial species may be administered so it produces the toxin in situ.

[0101] A “variant” is a strain having all the identifying characteristics of the NRRL or ATCC Accession Numbers as indicated in this text and can be identified as having a genome that hybridizes under conditions of high stringency to the genome of the NRRL or ATCC Accession Numbers.

[0102] “Hybridization” refers to a reaction in which one or more polynucleotides react to form a complex that is stabilized via hydrogen bonding between the bases of the nucleotide residues. The hydrogen bonding may occur by Watson-Crick base pairing, Hoogsteen binding, or in any other sequence-specific manner. The complex may comprise two strands forming a duplex structure, three or more strands

forming a multi-stranded complex, a single self-hybridizing strand, or any combination of these. Hybridization reactions can be performed under conditions of different “stringency”. In general, a low stringency hybridization reaction is carried out at about 40° C. in 10×SSC or a solution of equivalent ionic strength/temperature. A moderate stringency hybridization is typically performed at about 50° C. in 6×SSC, and a high stringency hybridization reaction is generally performed at about 60° C. in 1×SSC.

[0103] A variant of the indicated NRRL or ATCC Accession Number may also be defined as a strain having a genomic sequence that is greater than 85%, more preferably greater than 90% or more preferably greater than 95% sequence identity to the genome of the indicated NRRL or ATCC Accession Number. A polynucleotide or polynucleotide region (or a polypeptide or polypeptide region) has a certain percentage (for example, 80%, 85%, 90%, or 95%) of “sequence identity” to another sequence means that, when aligned, that percentage of bases (or amino acids) are the same in comparing the two sequences. This alignment and the percent homology or sequence identity can be determined using software programs known in the art, for example, those described in Current Protocols in Molecular Biology (F. M. Ausubel et al., eds., 1987).

[0104] NRRL is the abbreviation for the Agricultural Research Service Culture Collection, an international depositary authority for the purposes of depositing microorganism strains under the Budapest treaty on the international recognition of the deposit of microorganisms for the purposes of patent procedure, having the address National Center for Agricultural Utilization Research, Agricultural Research service, U.S. Department of Agriculture, 1815 North university Street, Peoria, Illinois 61604 USA.

[0105] ATCC is the abbreviation for the American Type Culture Collection, an international depositary authority for the purposes of depositing microorganism strains under the Budapest treaty on the international recognition of the deposit of microorganisms for the purposes of patent procedure, having the address ATCC Patent Depository, 10801 University Blvd., Manassas, VA 10110 USA.

[0106] Examples of biological control agents which may be combined with the compound and the composition of the invention are:

[0107] (A) Antibacterial agents selected from the group of:

[0108] (A1) bacteria, such as (A1.01) *Bacillus subtilis*, in particular strain QST713/AQ713 (available as SERENADE OPTI or SERENADE ASO from Bayer CropScience LP, US, having NRRL Accession No. B21661, U.S. Pat. No. 6,060,051); (A1.02) *Bacillus* sp., in particular strain D747 (available as DOUBLE NICKEL® from Kumiai Chemical Industry Co., Ltd.), having Accession No. FERM BP-8234, U.S. Pat. No. 7,094,592; (A1.03) *Bacillus pumilus*, in particular strain BU F-33, having NRRL Accession No. 50185 (available as part of the CARTISSA® product from BASF, EPA Reg. No. 71840-19); (A1.04) *Bacillus subtilis* var. *amyloliquefaciens* strain FZB24 having Accession No. DSM 10271 (available from Novozymes as TAEGRO® or TAEGRO® ECO (EPA Registration No. 70127-5)); (A1.05) a *Paenibacillus* sp. strain having Accession No. NRRL B-50972 or Accession No. NRRL B-67129, WO 2016/154297; (A1.06) *Bacillus subtilis* strain BU1814, (available as

VELONDIS® PLUS, VELONDIS® FLEX and VELONDIS® EXTRA from BASF SE); (A1.07) *Bacillus mojavensis* strain R3B (Accession No. NCAIM (P) B001389) (WO 2013/034938) from Certis USA LLC, a subsidiary of Mitsui & Co.; (A1.08) *Bacillus subtilis* CX-9060 from Certis USA LLC, a subsidiary of Mitsui & Co.; (A1.09) *Paenibacillus polymyxa*, in particular strain AC-1 (e.g. TOPSEED® from Green Biotech Company Ltd.); (A1.10) *Pseudomonas* proradix (e.g. PRORADIX® from Sourcon Padena); (A1.11) *Pantoea agglomerans*, in particular strain E325 (Accession No. NRRL B-21856) (available as BLOOMTIME BIOLOGICAL™ FD BIOPESTICIDE from Northwest Agri Products); and

[0109] (A2) fungi, such as (A2.01) *Aureobasidium pullulans*, in particular blastospores of strain DSM14940, blastospores of strain DSM 14941 or mixtures of blastospores of strains DSM14940 and DSM14941 (e.g., BOTECTOR® and BLOSSOM PROTECT® from bioferm, CH); (A2.02) *Pseudozyma aphidis* (as disclosed in WO2011/151819 by Yissum Research Development Company of the Hebrew University of Jerusalem); (A2.03) *Saccharomyces cerevisiae*, in particular strains CNCM No. I-3936, CNCM No. I-3937, CNCM No. I-3938 or CNCM No. I-3939 (WO 2010/086790) from Lesaffre et Compagnie, FR;

[0110] (B) biological fungicides selected from the group of:

[0111] (B1) bacteria, for example (B1.01) *Bacillus subtilis*, in particular strain QST713/AQ713 (available as SERENADE OPTI or SERENADE ASO from Bayer CropScience LP, US, having NRRL Accession No. B21661 and described in U.S. Pat. No. 6,060,051); (B1.02) *Bacillus pumilus*, in particular strain QST2808 (available as SONATA® from Bayer CropScience LP, US, having Accession No. NRRL B-30087 and described in U.S. Pat. No. 6,245,551); (B1.03) *Bacillus pumilus*, in particular strain GB34 (available as Yield Shield® from Bayer AG, DE); (B1.04) *Bacillus pumilus*, in particular strain BU F-33, having NRRL Accession No. 50185 (available as part of the CARTISSA product from BASF, EPA Reg. No. 71840-19); (B1.05) *Bacillus amyloliquefaciens*, in particular strain D747 (available as Double Nickel™ from Kumiai Chemical Industry Co., Ltd., having accession number FERM BP-8234, U.S. Pat. No. 7,094,592); (B1.06) *Bacillus subtilis* Y1336 (available as BIOBAC® WP from Bion-Tech, Taiwan, registered as a biological fungicide in Taiwan under Registration Nos. 4764, 5454, 5096 and 5277); (B1.07) *Bacillus subtilis* strain MBI 600 (available as SUBTILEX from BASF SE), having Accession Number NRRL B-50595, U.S. Pat. No. 5,061,495; (B1.08) *Bacillus subtilis* strain GB03 (available as Kodiak® from Bayer AG, DE); (B1.09) *Bacillus subtilis* var. *amyloliquefaciens* strain FZB24 having Accession No. DSM 10271 (available from Novozymes as TAEGRO® or TAEGRO® ECO (EPA Registration No. 70127-5)); (B1.10) *Bacillus mycoides*, isolate J, having Accession No. B-30890 (available as BMJ TGAI® or WG and LifeGard™ from Certis USA LLC, a subsidiary of Mitsui & Co.); (B1.11) *Bacillus licheniformis*, in particular strain SB3086, having Accession No. ATCC 55406, WO 2003/000051 (available as ECO-GUARD® Biofungicide and GREEN RELEAF™

- from Novozymes); (B1.12) a *Paenibacillus* sp. strain having Accession No. NRRL B-50972 or Accession No. NRRL B-67129, WO 2016/154297; (B1.13) *Bacillus subtilis* strain BU1814, (available as VELONDIS® PLUS, VELONDIS® FLEX and VELONDIS® EXTRA from BASF SE); (B1.14) *Bacillus subtilis* CX-9060 from Certis USA LLC, a subsidiary of Mitsui & Co.; (B1.15) *Bacillus amyloliquefaciens* strain F727 (also known as strain MBI110) (NRRL Accession No. B-50768; WO 2014/028521) (STARGUS® from Marrone Bio Innovations); (B1.16) *Bacillus amyloliquefaciens* strain FZB42, Accession No. DSM 23117 (available as RHIZOVITAL® from ABITEP, DE); (B1.17) *Bacillus licheniformis* FMCH001 and *Bacillus subtilis* FMCH002 (QUARTZO® (WG) and PRESENCE® (WP) from FMC Corporation); (B1.18) *Bacillus mojavensis* strain R3B (Accession No. NCAIM (P) B001389) (WO 2013/034938) from Certis USA LLC, a subsidiary of Mitsui & Co.; (B1.19) *Paenibacillus polymyxa* ssp. *plantarum* (WO 2016/020371) from BASF SE; (B1.20) *Paenibacillus* epiphyticus (WO 2016/020371) from BASF SE; (B1.21) *Pseudomonas chlororaphis* strain AFS009, having Accession No. NRRL B-50897, WO 2017/019448 (e.g., HOWLER™ and ZIO® from AgBiome Innovations, US); (B1.22) *Pseudomonas chlororaphis*, in particular strain MA342 (e.g. CEDOMON®, CERALL®, and CEDRESS® by Bioagri and Koppert); (B1.23) *Streptomyces lydicus* strain WYEC108 (also known as *Streptomyces lydicus* strain WYCD108US) (ACTINO-IRON® and ACTINO-VATE® from Novozymes); (B1.24) *Agrobacterium radiobacter* strain K84 (e.g. GALLTROL-A® from AgBioChem, CA); (B1.25) *Agrobacterium radiobacter* strain K1026 (e.g. NOGALL™ from BASF SE); (B1.26) *Bacillus subtilis* KTSB strain (FOLIACTIVE® from Donaghys); (B1.27) *Bacillus subtilis* IAB/BS03 (AVIV™ from STK Bio-Ag Technologies); (B1.28) *Bacillus subtilis* strain Y1336 (available as BIOBAC® WP from Bion-Tech, Taiwan, registered as a biological fungicide in Taiwan under Registration Nos. 4764, 5454, 5096 and 5277); (B1.29) *Bacillus amyloliquefaciens* isolate B246 (e.g. AVOGREEN™ from University of Pretoria); (B1.30) *Bacillus methylotrophicus* strain BAC-9912 (from Chinese Academy of Sciences' Institute of Applied Ecology); (B1.31) *Pseudomonas* proradix (e.g. PRORADIX® from Sourcon Padena); (B1.32) *Streptomyces griseoviridis* strain K61 (also known as *Streptomyces galbus* strain K61) (Accession No. DSM 7206) (MYCOSTOP® from Verdera; PREFENCE® from BioWorks; cf. Crop Protection 2006, 25, 468-475); (B1.33) *Pseudomonas fluorescens* strain A506 (e.g. BLIGHTBAN® A506 by NuFarm); and
- [0112] (B2) fungi, for example: (B2.01) *Coniothyrium minitans*, in particular strain CON/M/91-8 (Accession No. DSM-9660; e.g. Contans® from Bayer Crop-Science Biologics GmbH); (B2.02) *Metschnikowia fructicola*, in particular strain NRRL Y-30752; (B2.03) *Microsphaeropsis ochracea*; (B2.04) *Trichoderma atroviride*, in particular strain SC1 (having Accession No. CBS 122089, WO 2009/116106 and U.S. Pat. No. 8,431,120 (from Bi-PA)), strain 77B (T77 from Andermatt Biocontrol) or strain LU132 (e.g. Sentinel from Agrimm Technologies Limited); (B2.05) *Trichoderma harzianum* strain T-22 (e.g. Trianum-P from Andermatt Biocontrol or Koppert) or strain Cepa Simb-T5 (from Simbiose Agro); (B2.06) *Gliocladium roseum* (also known as *Clonostachys rosea* f. *rosea*), in particular strain 321U from Adjuvants Plus, strain ACM941 as disclosed in Xue (Efficacy of *Clonostachys rosea* strain ACM941 and fungicide seed treatments for controlling the root rot complex of field pea, Can Jour Plant Sci 83 (3): 519-524), or strain IK726 (Jensen D F, et al. Development of a biocontrol agent for plant disease control with special emphasis on the near commercial fungal antagonist *Clonostachys rosea* strain 'IK726'; Australas Plant Pathol. 2007; 36:95-101); (B2.07) *Talaromyces flavus*, strain V117b; (B2.08) *Trichoderma viride*, in particular strain B35 (Pietr et al., 1993, Zesz. Nauk. A R w Szczecinie 161:125-137); (B2.09) *Trichoderma asperellum*, in particular strain SKT-1, having Accession No. FERM P-16510 (e.g. ECO-HOPE® from Kumiai Chemical Industry), strain T34 (e.g. T34 Biocontrol by Biocontrol Technologies S.L., ES) or strain ICC 012 from Isagro; (B2.10) *Trichoderma atroviride*, strain CNCM I-1237 (e.g. Esquive® WP from Agrauxine, FR); (B2.11) *Trichoderma atroviride*, strain no. V08/002387; (B2.12) *Trichoderma atroviride*, strain NMI no. V08/002388; (B2.13) *Trichoderma atroviride*, strain NMI no. V08/002389; (B2.14) *Trichoderma atroviride*, strain NMI no. V08/002390; (B2.15) *Trichoderma atroviride*, strain LC52 (e.g. Tenet by Agrimm Technologies Limited); (B2.16) *Trichoderma atroviride*, strain ATCC 20476 (IMI 206040); (B2.17) *Trichoderma atroviride*, strain T11 (IMI352941/CECT20498); (B2.18) *Trichoderma harzianum*; (B2.19) *Trichoderma harzianum*; (B2.20) *Trichoderma harzianum* rifai T39 (e.g. Trichodex® from Makhteshim, US); (B2.21) *Trichoderma asperellum*, in particular, strain kd (e.g. T-Gro from Andermatt Biocontrol); (B2.22) *Trichoderma harzianum*, strain ITEM 908 (e.g. Trianum-P from Koppert); (B2.23) *Trichoderma harzianum*, strain TH35 (e.g. Root-Pro by Mycontrol); (B2.24) *Trichoderma virens* (also known as *Gliocladium virens*), in particular strain GL-21 (e.g. SoilGard by Certis, US); (B2.25) *Trichoderma viride*, strain TV1 (e.g. Trianum-P by Koppert); (B2.26) *Amelomyces quisqualis*, in particular strain AQ 10 (e.g. AQ 10® by IntrachemBio Italia); (B2.27) *Aureobasidium pullulans*, in particular blastospores of strain DSM14940; (B2.28) *Aureobasidium pullulans*, in particular blastospores of strain DSM 14941; (B2.29) *Aureobasidium pullulans*, in particular mixtures of blastospores of strains DSM14940 and DSM 14941 (e.g. Botector® by bio-ferm, CH); (B2.30) *Cladosporium cladosporioides*, strain H39, having Accession No. CBS122244, US 2010/0291039 (by Stichting Dienst Landbouwkundig Onderzoek); (B2.31) *Gliocladium catenulatum* (Synonym: *Clonostachys rosea* f. *catenulate*) strain J1446 (e.g. Prestop® by Lallemand); (B2.32) *Lecanicillium lecanii* (formerly known as *Verticillium lecanii*) conidia of strain KV01 (e.g. Vertalec® by Koppert/Arysta); (B2.33) *Penicillium vermiculatum*; (B2.34) *Pichia anomala*, strain WRL-076 (NRRL Y-30842), U.S. Pat. No. 7,579,183; (B2.35) *Trichoderma atroviride*, strain SKT-1 (FERM P-16510), JP Patent Publication (Kokai) 11-253151 A; (B2.36) *Trichoderma atroviride*, strain SKT-2 (FERM P-16511), JP Patent Publication (Kokai) 11-253151 A;

(B2.37) *Trichoderma atroviride*, strain SKT-3 (FERM P-17021), JP Patent Publication (Kokai) 11-253151 A; (B2.38) *Trichoderma gamsii* (formerly *T. viride*), strain ICC080 (IMI CC 392151 CABI, e.g. BioDerma by AGROBIOSOL DE MEXICO, S.A. DE C.V.); (B2.39) *Trichoderma harzianum*, strain DB 103 (available as T-GRO® 7456 by Dagutut Biolab); (B2.40) *Trichoderma polysporum*, strain IMI 206039 (e.g. Binab TF WP by BINAB Bio-Innovation AB, Sweden); (B2.41) *Trichoderma stromaticum*, having Accession No. Ts3550 (e.g. Tricovab by CEPLAC, Brazil); (B2.42) *Ulocladium oudemansii* strain U3, having Accession No. NM 99/06216 (e.g., BOTRY-ZEN® by Botry-Zen Ltd, New Zealand and BOTRYSTOP® from Bio-Works, Inc.); (B2.43) *Verticillium albo-atrum* (formerly *V. dahliae*), strain WCS850 having Accession No. WCS850, deposited at the Central Bureau for Fungi Cultures (e.g., DUTCH TRIG® by Tree Care Innovations); (B2.44) *Verticillium chlamydosporium*; (B2.45) mixtures of *Trichoderma asperellum* strain ICC 012 (also known as *Trichoderma harzianum* ICC012), having Accession No. CABI CC IMI 392716 and *Trichoderma gamsii* (formerly *T. viride*) strain ICC 080, having Accession No. IMI 392151 (e.g., BIO-TAM™ from Isagro USA, Inc. and BIODERMA® by Agrobiosol de Mexico, S.A. de C.V.); (B2.46) *Trichoderma asperelloides* JM41R (Accession No. NRRL B-50759) (TRICHO PLUS® from BASF SE); (B2.47) *Aspergillus flavus* strain NRRL 21882 AFLA-GUARD® from Syngenta/ChemChina); (B2.48) *Chaetomium cupreum* (Accession No. CABI 353812) (e.g. BOKUPRUM™ by AgriLife); (B2.49) *Saccharomyces cerevisiae*, in particular strain LASO2 (from Agro-Levures et Dérivés), strain LAS117 cell walls (CEREVISANE® from Lesaffre; ROMEO® from BASF SE), strains CNCM No. I-3936, CNCM No. I-3937, CNCM No. I-3938, CNCM No. I-3939 (WO 2010/086790) from Lesaffre et Compagnie, FR; (B2.50) *Trichoderma virens* strain G-41, formerly known as *Gliocladium virens* (Accession No. ATCC 20906) (e.g., ROOTSHIELD® PLUS WP and TURFSHIELD® PLUS WP from BioWorks, US); (B2.51) *Trichoderma hamatum*, having Accession No. ATCC 28012; (B2.52) *Ampelomyces quisqualis* strain AQ10, having Accession No. CNCM I-807 (e.g., AQ 10® by IntrachemBio Italia); (B2.53) *Phlebiopsis gigantea* strain VRA 1992 (ROTSTOP® C from Danstar Ferment); (B2.54) *Penicillium steckii* (DSM 27859; WO 2015/067800) from BASF SE; (B2.55) *Chaetomium globosum* (available as RIVADIOM® by Rivale); (B2.56) *Cryptococcus flavecens*, strain 3C (NRRL Y-50378); (B2.57) *Dactylaria candida*; (B2.58) *Dilophosphora alopecuri* (available as TWIST FUNGUS®); (B2.59) *Fusarium oxysporum*, strain Fo47 (available as FUSACLEAN® by Natural Plant Protection); (B2.60) *Pseudozyma flocculosa*, strain PF-A22 UL (available as SPORODEX® L by Plant Products Co., CA); (B2.61) *Trichoderma gamsii* (formerly *T. viride*), strain ICC 080 (IMI CC 392151 CABI) (available as BIODERMA® by AGROBIOSOL DE MEXICO, S.A. DE C.V.); (B2.62) *Trichoderma fertile* (e.g. product TrichoPlus from BASF); (B2.63) *Muscodor roseus*, in particular strain A3-5 (Accession No. NRRL 30548); (B2.64) *Simplicillium lanosoniveum*;

[0113] (C) biological control agents having an effect for improving plant growth and/or plant health which may be combined in the compound combinations according to the invention including

[0114] (C1) bacteria selected from the group consisting of (C1.01) *Bacillus pumilus*, in particular strain QST2808 (having Accession No. NRRL No. B-30087); (C1.02) *Bacillus subtilis*, in particular strain QST713/AQ713 (having NRRL Accession No. B-21661 and described in U.S. Pat. No. 6,060,051; available as SERENADE® OPTI or SERENADE® ASO from Bayer CropScience LP, US); (C1.03) *Bacillus subtilis*, in particular strain AQ30002 (having Accession Nos. NRRL B-50421 and described in U.S. patent application Ser. No. 13/330,576); (C1.04) *Bacillus subtilis*, in particular strain AQ30004 (and NRRL B-50455 and described in U.S. patent application Ser. No. 13/330,576); (C1.05) *Sinorhizobium meliloti* strain NRG-185-1 (NITRAGIN® GOLD from Bayer Crop-Science); (C1.06) *Bacillus subtilis* strain BU1814, (available as TEQUALIS® from BASF SE); (C1.07) *Bacillus subtilis* rm303 (RHIZOMAX® from Biofilm Crop Protection); (C1.08) *Bacillus amyloliquefaciens* pm414 (LOLI-PEPTA® from Biofilm Crop Protection); (C1.09) *Bacillus mycoides* BT155 (NRRL No. B-50921), (C1.10) *Bacillus mycoides* EE118 (NRRL No. B-50918), (C1.11) *Bacillus mycoides* EE141 (NRRL No. B-50916), (C1.12) *Bacillus mycoides* BT46-3 (NRRL No. B-50922), (C1.13) *Bacillus cereus* family member EE128 (NRRL No. B-50917), (C1.14) *Bacillus thuringiensis* BT013A (NRRL No. B-50924) also known as *Bacillus thuringiensis* 4Q7, (C1.15) *Bacillus cereus* family member EE349 (NRRL No. B-50928), (C1.16) *Bacillus amyloliquefaciens* SB3281 (ATCC #PTA-7542; WO 2017/205258), (C1.17) *Bacillus amyloliquefaciens* TJ1000 (available as QUIK-ROOTS® from Novozymes); (C1.18) *Bacillus firmus*, in particular strain CNMC I-1582 (e.g. VOTIVO® from BASF SE); (C1.19) *Bacillus pumilus*, in particular strain GB34 (e.g. YIELD SHIELD® from Bayer Crop Science, DE); (C1.20) *Bacillus amyloliquefaciens*, in particular strain IN937a; (C1.21) *Bacillus amyloliquefaciens*, in particular strain FZB42 (e.g. RHIZOVITAL® from ABITEP, DE); (C1.22) *Bacillus amyloliquefaciens* BS27 (Accession No. NRRL B-5015); (C1.23) a mixture of *Bacillus licheniformis* FMCH001 and *Bacillus subtilis* FMCH002 (available as QUARTZO® (WG), PRESENCE® (WP) from FMC Corporation); (C1.24) *Bacillus cereus*, in particular strain BP01 (ATCC 55675; e.g. MEPICHLOR® from Arysta Lifescience, US); (C1.25) *Bacillus subtilis*, in particular strain MBI 600 (e.g. SUBTILEX® from BASF SE); (C1.26) *Bradyrhizobium japonicum* (e.g. OPTIMIZE® from Novozymes); (C1.27) *Mesorhizobium cicer* (e.g., NODULATOR from BASF SE); (C1.28) *Rhizobium leguminosarum* biovar *viciae* (e.g., NODULATOR from BASF SE); (C1.29) *Delftia acidovorans*, in particular strain RAY209 (e.g. BIO-BOOST® from Brett Young Seeds); (C1.30) *Lactobacillus* sp. (e.g. LACTOPLANT® from LactoPAFI); (C1.31) *Paenibacillus polymyxa*, in particular strain AC-1 (e.g. TOPSEED® from Green Biotech Company Ltd.); (C1.32) *Pseudomonas prora*dix (e.g. PRORADIX® from Sourcon Padena); (C1.33) *Azospirillum*

brasilense (e.g., VIGOR® from KALO, Inc.); (C1.34) *Azospirillum lipoferum* (e.g., VERTEX-IF™ from TerraMax, Inc.); (C1.35) a mixture of *Azotobacter vinelandii* and *Clostridium pasteurianum* (available as INVIGORATE® from Agrinos); (C1.36) *Pseudomonas aeruginosa*, in particular strain PN1; (C1.37) *Rhizobium leguminosarum*, in particular bv. viceae strain Z25 (Accession No. CECT 4585); (C1.38) *Azorhizobium caulinodans*, in particular strain ZB-SK-5; (C1.39) *Azotobacter chroococcum*, in particular strain H23; (C1.40) *Azotobacter vinelandii*, in particular strain ATCC 12837; (C1.41) *Bacillus siamensis*, in particular strain KCTC 13613T; (C1.42) *Bacillus tequilensis*, in particular strain NII-0943; (C1.43) *Serratia marcescens*, in particular strain SRM (Accession No. MTCC 8708); (C1.44) *Thiobacillus* sp. (e.g. CROPAID® from Cropaid Ltd UK); and

[0115] (C2) fungi selected from the group consisting of (C2.01) *Purpureocillium lilacinum* (previously known as *Paecilomyces lilacinus*) strain 251 (AGAL 89/030550; e.g. BioAct from Bayer CropScience Biologics GmbH); (C2.02) *Penicillium bilaii*, strain ATCC 22348 (e.g. JumpStart® from Acceleron BioAg); (C2.03) *Talaromyces flavus*, strain V117b; (C2.04) *Trichoderma atroviride* strain CNCM I-1237 (e.g. Esquive® WP from Agrauxine, FR); (C2.05) *Trichoderma viride*, e.g. strain B35 (Pietr et al., 1993, Zesz. Nauk. A R w Szczecinie 161:125-137); (C2.06) *Trichoderma atroviride* strain LC52 (also known as *Trichoderma atroviride* strain LU132; e.g. Sentinel from Agrimm Technologies Limited); (C2.07) *Trichoderma atroviride* strain SC1 described in International Application No. PCT/IT2008/000196; (C2.08) *Trichoderma asperellum* strain kd (e.g. T-Gro from Andermatt Biocontrol); (C2.09) *Trichoderma asperellum* strain Eco-T (Plant Health Products, ZA); (C2.10) *Trichoderma harzianum* strain T-22 (e.g. Trianium-P from Andermatt Biocontrol or Koppert); (C2.11) *Myrothecium verrucaria* strain AARC-0255 (e.g. DiTera™ from Valent Biosciences); (C2.12) *Penicillium bilaii* strain ATCC ATCC20851; (C2.13) *Pythium oligandrum* strain M1 (ATCC 38472; e.g. Polyversum from Bioprepaty, CZ); (C2.14) *Trichoderma virens* strain GL-21 (e.g. SoilGard® from Certis, USA); (C2.15) *Verticillium albo-atrum* (formerly *V. dahliae*) strain WCS850 (CBS 276.92; e.g. Dutch Trig from Tree Care Innovations); (C2.16) *Trichoderma atroviride*, in particular strain no. V08/002387, strain no. NMI No. V08/002388, strain no. NMI No. V08/002389, strain no. NMI No. V08/002390; (C2.17) *Trichoderma harzianum* strain ITEM 908; (C2.18) *Trichoderma harzianum*, strain TSTh20; (C2.19) *Trichoderma harzianum* strain 1295-22; (C2.20) *Pythium oligandrum* strain DV74; (C2.21) *Rhizopogon amylopogon* (e.g. comprised in Myco-Sol from Helena Chemical Company); (C2.22) *Rhizopogon fulvigleba* (e.g. comprised in Myco-Sol from Helena Chemical Company); and (C2.23) *Trichoderma virens* strain GI-3;

[0116] (D) insecticidally active biological control agents selected from

[0117] (D1) bacteria selected from the group consisting of (D1.01) *Bacillus thuringiensis* subsp. *aizawai*, in

particular strain ABTS-1857 (SD-1372; e.g. XENTARI® from Valent BioSciences); (D1.02) *Bacillus mycoides*, isolate J.

[0118] (e.g. BmJ from Certis USA LLC, a subsidiary of Mitsui & Co.); (D1.03) *Bacillus sphaericus*, in particular Serotype H5a5b strain 2362 (strain ABTS-1743) (e.g. VECTOLEX® from Valent BioSciences, US); (D1.04) *Bacillus thuringiensis* subsp. *kurstaki* strain BMP 123 from Becker Microbial Products, IL; (D1.05) *Bacillus thuringiensis* subsp. *aizawai*, in particular serotype H-7 (e.g. FLORBAC® WG from Valent BioSciences, US); (D1.06) *Bacillus thuringiensis* subsp. *kurstaki* strain HD-1 (e.g. DIPEL® ES from Valent BioSciences, US); (D1.07) *Bacillus thuringiensis* subsp. *kurstaki* strain BMP 123 by Becker Microbial Products, IL; (D1.08) *Bacillus thuringiensis israelensis* strain BMP 144 (e.g. AQUABAC® by Becker Microbial Products IL); (D1.09) *Burkholderia* spp., in particular *Burkholderia rinojensis* strain A396 (also known as *Burkholderia rinojensis* strain MBI 305) (Accession No. NRRL B-50319; WO 2011/106491 and WO 2013/032693; e.g. MBI-206 TGA1 and ZELTO® from Marrone Bio Innovations); (D1.10) *Chromobacterium subtsugae*, in particular strain PRAA4-1T (MBI-203; e.g. GRANDEVOR from Marrone Bio Innovations); (D1.11) *Paenibacillus popilliae* (formerly *Bacillus popilliae*; e.g. MILKY SPORE POWDER™ and MILKY SPORE GRANULAR™ from St. Gabriel Laboratories); (D1.12) *Bacillus thuringiensis* subsp. *israelensis* (serotype H-14) strain AM65-52 (Accession No. ATCC 1276) (e.g. VECTOBAC® by Valent BioSciences, US); (D1.13) *Bacillus thuringiensis* var. *kurstaki* strain EVB-113-19 (e.g., BIOPROTEC® from AEF Global); (D1.14) *Bacillus thuringiensis* subsp. *tenebrionis* strain NB 176 (SD-5428; e.g. NOVODOR® FC from BioFa DE); (D1.15) *Bacillus thuringiensis* var. *japonensis* strain Buibui; (D1.16) *Bacillus thuringiensis* subsp. *kurstaki* strain ABTS 351; (D1.17) *Bacillus thuringiensis* subsp. *kurstaki* strain PB 54; (D1.18) *Bacillus thuringiensis* subsp. *kurstaki* strain SA 11; (D1.19) *Bacillus thuringiensis* subsp. *kurstaki* strain SA 12; (D1.20) *Bacillus thuringiensis* subsp. *kurstaki* strain EG 2348; (D1.21) *Bacillus thuringiensis* var. *Colmeri* (e.g. TIANBAOBT by Changzhou Jianghai Chemical Factory); (D1.22) *Bacillus thuringiensis* subsp. *aizawai* strain GC-91; (D1.23) *Serratia entomophila* (e.g. INVADE® by Wrightson Seeds); (D1.24) *Serratia marcescens*, in particular strain SRM (Accession No. MTCC 8708); and (D1.25) *Wolbachia pipientis* ZAP strain (e.g., ZAP MALES® from Mosquito-Mate); and

[0119] (D2) fungi selected from the group consisting of (D2.01) *Isaria fumosorosea* (previously known as *Paecilomyces fumosoroseus*) strain apopka 97; (D2.02) *Beauveria bassiana* strain ATCC 74040 (e.g. NATURALIS® from Intrachem Bio Italia); (D2.03) *Beauveria bassiana* strain GHA (Accession No. ATCC74250; e.g. BOTANIGUARD® ES and MYCONTROL-OR from Laverlam International Corporation); (D2.04) *Zoopthora radicans*; (D2.05) *Metarhizium robertsii* 15013-1 (deposited under NRRL accession number 67073), (D2.06) *Metarhizium robertsii* 23013-3 (deposited under NRRL accession number 67075), and (D2.07) *Metarhizium anisopliae* 3213-1 (deposited under NRRL accession number 67074) (WO 2017/066094; Pioneer Hi-Bred International); (D2.08) *Beauveria bassiana* strain ATP02 (Accession No. DSM 24665);

- [0120] (E) viruses selected from the group consisting of *Adoxophyes orana* (summer fruit tortrix) granulosus virus (GV), *Cydia pomonella* (codling moth) granulosus virus (GV), *Helicoverpa armigera* (cotton bollworm) nuclear polyhedrosis virus (NPV), *Spodoptera exigua* (beet armyworm) mNPV, *Spodoptera frugiperda* (fall armyworm) mNPV, and *Spodoptera littoralis* (African cotton leafworm) NPV;
- [0121] (F) bacteria and fungi which can be added as 'inoculant' to plants or plant parts or plant organs and which, by virtue of their particular properties, promote plant growth and plant health. Examples are: *Agrobacterium* spp., *Azorhizobium caulinodans*, *Azospirillum* spp., *Azotobacter* spp., *Bradyrhizobium* spp., *Burkholderia* spp., in particular *Burkholderia cepacia* (formerly known as *Pseudomonas cepacia*), *Gigaspora* spp., and *Gigaspora monosporum*, *Glomus* spp., *Laccaria* spp., *Lactobacillus buchneri*, *Paraglomus* spp., *Pisolithus tinctorius*, *Pseudomonas* spp., *Rhizobium* spp., in particular *Rhizobium trifolii*, *Rhizopogon* spp., *Scleroderma* spp., *Suillus* spp., and *Streptomyces* spp.; and
- [0122] (G) plant extracts and products formed by micro-organisms including proteins and secondary metabolites which can be used as biological control agents, such as *Allium sativum*, *Artemisia absinthium*, azadirachtin, Biokeeper WP, *Cassia nigricans*, *Celastrus angulatus*, *Chenopodium anthelminticum*, chitin, Armour-Zen, *Dryopteris filix-mas*, *Equisetum arvense*, Fortune Aza, Fungastop, Heads Up (*Chenopodium quinoa* saponin extract), *Pyrethrum*/Pyrethrins, *Quassia amara*, *Quercus*, *Quillaja*, Regalia, "Requiem™ Insecticide", rotenone, ryania/ryanodine, *Symphytum officinale*, *Tanacetum vulgare*, thymol, Triact 70, TriCon, *Tropaeolum majus*, *Urtica dioica*, Veratrin, *Viscum album*, *Brassicaceae* extract, in particular oilseed rape powder or mustard powder, as well as bioinsecticidal/acaricidal active substances obtained from olive oil, in particular unsaturated fatty/carboxylic acids having carbon chain lengths C₁₆-C₂₀ as active ingredients, such as, for example, contained in the product with the trade name FLIPPER®.
- [0123] The compound and the composition of the invention may be combined with one or more active ingredients selected from insecticides, acaricides and nematocides.
- [0124] "Insecticides" as well as the term "insecticidal" refers to the ability of a substance to increase mortality or inhibit growth rate of insects. As used herein, the term "insects" comprises all organisms in the class "Insecta".
- [0125] "Nematicide" and "nematicidal" refers to the ability of a substance to increase mortality or inhibit the growth rate of nematodes. In general, the term "nematode" comprises eggs, larvae, juvenile and mature forms of said organism.
- [0126] "Acaricide" and "acaricidal" refers to the ability of a substance to increase mortality or inhibit growth rate of ectoparasites belonging to the class Arachnida, sub-class Acari.
- [0127] Examples of insecticides, acaricides and nematocides, respectively, which could be mixed with the compound and the composition of the invention are:
- [0128] (1) Acetylcholinesterase (AChE) inhibitors, preferably carbamates selected from alanycarb, aldycarb, bendiocarb, benfuracarb, butocarboxim, butoxycarboxim, carbaryl, carbofuran, carbosulfan, ethiofencarb, fenobucarb, formetanate, furathiocarb, isoprocarb, methiocarb, methomyl, metolcarb, oxamyl, pirimicarb, propoxur, thiodicarb, thiofanox, triazamate, trimethacarb, XMC and xylylcarb, or organophosphates selected from acephate, azamephos, azinphos-ethyl, azinphos-methyl, cadusafos, chlorethoxyfos, chlorfenvinphos, chlormephos, chlorpyrifos-methyl, coumaphos, cyanophos, demeton-S-methyl, diazinon, dichlorvos/DDVP, dicrotophos, dimethoate, dimethylvinphos, disulfoton, EPN, ethion, ethoprophos, famphur, fenamiphos, fenitrothion, fenthion, fosthiazate, imicyafos, heptenophos, isofenphos, isopropyl O-(methoxyaminothiophosphoryl) salicylate, isoxathion, malathion, mecarbam, methamidophos, methidathion, mevinphos, monocrotophos, naled, omethoate, oxydemeton-methyl, parathion-methyl, phenthoate, phorate, phosalone, phosmet, phosphamidon, phoxim, pirimiphos-methyl, profenofos, propetamphos, prothiofos, pyraclofos, pyridaphenthion, quinalphos, sulfotep, tebufos, temephos, terbufos, tetrachlorvinphos, thiometon, triazophos, trichlorfon and vamidothion.
- [0129] (2) GABA-gated chloride channel blockers, preferably cyclodiene-organochlorines selected from chlordane and endosulfan, or phenylpyrazoles (fiproles) selected from ethiprole and fipronil.
- [0130] (3) Sodium channel modulators, preferably pyrethroids selected from acrinathrin, allethrin, d-cis-trans allethrin, d-trans allethrin, bifenthrin, bioallethrin, bioallethrin s-cyclopentenyl isomer, bioresmethrin, cycloprothrin, cyfluthrin, beta-cyfluthrin, cyhalothrin, lambda-cyhalothrin, gamma-cyhalothrin, cypermethrin, alpha-cypermethrin, beta-cypermethrin, theta-cypermethrin, zeta-cypermethrin, cyphenothrin [(1R)-trans-isomer], deltamethrin, empenethrin [(EZ)-(1R)-isomer], esfenvalerate, etofenprox, fenpropathrin, fenvalerate, flucythrinate, flumethrin, tau-fluvalinate, halfenprox, imiprothrin, kadethrin, momfluorothrin, permethrin, phenothrin [(1R)-trans-isomer], prallethrin, pyrethrins (pyrethrum), resmethrin, silafluofen, tefluthrin, tetramethrin, tetramethrin [(1R)-isomer], tralomethrin and transfluthrin, or DDT or methoxychlor.
- [0131] (4) Nicotinic acetylcholine receptor (nAChR) competitive modulators, preferably neonicotinoids selected from acetamiprid, clothianidin, dinotefuran, imidacloprid, nitenpyram, thiacloprid and thiamethoxam, or nicotine, or sulfoximines selected from sulfoxaflor, or butenolids selected from flupyradifurone, or mesoionics selected from triflumezopyrim.
- [0132] (5) Nicotinic acetylcholine receptor (nAChR) allosteric modulators (Site I), preferably spinosyns selected from spinetoram and spinosad.
- [0133] (6) Glutamate-gated chloride channel (GluCl) allosteric modulators, preferably avermectins/milbemycins selected from abamectin, emamectin benzoate, lepimectin and milbemectin.
- [0134] (7) Juvenile hormone mimics, preferably juvenile hormone analogues selected from hydroprene, kinoprene and methoprene, or fenoxycarb or pyriproxyfen.
- [0135] (8) Miscellaneous non-specific (multi-site) inhibitors, preferably alkyl halides selected from methyl bromide and other alkyl halides, or chloropicrine or sulphuryl fluoride or borax or tartar emetic or methyl isocyanate generators selected from diazomet and metam.

[0136] (9) Chordotonal organ TRPV channel modulators, preferably pyridine azomethanes selected from pymetrozine and pyrifluquinazone, or pyropenes selected from afidopyropen.

[0137] (10) Mite growth inhibitors affecting CHS1 selected from clofentezine, hexythiazox, diflovidazin and etoxazole.

[0138] (11) Microbial disruptors of the insect gut membranes selected from *Bacillus thuringiensis* subspecies *israelensis*, *Bacillus sphaericus*, *Bacillus thuringiensis* subspecies *aizawai*, *Bacillus thuringiensis* subspecies *kurstaki*, *Bacillus thuringiensis* subspecies *tenebrionis*, and B.t. plant proteins selected from CryIAb, CryIAc, CryIFa, CryIA.105, Cry2Ab, Vip3A, mCry3A, Cry3Ab, Cry3Bb and Cry34Ab1/35Ab1.

[0139] (12) Inhibitors of mitochondrial ATP synthase, preferably ATP disruptors selected from diafenthiuron, or organotin compounds selected from azocyclotin, cyhexatin and fenbutatin oxide, or propargite or tetradifon.

[0140] (13) Uncouplers of oxidative phosphorylation via disruption of the proton gradient selected from chlorfenapyr, DNOC and sulfuramid.

[0141] (14) Nicotinic acetylcholine receptor channel blockers selected from bensultap, cartap hydrochloride, thiocylam and thiosultap-sodium.

[0142] (15) Inhibitors of chitin biosynthesis affecting CHS1, preferably benzoylureas selected from bistrifluron, chlorfluazuron, diflubenzuron, flucycloxuron, flufenoxuron, hexaflumuron, lufenuron, novaluron, noviflumuron, teflubenzuron and triflumuron.

[0143] (16) Inhibitors of chitin biosynthesis, type 1 selected from buprofezin.

[0144] (17) Moulting disruptor (in particular for Diptera, i.e. dipterans) selected from cyromazine.

[0145] (18) Ecdysone receptor agonists, preferably diacylhydrazines selected from chromafenozide, halofenozide, methoxyfenozide and tebufenozide.

[0146] (19) Octopamine receptor agonists selected from amitraz.

[0147] (20) Mitochondrial complex III electron transport inhibitors selected from hydramethylnone, acequinocyl, fluacrypyrim and bifenazate.

[0148] (21) Mitochondrial complex I electron transport inhibitors, preferably METI acaricides and insecticides selected from fenazaquin, fenpyroximate, pyrimidifen, pyridaben, tebufenpyrad and tolfenpyrad, or rotenone (Deris).

[0149] (22) Voltage-dependent sodium channel blockers, preferably oxadiazines selected from indoxacarb, or semicarbazones selected from metaflumizone.

[0150] (23) Inhibitors of acetyl COA carboxylase, preferably tetrone and tetramic acid derivatives selected from spirodiclofen, spiromesifen, spiropidion and spirotetramat.

[0151] (24) Mitochondrial complex IV electron transport inhibitors, preferably phosphides selected from aluminium phosphide, calcium phosphide, phosphine and zinc phosphide, or cyanides selected from calcium cyanide, potassium cyanide and sodium cyanide.

[0152] (25) Mitochondrial complex II electron transport inhibitors, preferably beta-ketonitrile derivatives selected from cyenopyrafen and cyflumetofen, or carboxanilides selected from pyflubumide.

[0153] (28) Ryanodine receptor modulators, preferably diamides selected from chlorantraniliprole, cyantraniliprole, cyclaniliprole, flubendiamide and tetraniliprole.

[0154] (29) Chordotonal organ Modulators (with undefined target site) selected from flonicamid.

[0155] (30) GABA-gated chlorid channel allosteric modulators, preferably meta-diamides selected from broflanilide, or isoxazoles selected from fluxametamide.

[0156] (31) Baculoviruses, preferably Granuloviruses (GVs) selected from *Cydia pomonella* GV and *Thaumetotibia leucotreta* (GV), or Nucleopolyhedroviruses (NPVs) selected from *Anticarsia gemmatilis* MNPV, Flucypyriprole and *Helicoverpa armigera* NPV.

[0157] (32) Nicotinic acetylcholine receptor allosteric modulators (Site II) selected from GS-omega/kappa HXTX-Hv1a peptide.

[0158] (33) further active compounds selected from Acynonapyr, Afoxolaner, Azadirachtin, Benclothiaz, Benzoximate, Benzpyrimoxan, Bromopropylate, Chinomethionat, Chloroprallethrin, Cryolite, Cyclobutirfluram, Cycloxaprid, Cyetpyrafen, Cyhalodiamide, Cyproflanilide (CAS 2375110-88-4), Dicloromezotiaz, Dicofol, Dimpropyridaz, epsilon-Metofluthrin, epsilon-Momfluthrin, Flometoquin, Fluzaindolizine, Flucypyriprole (CAS 1771741-86-6), Flusulfone, Flufenerim, Flufenoxystrobin, Flufiprole, Fluhexafon, Fluopyram, Flupyrimin, Fluralaner, Fufenozide, Flupentiofenox, Guadipyr, Heptafluthrin, Imidaclothiz, Iprodione, Isocycloseram, kappa-Bifenthrin, kappa-Tefluthrin, Lotilaner, Meperfluthrin, Nicofluprole (CAS 1771741-86-6), Oxazosulfyl, Paichongding, Pyridalyl, Pyrifluquinazon, Pyriminostrobin, Sarolaner, Spidoxamat, Spirobudiclofen, Tetramethylfluthrin, Tetrachlorantraniliprole, Tigolaner, Tioxazafen, Thiofluoximate, Tyclopyrazoflor, Iodomethane; furthermore preparations based on *Bacillus firmus* (I-1582, Votivo) and azadirachtin (BioNeem), and also the following compounds: 1-{2-fluoro-4-methyl-5-[(2,2,2-trifluoroethyl) sulphonyl]phenyl}-3-(trifluoromethyl)-1H-1,2,4-triazole-5-amine (known from WO2006/043635) (CAS 885026-50-6), 2-chloro-N-[2-{1-[(2E)-3-(4-chlorophenyl) prop-2-en-1-yl]piperidin-4-yl}-4-(trifluoromethyl) phenyl]isonicotinamide (known from WO2006/003494) (CAS 872999-66-1), 3-(4-chloro-2,6-dimethylphenyl)-4-hydroxy-8-methoxy-1,8-diazaspiro[4.5]dec-3-en-2-one (known from WO 2010052161) (CAS 1225292-17-0), 3-(4-chloro-2,6-dimethylphenyl)-8-methoxy-2-oxo-1,8-diazaspiro[4.5]dec-3-en-4-yl ethyl carbonate (known from EP2647626) (CAS 1440516-42-6), PF1364 (known from JP2010/018586) (CAS 1204776-60-2), (3E)-3-[1-[(6-chloro-3-pyridyl)methyl]-2-pyridylidene]-1,1,1-trifluoropropan-2-one (known from WO2013/144213) (CAS 1461743-15-6), N-[3-(benzylcarbamoyl)-4-chlorophenyl]-1-methyl-3-(pentafluoroethyl)-4-(trifluoromethyl)-1H-pyrazole-5-carboxamide (known from WO2010/051926) (CAS 1226889-14-0), 5-bromo-4-chloro-N-[4-chloro-2-methyl-6-(methylcarbamoyl)phenyl]-2-(3-chloro-2-pyridyl) pyrazole-3-carboxamide (known from CN103232431) (CAS 1449220-44-3), 4-[5-(3,5-dichlorophenyl)-4,5-dihydro-5-(trifluoromethyl)-3-isoxazolyl]-2-methyl-N-(cis-1-oxido-3-thietanyl)-benzamide, 4-[5-(3,5-dichlorophenyl)-4,5-dihydro-5-(trifluoromethyl)-3-isoxazolyl]-2-methyl-N-(trans-1-oxido-3-thietanyl)-benzamide and 4-[(5S)-5-(3,5-dichlorophenyl)-4,5-dihydro-5-(trifluoromethyl)-3-isoxazolyl]-2-methyl-N-(cis-1-oxido-3-thietanyl)benzamide (known from WO 2013/050317 A1) (CAS 1332628-83-7),

N-[3-chloro-1-(3-pyridinyl)-1H-pyrazol-4-yl]-N-ethyl-3-[(3,3,3-trifluoropropyl) sulfinyl]-propanamide, (+)-N-[3-chloro-1-(3-pyridinyl)-1H-pyrazol-4-yl]-N-ethyl-3-[(3,3,3-trifluoropropyl) sulfinyl]-propanamide and (-)-N-[3-chloro-1-(3-pyridinyl)-1H-pyrazol-4-yl]-N-ethyl-3-[(3,3,3-trifluoropropyl) sulfinyl]-propanamide (known from WO 2013/162715 A2, WO 2013/162716 A2, US 2014/0213448 A1) (CAS 1477923-37-7), 5-[[[(2E)-3-chloro-2-propen-1-yl] amino]-1-[2,6-dichloro-4-(trifluoromethyl)phenyl]-4-[(trifluoromethyl) sulfinyl]-1H-pyrazole-3-carbonitrile (known from CN 101337937 A) (CAS 1105672-77-2), 3-bromo-N-[4-chloro-2-methyl-6-[(methylamino)thioxomethyl]phenyl]-1-(3-chloro-2-pyridinyl)-1H-pyrazole-5-carboxamide, (Liudaibenjiaxuanan, known from CN 103109816 A) (CAS 1232543-85-9); N-[4-chloro-2-[(1,1-dimethylethyl)amino] carbonyl]-6-methylphenyl]-1-(3-chloro-2-pyridinyl)-3-(fluoromethoxy)-1H-pyrazole-5-carboxamide (known from WO 2012/034403 A1) (CAS 1268277-22-0), N-[2-(5-amino-1,3,4-thiadiazol-2-yl)-4-chloro-6-methylphenyl]-3-bromo-1-(3-chloro-2-pyridinyl)-1H-pyrazole-5-carboxamide (known from WO 2011/085575 A1) (CAS 1233882-22-8), 4-[3-[2,6-dichloro-4-[(3,3-dichloro-2-propen-1-yl)oxy] phenoxy]propoxy]-2-methoxy-6-(trifluoromethyl)-pyrimidine (known from CN 101337940 A) (CAS 1108184-52-6); (2E)- and 2 (Z)-2-[2-(4-cyanophenyl)-1-[3-(trifluoromethyl)phenyl]ethylidene]-N-[4-(difluoromethoxy)phenyl]-hydrazinecarboxamide (known from CN 101715774 A) (CAS 1232543-85-9); 3-(2,2-dichloroethenyl)-2,2-dimethyl-4-(1H-benzimidazol-2-yl)phenyl-cyclopropanecarboxylic acid ester (known from CN 103524422 A) (CAS 1542271-46-4); (4aS)-7-chloro-2,5-dihydro-2-[[[(methoxycarbonyl) 4-[(trifluoromethyl)thio] phenyl]amino]carbonyl]-indeno[1,2-e][1,3,4]oxadiazine-4a (3H)-carboxylic acid methyl ester (known from CN 102391261 A) (CAS 1370358-69-2); 6-deoxy-3-O-ethyl-2,4-di-O-methyl-, 1-[N-[4-[1-[4-(1,1,2,2,2-pentafluoroethoxy)phenyl]-1H-1,2,4-triazol-3-yl]phenyl]carbamate]- α -L-mannopyranose (known from US 2014/0275503 A1) (CAS 1181213-14-8); 8-(2-cyclopropylmethoxy-4-trifluoromethyl-phenoxy)-3-(6-trifluoromethyl-pyridazin-3-yl)-3-aza-bicyclo[3.2.1]octane (CAS 1253850-56-4), (8-anti)-8-(2-cyclopropylmethoxy-4-trifluoromethyl-phenoxy)-3-(6-trifluoromethyl-pyridazin-3-yl)-3-aza-bicyclo[3.2.1]octane (CAS 933798-27-7), (8-syn)-8-(2-cyclopropylmethoxy-4-trifluoromethyl-phenoxy)-3-(6-trifluoromethyl-pyridazin-3-yl)-3-aza-bicyclo[3.2.1]octane (known from WO 2007040280 A1, WO 2007040282 A1) (CAS 934001-66-8), N-[4-(aminothioxomethyl)-2-methyl-6-[(methylamino) carbonyl]phenyl]-3-bromo-1-(3-chloro-2-pyridinyl)-1H-pyrazole-5-carboxamide (known from CN 103265527 A) (CAS 1452877-50-7), 3-(4-chloro-2,6-dimethylphenyl)-8-methoxy-1-methyl-1,8-diazaspiro[4.5]decane-2,4-dione (known from WO 2014/187846 A1) (CAS 1638765-58-8), 3-(4-chloro-2,6-dimethylphenyl)-8-methoxy-1-methyl-2-oxo-1,8-diazaspiro[4.5]dec-3-en-4-yl-carbonic acid ethyl ester (known from WO 2010/066780 A1, WO 2011151146 A1) (CAS 1229023-00-0), N-[1-(2,6-difluorophenyl)-1H-pyrazol-3-yl]-2-(trifluoromethyl)benzamide (known from WO 2014/053450 A1) (CAS 1594624-87-9), N-[2-(2,6-difluorophenyl)-2H-1,2,3-triazol-4-yl]-2-(trifluoromethyl) benzamide (known from WO 2014/053450 A1) (CAS 1594637-65-6), N-[1-(3,5-difluoro-2-pyridinyl)-1H-pyrazol-3-yl]-2-(trifluoromethyl)benzamide (known from WO 2014/053450 A1) (CAS 1594626-19-3), (3R)-3-(2-chloro-

5-thiazolyl)-2,3-dihydro-8-methyl-5,7-dioxo-6-phenyl-5H-thiazolo[3,2-a]pyrimidinium inner salt (known from WO 2018/177970 A1) (CAS 2246757-58-2); 3-(2-chloro-5-thiazolyl)-2,3-dihydro-8-methyl-5,7-dioxo-6-phenyl-5H-thiazolo[3,2-a]pyrimidinium inner salt (known from WO 2018/177970 A1) (CAS 2246757-56-0); N-[3-chloro-1-(3-pyridinyl)-1H-pyrazol-4-yl]-2-(methylsulfonyl)-propanamide (known from WO 2019/236274 A1) (CAS 2396747-83-2), N-[2-bromo-4-[1,2,2,2-tetrafluoro-1-(trifluoromethyl)ethyl]-6-(trifluoromethyl)phenyl]-2-fluoro-3-[(4-fluorobenzoyl)amino]-benzamide (known from WO 2019059412 A1) (CAS 1207977-87-4), 3-Bromo-1-(3-chloro-2-pyridinyl)-N-[4,6-dichloro-3-fluoro-2-[(methylamino) carbonyl]phenyl]-1H-Pyrazole-5-carboxamide (Fluochlorodiamide; known from CN110835330 A, CN106977494 A) (CAS: 2129147-03-9).

[0159] Examples of nematocides which could be mixed with the compound and the composition of the invention are:

[0160] (Group N-1) Acetylcholinesterase (AChE) inhibitors, preferably (N-1A) carbamates selected from aldicarb, benfuracarb, carbofuran, carbosulfan and thiodicarb, or (N-1B) organophosphates selected from cadusafos, ethoprophos, fenamiphos, fosthiazate, imicyafos, phorate and terbufos.

[0161] (Group N-2) Glutamate-gated chloride channel (GluCl) allosteric modulators, preferably avermectins selected from abamectin and emamectin benzoate.

[0162] (Group N-3) Mitochondrial complex II electron transport inhibitors, especially inhibitors of succinate-coenzyme Q reductase, preferably pyridinylmethyl-benzamides selected from fluopyram.

[0163] (Group N-4) Lipid synthesis/growth regulation modulators, especially inhibitors of acetyl CoA carboxylase, preferably tetrone and tetramic acid derivatives selected from spirotetramat.

[0164] (Group N-UN) Compounds of unknown or uncertain mode of action with various chemistries, selected from fluensulfone, fluzaindolizine, furfural, iprodione and tiozafen.

[0165] (Group N-UNX) Compounds of unknown or uncertain mode of action: Presumed multi-site inhibitors, preferably volatile sulphur generators selected from carbon disulphide and dimethyl disulphide (DMS), or carbon disulphide liberators selected from sodium tetrathiocarbonate, or alkyl halides selected from methyl bromide and methyl iodide (iodomethane), or halogenated hydrocarbons selected from 1,2-dibromo-3-chloropropane (DBCP) and 1,3-dichloropropene, or chloropicrin, or methyl isothiocyanate generators selected from allyl isothiocyanate, diazomet, metam potassium and metam sodium.

[0166] (Group N-UNB) Bacterial agents (non-Bt) of unknown or uncertain mode of action, preferably bacterium or bacterium-derived, selected from *Burkholderia* spp., e.g. *rinjensis* A396, *Bacillus* spp., e.g. *firmus*, *licheniformis*, *amyloliquefaciens* or *subtilis*, *Pasteuria* spp., e.g. *penetrans* or *nishizawae*, *Pseudomonas* spp., e.g. *chlororaphis* or *fluorescens*, and *Streptomyces* spp., e.g. *lydicus*, *dicklowii* or *albogriseolus*.

[0167] (Group N-UNF) Fungal agents of unknown or uncertain mode of action, preferably fungus or fungus-derived, selected from *Actinomyces* spp., e.g. *streptococcus*, *Arthrobotrys* spp., e.g. *oligospora*, *Aspergillus* spp., e.g. *niger*, *Muscodor* spp., e.g. *albus*, *Myrothecium* spp., e.g. *verrucaria*, *Paecilomyces* spp., e.g. *lilacinus* (*Purpureocill-*

lium lilacinum), *carneus* or *fumosoroseus*, *Pochonia* spp., e.g. *chlamydosporia*, and *Trichoderma* spp., e.g. *harzianum*, *virens*, *atroviride* or *viride*.

[0168] (Group N-UNE) Botanical or animal derived agents, including synthetic extracts and unrefined oils, with unknown or uncertain mode of action, preferably botanical or animal derived agents selected from azadirachtin, *camellia* seed cake, essential oils, garlic extract, pongamia oil, terpenes, e.g. carvacrol, and *Quillaja saponaria* extract.

[0169] Examples of herbicides which could be mixed with the compound and the composition of the invention are:

[0170] acetochlor, acifluorfen, acifluorfen-methyl, acifluorfen-sodium, aclonifen, alachlor, allidochlor, alloxydim, alloxydim-sodium, ametryn, amicarbazone, amidochlor, amidosulfuron, 4-amino-3-chloro-6-(4-chloro-2-fluoro-3-methylphenyl)-5-fluoropyridine-2-carboxylic acid, aminocyclopyrachlor, aminocyclopyrachlor-potassium, aminocyclopyrachlor-methyl, aminopyralid, aminopyralid-dimethylammonium, aminopyralid-tripromine, amitrole, ammoniumsulfamate, anilofos, asulam, asulam-potassium, asulam sodium, atrazine, azafenidin, azimsulfuron, beflubutamid, (S)-(-)-beflubutamid, beflubutamid-M, benazolin, benazolin-ethyl, benazolin-dimethylammonium, benazolin-potassium, benfluralin, benfuresate, bensulfuron, bensulfuron-methyl, bensulide, bentazone, bentazone-sodium, benzobicyclon, benzofenap, bicyclopyrone, bifenox, bilanafos, bilanafos-sodium, bipyrazone, bispyribac, bispyribac-sodium, bixlozone, bromacil, bromacil-lithium, bromacil-sodium, bromobutide, bromofenoxim, bromoxynil, bromoxynil-butyrate, potassium, heptanoate und-octanoate, busoxinone, butachlor, butafenacil, butamifos, butenachlor, butralin, butoxydim, butylate, cafenstrole, cambendichlor, carbetamide, carfentrazone, carfentrazone-ethyl, chloramben, chloramben-ammonium, chloramben-diolamine, chloamben-methyl, chloramben-methylammonium, chloramben-sodium, chlorbromuron, chlorfenac, chlorfenac-ammonium, chlorfenac-sodium, chlorfenprop, chlorfenprop-methyl, chlorflurenol, chlorflurenol-methyl, chloridazon, chlorimuron, chlorimuron-ethyl, chlorophthalim, chlorotoluron, chlorsulfuron, chlorthal, chlorthal-dimethyl, chlorthalmonomethyl, cinidon, cinidon-ethyl, cinmethylin, exo-(+)-cinmethylin, i.e. (1R,2S,4S)-4-isopropyl-1-methyl-2-[(2-methylbenzyl)oxy]-7-oxabicyclo[2.2.1]heptane, exo-(-)-cinmethylin, i.e. (1R,2S,4S)-4-isopropyl-1-methyl-2-[(2-methylbenzyl)oxy]-7-oxabicyclo[2.2.1]heptane, cinosulfuron, clacyfos, clethodim, clodinafop, clodinafop-ethyl, clodinafop-propargyl, clomazone, clomeprop, clopyralid, clopyralid-methyl, clopyralid-olamine, clopyralid-potassium, clopyralid-tripromine, cloransulam, cloransulam-methyl, cumyluron, cyanamide, cyanazine, cycloate, cyclopyranil, cyclopyrimorate, cyclosulfamuron, cycloxydim, cyhalofop, cyhalofop-butyl, cyprazine, 2,4-D (including thea mmonium, butotyl-, butyl-, choline, diethylammonium-, dimethylammonium, -diolamine, -doboxyl, -dodecylammonium, etexyl, ethyl, 2-ethylhexyl, heptylammonium, isobutyl, isooctyl, isopropyl, isopropylammonium, lithium, meptyl, methyl, potassium, tetradecylammonium, triethylammonium, triisopropanolammonium, tripromine and trolamine salt thereof), 2,4-DB, 2,4-DB-butyl-, dimethylammonium,

isooctyl-, potassium und-sodium, daimuron (dymron), dalapon, dalapon-calcium, dalapon-magnesium, dalapon-sodium, dazomet, dazomet-sodium, n-decanol, 7-deoxy-D-sedoheptulose, desmedipham, detosylpyrazolate (DTP), dicamba and its salts, e. g. dicamba-biproamine, dicamba-N,N-Bis(3-aminopropyl) methylamine, dicamba-butotyl, dicamba-choline, dicamba-diglycolamine, dicamba-dimethylammonium, dicamba-diethanolamine ammonium, dicamba-diethylammonium, dicamba-isopropylammonium, dicamba-methyl, dicamba-monoethanolamine, dicamba-olamine, dicamba-potassium, dicamba-sodium, dicamba-triethanolamine, dichlobenil, 2-(2,4-dichlorobenzyl)-4,4-dimethyl-1,2-oxazolidin-3-one, 2-(2,5-dichlorobenzyl)-4,4-dimethyl-1,2-oxazolidin-3-one, dichlorprop, dichlorprop-butotyl, dichlorprop-dimethylammonium, dichlorprop-etexyl, dichlorprop-ethylammonium, dichlorprop-isooctyl, dichlorprop-methyl, dichlorprop-potassium, dichlorprop-sodium, dichlorprop-P, dichlorprop-P-dimethyl-ammonium, dichlorprop-P-etexyl, dichlorprop-P-potassium, dichlorprop-sodium, diclofop, diclofop-methyl, diclofop-P, diclofop-P-methyl, diclosulam, difenzoquat, difenzoquat-metilsulfate, diflufenican, diflufenzopyr, diflufenzopyr-sodium, dimefuron, dimepiperate, dimesulfazet, dimethachlor, dimethametryn, dimethenamid, dimethenamid-P, dimetrasulfuron, dinitramine, dinoterb, dinoterb-acetate, diphenamid, diquat, diquat-dibromid, diquat-dichloride, dithiopyr, diuron, DNOC, DNOC-ammonium, DNOC-potassium, DNOC-sodium, endothal, endothal-diammonium, endothal-dipotassium, endothal-disodium, Epyrifenacil (S-3100), EPTC, esprocarb, ethalfluralin, ethametsulfuron, ethametsulfuron-methyl, ethiozin, ethofumesate, ethoxyfen, ethoxyfen-ethyl, ethoxysulfuron, etobenzanid, F-5231, i.e. N-[2-Chlor-4-fluor-5-[4-(3-fluorpropyl)-4,5-dihydro-5-oxo-1H-tetrazol-1-yl]-phenyl]-ethansulfonamid, F-7967, i.e. 3-[7-Chlor-5-fluor-2-(trifluormethyl)-1H-benzimidazol-4-yl]-1-methyl-6-(trifluoromethyl)pyrimidin-2,4 (1H,3H)-dione, fenoxaprop, fenoxaprop-P, fenoxaprop-ethyl, fenoxaprop-P-ethyl, fenoxasulfone, fenpyrazone, fenquinotrione, fentrazamide, flamprop, flamprop-isopropyl, flamprop-methyl, flamprop-M-isopropyl, flamprop-M-methyl, flazasulfuron, florasulam, florpyrauxifen, florpyrauxifen-benzyl, fluazifop, fluazifop-butyl, fluazifop-methyl, fluazifop-P, fluazifop-P-butyl, flucarbazone, flucarbazone-sodium, flucetosulfuron, fluchloralin, flufenacet, flufenpyr, flufenpyr-ethyl, flumetsulam, flumiclorac, flumiclorac-pentyl, flumioxazin, fluometuron, flurenol, flurenol-butyl-, dimethylammonium und-methyl, fluoroglycofen, fluoroglycofen-ethyl, flupropanate, flupropanate-sodium, flupyrsulfuron, flupyrsulfuron-methyl, flupyrsulfuron-methyl-sodium, fluridone, flurochloridone, fluroxypyr, fluroxypyr-butometyl, fluroxypyr-meptyl, flurtamone, fluthiacet, fluthiacet-methyl, fomesafen, fomesafen-sodium, foramsulfuron, foramsulfuron sodium salt, fosamine, fosamine-ammonium, glufosinate, glufosinate-ammonium, glufosinate-sodium, L-glufosinate-ammonium, L-glufosinate-sodium, glufosinate-P-sodium, glufosinate-P-ammonium, glyphosate, glyphosate-ammonium, isopropylammonium, -diammonium, -dimethylammonium, -potassium, -

sodium, sesquisodium and-trimesium, H-9201, i.e. O-(2,4-Dimethyl-6-nitrophenyl)-O-ethyl-isopropylphosphoramidothioat, halauxifen, halauxifen-methyl, halosafen, halosulfuron, halosulfuron-methyl, haloxyfop, haloxyfop-P, haloxyfop-ethoxyethyl, haloxyfop-P-ethoxyethyl, haloxyfop-methyl, haloxyfop-P-methyl, haloxyfop-sodium, hexazinone, HNPC-A8169, i.e. prop-2-yn-1-yl (2S)-2-{3-[(5-tert-butylpyridin-2-yl)oxy]phenoxy}propanoate, HW-02, i.e. 1-(Dimethoxyphosphoryl)-ethyl-(2,4-dichlorophenoxy) acetat, hydantocidin, imazamethabenz, imazamethabenz-methyl, imazamox, imazamox-ammonium, imazapic, imazapic-ammonium, imazapyr, imazapyr-isopropylammonium, imazaquin, imazaquin-ammonium, imazaquin-methyl, imazethapyr, imazethapyr-immonium, imazosulfuron, indanofan, indaziflam, iodosulfuron, iodosulfuron-methyl, iodosulfuron-methyl-sodium, ioxynil, ioxynil-lithium,-octanoate,-potassium und sodium, ipfencarbazone, isoproturon, isouron, isoxaben, isoxaflutole, karbutilate, KUH-043, i.e. 3-({[5-(Difluormethyl)-1-methyl-3-(trifluormethyl)-1H-pyrazol-4-yl]methyl}sulfonyl)-5,5-dimethyl-4,5-dihydro-1,2-oxazol, ketospiradox, ketospiradox-potassium, lactofen, lancotrione, lenacil, linuron, MCPA, MCPA-butyl, butyl,-dimethylammonium,-diolamine,-2-ethylhexyl,-ethyl,-isobutyl, isocetyl,-isopropyl,-isopropylammonium,-methyl, olamine,-potassium,-sodium and-trolamine, MCPB, MCPB-methyl,-ethyl und-sodium, mecoprop, mecoprop-butyl, mecoprop-demethylammonium, mecoprop-diolamine, mecoprop-etexyl, mecoprop-ethadyl, mecoprop-isocetyl, mecoprop-methyl, mecoprop-potassium, mecoprop-sodium, and mecoprop-trolamine, mecoprop-P, mecoprop-P-butyl,-dimethylammonium,-2-ethylhexyl and-potassium, mefenacet, mefluidide, mefluidide-diolamine, mefluidide-potassium, mesosulfuron, mesosulfuron-methyl, mesosulfuron sodium salt, mesotrione, methabenzthiazuron, metam, metamifop, metamitron, metazachlor, metazosulfuron, methabenzthiazuron, methiopyrsulfuron, methiozolin, methyl isothiocyanate, metobromuron, metolachlor, S-metolachlor, metosulam, metoxuron, metribuzin, metsulfuron, metsulfuron-methyl, molinate, monolinuron, monosulfuron, monosulfuron-methyl, MT-5950, i.e. N-[3-chlor-4-(1-methylethyl)-phenyl]-2-methylpentanamid, NGGC-011, napropamide, NC-310, i.e. 4-(2,4-Dichlorbenzoyl)-1-methyl-5-benzyl-oxypyrazol, NC-656, i.e. 3-[(isopropylsulfonyl)methyl]-N-(5-methyl-1,3,4-oxadiazol-2-yl)-5-(trifluoro-methyl) [1,2,4]triazolo[4,3-a]pyridine-8-carboxamide, neburon, nicosulfuron, nonanoic acid (pelargonic acid), norflurazon, oleic acid (fatty acids), orbencarb, orthosulfamuron, oryzalin, oxadiargyl, oxadiazon, oxasulfuron, oxaziclomefone, oxyfluorfen, paraquat, paraquat-dichloride, paraquat-dimethylsulfate, pebulate, pendimethalin, penoxsulam, pentachlorophenol, pentoxazone, pethoxamid, petroleum oils, phenmedipham, phenmedipham-ethyl, picloram, picloram-dimethylammonium, picloram-etexyl, picloram-isocetyl, picloram-methyl, picloram-olamine, picloram-potassium, picloram-triethylammonium, picloram-tripromine, picloram-trolamine, picolinafen, pinoxaden, piperophos, pretilachlor, primisulfuron, primisulfuron-methyl, prodiamine, profoxydim, prometon, prom-

etryn, propachlor, propanil, propaquizafop, propazine, propham, propisochlor, propoxycarbazone, propoxycarbazone-sodium, propyrisulfuron, propyzamide, prosulfocarb, prosulfuron, pyraclostrobin, pyraflufen, pyraflufen-ethyl, pyrasulfotole, pyrazolynate (pyrazolate), pyrazosulfuron, pyrazosulfuron-ethyl, pyrazoxyfen, pyribambenz, pyribambenz-isopropyl, pyribambenz-propyl, pyribenzoxim, pyributicarb, pyridafol, pyridate, pyriftalid, pyriminobac, pyriminobac-methyl, pyrimisulfan, pyriothiac, pyriothiac-sodium, pyroxasulfone, pyroxsulam, quinclorac, quinclorac-dimethylammonium, quinclorac-methyl, quinmerac, quincolamine, quizalofop, quizalofop-ethyl, quizalofop-P, quizalofop-P-ethyl, quizalofop-P-tefuryl, QYM201, i.e. 1-{2-chloro-3-[(3-cyclopropyl-5-hydroxy-1-methyl-1H-pyrazol-4-yl) carbonyl]-6-(trifluoromethyl)phenyl}piperidin-2-one, rimsulfuron, saflufenacil, sethoxydim, siduron, simazine, simetryn, SL-261, sulcotrione, sulfentrazone, sulfometuron, sulfometuron-methyl, sulfosulfuron, SYP-249, i.e. 1-Ethoxy-3-methyl-1-oxobut-3-en-2-yl-5-[2-chlor-4-(trifluoromethyl) phenoxy]-2-nitrobenzoat, SYP-300, i.e. 1-[7-Fluor-3-oxo-4-(prop-2-in-1-yl)-3,4-dihydro-2H-1,4-benzoxazin-6-yl]-3-propyl-2-thioxoimidazolidin-4,5-dion, 2,3,6-TBA, TCA (trichloro acetic acid) and its salts, e.g. TCA-ammonium, TCA-calcium, TCA-ethyl, TCA-magnesium, TCA-sodium, tebuthiuron, tefuryltrione, tembotrione, tepaloxymid, terbacil, terbucarb, terbutolone, terbuthylazine, terbutryn, tetflupyrrolimet, thaxtomin, thenylchlor, thiazopyr, thiencarbazone, thiencarbazone-methyl, thifensulfuron, thifensulfuron-methyl, thiobencarb, tiafenacil, topralate, topramezone, tralkoxydim, triafamone, triallate, triasulfuron, triaziflam, tribenuron, tribenuron-methyl, triclopyr, triclopyr-butyl, triclopyr-choline, triclopyr-ethyl, triclopyr-triethylammonium, trietazine, trifloxysulfuron, trifloxysulfuron-sodium, trifludimoxazin, trifluralin, triflusulfuron, triflusulfuron-methyl, tritosulfuron, urea sulfate, vernolate, XDE-848, ZJ-0862, i.e. 3,4-Dichlor-N-{2-[(4,6-dimethoxypyrimidin-2-yl)oxy]benzyl}anilin, 3-(2-chloro-4-fluoro-5-(3-methyl-2,6-dioxo-4-trifluoromethyl-3,6-dihydro-1,2,4-triazol-5-yl)phenyl)-5-methyl-4,5-dihydroisoxazole-5-carboxylic acid ethyl ester, ethyl-[(3-{2-chlor-4-fluor-5-[3-methyl-2,6-dioxo-4-(trifluoromethyl)-3,6-dihydro-1,2,4-triazol-5-yl]phenyl}oxy)pyridin-2-yl]oxy]acetate, 3-chloro-2-[3-(difluoromethyl) isoxazolyl-5-yl]phenyl-5-chloropyrimidin-2-yl ether, 2-(3,4-dimethoxyphenyl)-4-[(2-hydroxy-6-oxocyclohex-1-en-1-yl) carbonyl]-6-methylpyridazine-3 (2H)-one, 2-[(2-methoxyethoxy)methyl]-6-methylpyridin-3-yl]carbonyl)cyclohexane-1,3-dione, (5-hydroxy-1-methyl-1H-pyrazol-4-yl) (3,3,4-trimethyl-1,1-dioxido-2,3-dihydro-1-benzothiophen-5-yl) methanone, 1-methyl-4-[(3,3,4-trimethyl-1,1-dioxido-2,3-dihydro-1-benzothiophen-5-yl) carbonyl]-1H-pyrazol-5-yl propane-1-sulfonate, 4-{2-chloro-3-[(3,5-dimethyl-1H-pyrazol-1-yl)methyl]-4-(methylsulfonyl)benzoyl}-1-methyl-1H-pyrazol-5-yl-1,3-dimethyl-1H-pyrazole-4-carboxylate, cyanomethyl 4-amino-3-chloro-5-fluoro-6-(7-fluoro-1H-indol-6-yl)pyridine-2-carboxylate, prop-2-yn-1-yl 4-amino-3-chloro-5-fluoro-6-(7-fluoro-1H-indol-6-yl)pyridine-2-carboxylate, methyl

4-amino-3-chloro-5-fluoro-6-(7-fluoro-1H-indol-6-yl)pyridine-2-carboxylate, 4-amino-3-chloro-5-fluoro-6-(7-fluoro-1H-indol-6-yl)pyridine-2-carboxylic acid, benzyl 4-amino-3-chloro-5-fluoro-6-(7-fluoro-1H-indol-6-yl)pyridine-2-carboxylate, ethyl 4-amino-3-chloro-5-fluoro-6-(7-fluoro-1H-indol-6-yl)pyridine-2-carboxylate, methyl 4-amino-3-chloro-5-fluoro-6-(7-fluoro-1-isobutyl-1H-indol-6-yl)pyridine-2-carboxylate, methyl 6-(1-acetyl-7-fluoro-1H-indol-6-yl)-4-amino-3-chloro-5-fluoropyridine-2-carboxylate, methyl 4-amino-3-chloro-6-[1-(2,2-dimethylpropanoyl)-7-fluoro-1H-indol-6-yl]-5-fluoropyridine-2-carboxylate, methyl 4-amino-3-chloro-5-fluoro-6-[7-fluoro-1-(methoxyacetyl)-1H-indol-6-yl]pyridine-2-carboxylate, potassium 4-amino-3-chloro-5-fluoro-6-(7-fluoro-1H-indol-6-yl)pyridine-2-carboxylate, sodium 4-amino-3-chloro-5-fluoro-6-(7-fluoro-1H-indol-6-yl)pyridine-2-carboxylate, butyl 4-amino-3-chloro-5-fluoro-6-(7-fluoro-1H-indol-6-yl)pyridine-2-carboxylate, 4-hydroxy-1-methyl-3-[4-(trifluoromethyl)pyridin-2-yl]imidazolidin-2-one, 3-(5-tert-butyl-1,2-oxazol-3-yl)-4-hydroxy-1-methylimidazolidin-2-one, 3-[5-chloro-4-(trifluoromethyl)pyridin-2-yl]-4-hydroxy-1-methylimidazolidin-2-one, 4-hydroxy-1-methoxy-5-methyl-3-[4-(trifluoromethyl)pyridin-2-yl]imidazolidin-2-one, 6-[(2-hydroxy-6-oxocyclohex-1-en-1-yl) carbonyl]-1,5-dimethyl-3-(2-methylphenyl)quinazolin-2,4 (1H,3H)-dione, 3-(2,6-dimethylphenyl)-6-[(2-hydroxy-6-oxocyclohex-1-en-1-yl) carbonyl]-1-methylquinazolin-2,4 (1H,3H)-dione, 2-[2-chloro-4-(methylsulfonyl)-3-(morpholin-4-ylmethyl) benzoyl]-3-hydroxycyclohex-2-en-1-one, 1-(2-carboxyethyl)-4-(pyrimidin-2-yl)pyridazin-1-ium salt (with anions such as chloride, acetate or trifluoroacetate), 1-(2-carboxyethyl)-4-(pyridazin-3-yl)pyridazin-1-ium salt (with anions such as chloride, acetate or trifluoroacetate), 4-(pyrimidin-2-yl)-1-(2-sulfoethyl)pyridazin-1-ium salt (with anions such as chloride, acetate or trifluoroacetate), 4-(pyridazin-3-yl)-1-(2-sulfoethyl)pyridazin-1-ium salt (with anions such as chloride, acetate or trifluoroacetate), 1-(2-Carboxyethyl)-4-(1,3-thiazol-2-yl)pyridazin-1-ium salt (with anions such as chloride, acetate or trifluoroacetate), 1-(2-Carboxyethyl)-4-(1,3-thiazol-2-yl)pyridazin-1-ium salt (with anions such as chloride, acetate or trifluoroacetate).

[0171] Examples of plant growth regulators which could be mixed with the compound and the composition of the invention are:

[0172] Absciscic acid and related analogues [e.g. (2Z,4E)-5-[6-Ethynyl-1-hydroxy-2,6-dimethyl-4-oxocyclohex-2-en-1-yl]-3-methylpenta-2,4-dienoic acid, methyl-(2Z,4E)-5-[6-ethynyl-1-hydroxy-2,6-dimethyl-4-oxocyclohex-2-en-1-yl]-3-methylpenta-2,4-dienoate, (2Z,4E)-3-ethyl-5-(1-hydroxy-2,6,6-trimethyl-4-oxocyclohex-2-en-1-yl) penta-2,4-dienoic acid, (2E,4E)-5-(1-hydroxy-2,6,6-trimethyl-4-oxocyclohex-2-en-1-yl)-3-(trifluoromethyl) penta-2,4-dienoic acid, methyl (2E,4E)-5-(1-hydroxy-2,6,6-trimethyl-4-oxocyclohex-2-en-1-yl)-3-(trifluoromethyl) penta-2,4-dienoate, (2Z,4E)-5-(2-hydroxy-1,3-dimethyl-5-oxobicyclo[4.1.0]hept-3-en-2-yl)-3-methylpenta-2,4-dienoic acid], acibenzolar, acibenzolar-S-methyl, S-adenosylhomocysteine, allantoin, 2-Aminoethoxyvinylglycine (AVG), aminooxyacetic acid

and related esters [e.g. (Isopropylidene)-aminooxyacetic acid-2-(methoxy)-2-oxoethylester, (Isopropylidene)-aminooxyacetic acid-2-(hexyloxy)-2-oxoethylester, (Cyclohexylidene)-aminooxyacetic acid-2-(isopropoxy)-2-oxoethylester], 1-aminocycloprop-1-yl carboxylic acid and derivatives thereof, e.g. disclosed in DE3335514, EP30287, DE2906507 or U.S. Pat. No. 5,123,951, 5-aminolevulinic acid, ancymidol, 6-benzylaminopurine, bixin, brassinolide, brassinolide-ethyl, L-canaline, catechin and catechines (e.g. (2S,3R)-2-(3,4-Dihydroxyphenyl)-3,4-dihydro-2H-chromen-3,5,7-triol), chitoooligosaccharides (CO; COs differ from LCOs in that they lack the pendant fatty acid chain that is characteristic of LCOs. COs, sometimes referred to as N-acetylchitoooligosaccharides, are also composed of GlcNAc residues but have side chain decorations that make them different from chitin molecules [(C₈H₁₃NO₅)_n, CAS No. 1398-61-4] and chitosan molecules [(C₅H₁₁NO₄)_n, CAS No. 9012-76-4]), chitinous compounds, chlormequat chloride, cloprop, cyclanilide, 3-(Cycloprop-1-enyl) propionic acid, 1-[2-(4-cyano-3,5-dicyclo-propylphenyl) acetamido] cyclohexanecarboxylic acid, 1-[2-(4-cyano-3-cyclopropylphenyl) acetamido]cyclo-hexanecarboxylic acid, daminozide, dazomet, dazomet-sodium, n-decanol, dikegulac, dikegulac-sodium, endothal, endothal-dipotassium, disodium, and mono(N,N-dimethylalkylammonium), ethephon, flumetralin, flurenol, flurenol-butyl, flurenol-methyl, flurprimidol, forchlorfenuron, gibberellic acid, inabenfide, indol-3-acetic acid (IAA), 4-indol-3-ylbutyric acid, isoprotiolane, probenazole, jasmonic acid, Jasmonic acid or derivatives thereof (e.g. jasmonic acid methyl ester, jasmonic acid ethyl ester), lipo-chitoooligosaccharides (LCO, sometimes referred to as symbiotic nodulation (Nod) signals (or Nod factors) or as Myc factors, consist of an oligosaccharide backbone of β-1,4-linked N-acetyl-D-glucosamine ("GlcNAc") residues with an N-linked fatty acyl chain condensed at the non-reducing end. As understood in the art, LCOs differ in the number of GlcNAc residues in the backbone, in the length and degree of saturation of the fatty acyl chain and in the substitutions of reducing and non-reducing sugar residues), linoleic acid or derivatives thereof, linolenic acid or derivatives thereof, maleic hydrazide, mepiquat chloride, mepiquat pentaborate, 1-methylcyclopropene, 3-methylcyclopropene, 1-ethylcyclopropene, 1-n-propylcyclopropene, 1-cyclopropenylmethanol, methoxyvinylglycine (MVG), 3'-methyl abscisic acid, 1-(4-methylphenyl)-N-(2-oxo-1-propyl-1,2,3,4-tetrahydroquinolin-6-yl) methanesulfonamide and related substituted tetrahydroquinolin-6-yl) methanesulfonamides, (3E,3aR,8bS)-3-({[(2R)-4-Methyl-5-oxo-2,5-dihydrofuran-2-yl]oxy}methylene)-3,3a,4,8b-tetrahydro-2H-indeno[1,2-b]furan-2-one and related lactones as outlined in EP2248421, 2-(1-naphthyl) acetamide, 1-naphthylacetic acid, 2-naphthylxyacetic acid, nitrophenolate-mixture, 4-Oxo-4 [(2-phenylethyl)amino]butyric acid, paclobutrazol, 4-phenylbutyric acid and its related salts (e.g. sodium-4-phenylbutanoate, potassium-4-phenylbutanoate), phenylalanine, N-phenylphthalamic acid, prohexadione, prohexadione-calcium, putrescine, prohydrojasmon, rhizobitoxin, salicylic acid, salicylic acid methyl ester, sarcosine, sodium cycloprop-1-en-1-yl acetate, sodium cycloprop-2-en-1-yl acetate, sodium-3-(cycloprop-2-en-1-yl) propanoate, sodium-3-(cycloprop-1-en-1-yl) propanoate, sidefungin, spermidine, spermine, strigolactone, tecnazene, thidiazuron, triacenta-

nol, trinexapac, trinexapac-ethyl, tryptophan, tsitodef, uni-conazole, uniconazole-P, 2-fluoro-N-(3-methoxyphenyl)-9H-purin-6-amine.

[0173] Examples of safeners which could be mixed with the compound and the composition of the invention are:

[0174] S1) Compounds from the group of heterocyclic carboxylic acid derivatives:

[0175] S1^a) Compounds of the dichlorophenylpyrazoline-3-carboxylic acid type (S1^a), preferably compounds such as 1-(2,4-dichlorophenyl)-5-(ethoxycarbonyl)-5-methyl-2-pyrazoline-3-carboxylic acid, ethyl 1-(2,4-dichlorophenyl)-5-(ethoxycarbonyl)-5-methyl-2-pyrazoline-3-carboxylate (S1-1) ("mefenpyr-diethyl"), and related compounds as described in WO-A-91/07874;

[0176] S1^b) Derivatives of dichlorophenylpyrazolecarboxylic acid (S1^b), preferably compounds such as ethyl 1-(2,4-dichlorophenyl)-5-methylpyrazole-3-carboxylate (S1-2), ethyl 1-(2,4-dichlorophenyl)-5-isopropylpyrazole-3-carboxylate (S1-3), ethyl 1-(2,4-dichlorophenyl)-5-(1,1-dimethylethyl) pyrazole-3-carboxylate (S1-4) and related compounds as described in EP-A-333131 131 and EP-A-269806;

[0177] S1^c) Derivatives of 1,5-diphenylpyrazole-3-carboxylic acid (S1^c), preferably compounds such as ethyl 1-(2,4-dichlorophenyl)-5-phenylpyrazole-3-carboxylate (S1-5), methyl 1-(2-chlorophenyl)-5-phenylpyrazole-3-carboxylate (S1-6) and related compounds as described, for example, in EP-A-268554;

[0178] S1^d) Compounds of the triazolecarboxylic acid type (S1^d), preferably compounds such as fenchlorazole (ethyl ester), i.e. ethyl 1-(2,4-dichlorophenyl)-5-trichloromethyl-1H-1,2,4-triazole-3-carboxylate (S1-7), and related compounds, as described in EP-A-174562 and EP-A-346620;

[0179] S1^e) Compounds of the 5-benzyl- or 5-phenyl-2-isoxazoline-3-carboxylic acid or of the 5,5-diphenyl-2-isoxazoline-3-carboxylic acid type (S1^e), preferably compounds such as ethyl 5-(2,4-dichlorobenzyl)-2-isoxazoline-3-carboxylate (S1-8) or ethyl 5-phenyl-2-isoxazoline-3-carboxylate (S1-9) and related compounds as described in WO-A-91/08202, or 5,5-diphenyl-2-isoxazolinecarboxylic acid (S1-10) or ethyl 5,5-diphenyl-2-isoxazoline-3-carboxylate (S1-11) ("isoxadifen-ethyl") or n-propyl 5,5-diphenyl-2-isoxazoline-3-carboxylate (S1-12) or ethyl 5-(4-fluorophenyl)-5-phenyl-2-isoxazoline-3-carboxylate (S1-13), as described in patent application WO-A-95/07897.

[0180] S2) Compounds from the group of the 8-quinolinoxy derivatives (S2):

[0181] S2^a) Compounds of the 8-quinolinoxyacetic acid type (S2^a), preferably 1-methylhexyl (5-chloro-8-quinolinoxy)acetate ("cloquintocet-mexyl") (S2-1), 1,3-dimethylbut-1-yl (5-chloro-8-quinolinoxy)acetate (S2-2), 4-allyloxybutyl (5-chloro-8-quinolinoxy)acetate (S2-3), 1-allyloxyprop-2-yl (5-chloro-8-quinolinoxy)acetate (S2-4), ethyl (5-chloro-8-quinolinoxy)acetate (S2-5), methyl 5-chloro-8-quinolinoxyacetate (S2-6), allyl (5-chloro-8-quinolinoxy)acetate (S2-7), 2-(2-propylideneiminoxy)-1-ethyl (5-chloro-8-quinolinoxy)acetate (S2-8), 2-oxoprop-1-yl (5-chloro-8-quinolinoxy)acetate (S2-9) and related compounds, as described in EP-A-86750, EP-A-94349 and EP-A-191736 or EP-A-0 492 366, and also (5-chloro-8-quinolinoxy) acetic

acid (S2-10), hydrates and salts thereof, for example the lithium, sodium, potassium, calcium, magnesium, aluminum, iron, ammonium, quaternary ammonium, sulfonium or phosphonium salts thereof, as described in WO-A-2002/34048;

[0182] S2^b) Compounds of the (5-chloro-8-quinolinoxy) malonic acid type (S2^b), preferably compounds such as diethyl (5-chloro-8-quinolinoxy) malonate, diallyl (5-chloro-8-quinolinoxy) malonate, methyl ethyl (5-chloro-8-quinolinoxy) malonate and related compounds, as described in EP-A-0 582 198.

[0183] S3) Active compounds of the dichloroacetamide type (S3), which are frequently used as pre-emergence safeners (soil-acting safeners), for example "dichloromid" (N,N-diallyl-2,2-dichloroacetamide) (S3-1), "R-29148" (3-dichloroacetyl-2,2,5-trimethyl-1,3-oxazolidine) from Stauffer (S3-2), "R-28725" (3-dichloroacetyl-2,2-dimethyl-1,3-oxazolidine) from Stauffer (S3-3), "benoxacor" (4-dichloroacetyl-3,4-dihydro-3-methyl-2H-1,4-benzoxazine) (S3-4), "PPG-1292" (N-allyl-N-[(1,3-dioxolan-2-yl)methyl]dichloroacetamide) from PPG Industries (S3-5), "DKA-24" (N-allyl-N-[(allylaminocarbonyl)methyl]dichloroacetamide) from Sagro-Chem (S3-6), "AD-67" or "MON 4660" (3-dichloroacetyl-1-oxa-3-azaspiro[4.5]decane) from Nitrokemia or Monsanto (S3-7), "TI-35" (1-dichloroacetylazepane) from TRI-Chemical RT (S3-8), "Diclonon" (Dicyclonon) or "BAS145138" or "LAB145138" (S3-9) ((RS)-1-dichloroacetyl-3,3,8a-trimethylperhydropyrrolo[1,2-a]pyrimidin-6-one) from BASF, "furilazole" or "MON 13900" ((RS)-3-dichloroacetyl-5-(2-furyl)-2,2-dimethylloxazolidine) (S3-10), and the (R) isomer thereof (S3-11).

[0184] S4) Compounds from the class of the acylsulfonamides (S4):

[0185] S4^a) N-Acylsulfonamides and salts thereof, as described in WO-A-97/45016,

[0186] S4^b) Compounds of the 4-(benzoylsulfamoyl) benzamide type and salts thereof, as described in WO-A-99/16744,

[0187] S4^c) Compounds from the class of the benzoylsulfamoylphenylureas as described in EP-A-365484, for example 1-[4-(N-2-methoxybenzoylsulfamoyl)phenyl]-3-methylurea, 1-[4-(N-2-methoxybenzoyl-sulfamoyl)phenyl]-3,3-dimethylurea and 1-[4-(N-4,5-dimethylbenzoylsulfamoyl)phenyl]-3-methylurea;

[0188] S4^d) Compounds of the N-phenylsulfonylterephthalamide type and salts thereof, which are known, for example, from CN 101838227.

[0189] S5) Active compounds from the class of the hydroxyaromatics and the aromatic-aliphatic carboxylic acid derivatives (S5), for example ethyl 3,4,5-triacetoxybenzoate, 3,5-dimethoxy-4-hydroxybenzoic acid, 3,5-dihydroxybenzoic acid, 4-hydroxysalicylic acid, 4-fluorosalicylic acid, 2-hydroxycinnamic acid, 2,4-dichlorocinnamic acid, as described in WO-A-2004/084631, WO-A-2005/015994, WO-A-2005/016001.

[0190] S6) Active compounds from the class of the 1,2-dihydroquinoxalin-2-ones (S6), for example 1-methyl-3-(2-thienyl)-1,2-dihydroquinoxalin-2-one, 1-methyl-3-(2-thienyl)-1,2-dihydroquinoxaline-2-thione, 1-(2-aminoethyl)-3-(2-thienyl)-1,2-dihydroquinoxalin-2-one hydrochloride, 1-(2-methylsulfonylami-

- noethyl)-3-(2-thienyl)-1,2-dihydroquinoxalin-2-one, as described in WO-A-2005/112630.
- [0191] S7) Compounds from the class of the diphenylmethoxyacetic acid derivatives (S7), e.g. methyl diphenylmethoxyacetate (CAS Reg. No. 41858-19-9) (S7-1), ethyl diphenylmethoxyacetate or diphenylmethoxyacetic acid, as described in WO-A-98/38856.
- [0192] S8) 2-fluoroacrylic acid derivatives as described in WO-A-98/27049.
- [0193] S9) active compounds from the class of the 3-(5-tetrazolylcarbonyl)-2-quinolones (S9), for example 1,2-dihydro-4-hydroxy-1-ethyl-3-(5-tetrazolylcarbonyl)-2-quinolone (CAS Reg. No. 219479-18-2), 1,2-dihydro-4-hydroxy-1-methyl-3-(5-tetrazolylcarbonyl)-2-quinolone (CAS Reg. No. 95855-00-8), as described in WO-A-199/000020;
- [0194] S10) N-acylsulfonamides as described in WO-A-2007/023719 and WO-A-2007/023764.
- [0195] S11) Active compounds of the oxyimino compound type (S11), which are known as seed-dressing agents, for example
- [0196] “oxabetrinil” ((Z)-1,3-dioxolan-2-ylmethoxyimino (phenyl) acetonitrile) (S11-1), which is known as a seed-dressing safener for millet/sorghum against metolachlor damage,
- [0197] “fluxofenim” (1-(4-chlorophenyl)-2,2,2-trifluoro-1-ethanone O-(1,3-dioxolan-2-ylmethyl) oxime) (S11-2), which is known as a seed-dressing safener for millet/sorghum against metolachlor damage, and
- [0198] “cyometrinil” or “CGA-43089” ((Z)-cyanomethoxyimino (phenyl) acetonitrile) (S11-3), which is known as a seed-dressing safener for millet/sorghum against metolachlor damage.
- [0199] S12) active compounds from the class of the isothiochromanones (S12), for example methyl [(3-oxo-1H-2-benzothiopyran-4 (3H)-ylidene) methoxy] acetate (CAS Reg. No. 205121-04-6) (S12-1) and related compounds from WO-A-1998/13361.
- [0200] S13) One or more compounds from group (S13):
- [0201] “naphthalic anhydride” (1,8-naphthalenedicarboxylic anhydride) (S13-1), which is known as a seed-dressing safener for corn against thiocarbamate herbicide damage,
- [0202] “fenclorim” (4,6-dichloro-2-phenylpyrimidine) (S13-2), which is known as a safener for pretilachlor in sown rice,
- [0203] “flurazole” (benzyl 2-chloro-4-trifluoromethyl-1,3-thiazole-5-carboxylate) (S13-3), which is known as a seed-dressing safener for millet/sorghum against alachlor and metolachlor damage,
- [0204] “CL 304415” (CAS Reg. No. 31541-57-8)
- [0205] (4-carboxy-3,4-dihydro-2H-1-benzopyran-4-acetic acid) (S13-4) from American Cyanamid, which is known as a safener for corn against damage by imidazolinones,
- [0206] “MG 191” (CAS Reg. No. 96420-72-3) (2-dichloromethyl-2-methyl-1,3-dioxolane) (S13-5) from Nitrokemia, which is known as a safener for corn,
- [0207] “MG 838” (CAS Reg. No. 133993-74-5)
- [0208] (2-propenyl 1-oxa-4-azaspiro[4.5]decane-4-carbodithioate) (S13-6) from Nitrokemia
- [0209] “disulfoton” (O,O-diethyl S-2-ethylthioethyl phosphorodithioate) (S13-7),
- [0210] “dietholate” (O,O-diethyl O-phenyl phosphorothioate) (S13-8),
- [0211] “mephenate” (4-chlorophenyl methylcarbamate) (S13-9).
- [0212] S14) active compounds which, in addition to herbicidal action against weeds, also have safener action on crop plants such as rice, for example
- [0213] “dimepiperate” or “MY-93” (S-1-methyl 1-phenylethylpiperidine-1-carbothioate), which is known as a safener for rice against damage by the herbicide molinate,
- [0214] “daimuron” or “SK 23” (1-(1-methyl-1-phenylethyl)-3-p-tolylurea), which is known as safener for rice against imazosulfuron herbicide damage,
- [0215] “cumyluron”=“JC-940” (3-(2-chlorophenylmethyl)-1-(1-methyl-1-phenylethyl) urea, see JP-A-60087254), which is known as safener for rice against damage by some herbicides,
- [0216] “methoxyphenone” or “NK 049” (3,3'-dimethyl-4-methoxybenzophenone), which is known as a safener for rice against damage by some herbicides,
- [0217] “CSB” (1-bromo-4-(chloromethylsulfonyl)benzene) from Kumiai, (CAS Reg. No. 54091-06-4), which is known as a safener against damage by some herbicides in rice.
- [0218] S15) Pyridine-2-oxy-3-carbonamides as described in WO-A-2008/131861 and WO-A-2008/131860.
- [0219] S16) Active compounds which are used primarily as herbicides but also have safener action on crop plants, for example (2,4-dichlorophenoxy) acetic acid (2,4-D), (4-chlorophenoxy) acetic acid, (R,S)-2-(4-chloro-o-tolylloxy) propionic acid (mecoprop), 4-(2,4-dichlorophenoxy) butyric acid (2,4-DB), (4-chloro-o-tolylloxy) acetic acid (MCPA), 4-(4-chloro-o-tolylloxy) butyric acid, 4-(4-chlorophenoxy) butyric acid, 3,6-dichloro-2-methoxybenzoic acid (dicamba), 1-(ethoxycarbonyl)ethyl 3,6-dichloro-2-methoxybenzoate (lactidichlor-ethyl).
- [0220] Examples of nitrification inhibitors which can be mixed with the compound and the composition of the invention are selected from the group consisting of 2-(3,4-dimethyl-1H-pyrazol-1-yl) succinic acid, 2-(4,5-dimethyl-1H-pyrazol-1-yl) succinic acid, 3,4-dimethyl pyrazolium glycolate, 3,4-dimethyl pyrazolium citrate, 3,4-dimethyl pyrazolium lactate, 3,4-dimethyl pyrazolium mandelate, 1,2,4-triazole, 4-Chloro-3-methylpyrazole, N-((3 (5)-methyl-1H-pyrazole-1-yl)methyl) acetamide, N-((3 (5)-methyl-1H-pyrazole-1-yl)methyl) formamide, N-((3 (5),4-dimethylpyrazole-1-yl)methyl) formamide, N-((4-chloro-3 (5)-methyl-pyrazole-1-yl)methyl) form-amide; reaction adducts of dicyandiamide, urea and formaldehyde, triazonyl-formaldehyde-dicyandiamide adducts, 2-cyano-1-((4-oxo-1,3,5-triazinan-1-yl)methyl) guanidine, 1-((2-cyanoguanidino)methyl) urea, 2-cyano-1-((2-cyanoguanidino)methyl) guanidine, 2-chloro-6-(trichloromethyl)-pyridine (nitrapyrin or N-serve), dicyandiamide, 3,4-dimethyl pyrazole phosphate, 4,5-dimethyl pyrazole phosphate, 3,4-dimethylpyrazole, 4,5-dimethyl pyrazole, ammoniumthiosulfate, neem, products based on ingredients of neem, linoleic acid, alpha-linolenic acid, methyl p-coumarate, methyl ferulate, methyl 3-(4-hydroxyphenyl) propionate, karanjin, brachialacton, p-benzoquinone sorgoleone, 4-amino-1,2,4-triazole hydrochloride, 1-amido-2-thiourea, 2-amino-4-chloro-

6-methylpyrimidine, 2-mercapto-benzothiazole, 5-ethoxy-3-trichloromethyl-1,2,4-thiadiazole (terrazole, etridiazole), 2-sulfanilamidothiazole, 3-methylpyrazol, 1,2,4-triazol thio-urea, cyan amide, melamine, zeolite powder, catechol, benzoquinone, sodium tetraborate, allylthiourea, chlorate salts, and zinc sulfate.

[0221] The compound and the composition of the invention may be combined with one or more agriculturally beneficial agents.

[0222] Examples of agriculturally beneficial agents include biostimulants, plant growth regulators, plant signal molecules, growth enhancers, microbial stimulating molecules, biomolecules, soil amendments, nutrients, plant nutrient enhancers, etc., such as lipo-chitoooligosaccharides (LCO), chitoooligosaccharides (CO), chitinous compounds, flavonoids, jasmonic acid or derivatives thereof (e.g., jasmonates), cytokinins, auxins, gibberellins, abscisic acid, ethylene, brassinosteroids, salicylates, macro- and micro-nutrients, linoleic acid or derivatives thereof, linolenic acid or derivatives thereof, karrikins, and beneficial microorganisms (e.g., *Rhizobium* spp., *Bradyrhizobium* spp., *Sinorhizobium* spp., *Azorhizobium* spp., *Glomus* spp., *Gigaspora* spp., *Hymenoscyphus* spp., *Oidiodendron* spp., *Laccaria* spp., *Pisolithus* spp., *Rhizopogon* spp., *Scleroderma* spp., *Rhizoctonia* spp., *Acinetobacter* spp., *Arthrobacter* spp., *Arthrobotrys* spp., *Aspergillus* spp., *Azospirillum* spp., *Bacillus* spp., *Burkholderia* spp., *Candida* spp., *Chryseomonas* spp., *Enterobacter* spp., *Eupenicillium* spp., *Exiguobacterium* spp., *Klebsiella* spp., *Kluyvera* spp., *Microbacterium* spp., *Mucor* spp., *Paecilomyces* spp., *Paenibacillus* spp., *Penicillium* spp., *Pseudomonas* spp., *Serratia* spp., *Stenotrophomonas* spp., *Streptomyces* spp., *Streptosporangium* spp., *Swaminathania* spp., *Thiobacillus* spp., *Torulospira* spp., *Vibrio* spp., *Xanthobacter* spp., *Xanthomonas* spp., etc.), and combinations thereof.

[0223] According to some embodiments, the compound and the composition of the invention may be combined with one or more biostimulants. Biostimulants may enhance metabolic or physiological processes such as respiration, photosynthesis, nucleic acid uptake, ion uptake, nutrient delivery, or a combination thereof. Non-limiting examples of biostimulants that may be included or used in the composition of the present invention may include seaweed extracts (e.g., *ascophyllum nodosum*; BAYFOLAN ALGAE, Aglukon gmbH, Germany), bacterial extracts (e.g., extracts of one or more diazotrophs, phosphate-solubilizing microorganism japonisms and/or biopesticides), fungal extracts, humic acids (e.g., potassium humate), fulvic acids, myo-inositol, and/or glycine, protein hydrolysates and amino-acids both from animal BAYFOLAN AMBITION & BAYFOLAN cobre, SICIT, Italy) and plant origin, inorganic compounds (e.g. silica) and any combinations thereof. According to some embodiments, the biostimulants may comprise one or more *Azospirillum* extracts (e.g., an extract of media comprising *A. brasilense* INTA Az-39), one or more *Bradyrhizobium* extracts (e.g., an extract of media comprising *B. elkanii* SEMIA 501, *B. elkanii* SEMIA 587, *B. elkanii* SEMIA 5019, *B. japonicum* NRRL B-50586 (also deposited as NRRL B-59565), *B. japonicum* NRRL B-50587 (also deposited as NRRL B-59566), *B. japonicum* NRRL B-50588 (also deposited as NRRL B-59567), *B. japonicum* NRRL B-50589 (also deposited as NRRL B-59568), *B. japonicum* NRRL B-50590 (also deposited as NRRL B-59569), *B. japonicum* NRRL B-50591 (also deposited as

NRRL B-59570), *B. japonicum* NRRL B-50592 (also deposited as NRRL B-59571), *B. japonicum* NRRL B-50593 (also deposited as NRRL B-59572), *B. japonicum* NRRL B-50594 (also deposited as NRRL B-50493), *B. japonicum* NRRL B-50608, *B. japonicum* NRRL B-50609, *B. japonicum* NRRL B-50610, *B. japonicum* NRRL B-50611, *B. japonicum* NRRL B-50612, *B. japonicum* NRRL B-50726, *B. japonicum* NRRL B-50727, *B. japonicum* NRRL B-50728, *B. japonicum* NRRL B-50729, *B. japonicum* NRRL B-50730, *B. japonicum* SEMIA 566, *B. japonicum* SEMIA 5079, *B. japonicum* SEMIA 5080, *B. japonicum* USDA 6, *B. japonicum* USDA 110, *B. japonicum* USDA 122, *B. japonicum* USDA 123, *B. japonicum* USDA 127, *B. japonicum* USDA 129 and/or *B. japonicum* USDA 532C), one or more *Rhizobium* extracts (e.g., an extract of media comprising *R. leguminosarum* SO12A-2), one or more *Sinorhizobium* extracts (e.g., an extract of media comprising *S. fredii* CCBAU114 and/or *S. fredii* USDA 205), one or more *Penicillium* extracts (e.g., an extract of media comprising *P. bilaiae* ATCC 18309, *P. bilaiae* ATCC 20851, *P. bilaiae* ATCC 22348, *P. bilaiae* NRRL 50162, *P. bilaiae* NRRL 50169, *P. bilaiae* NRRL 50776, *P. bilaiae* NRRL 50777, *P. bilaiae* NRRL 50778, *P. bilaiae* NRRL 50777, *P. bilaiae* NRRL 50778, *P. bilaiae* NRRL 50779, *P. bilaiae* NRRL 50780, *P. bilaiae* NRRL 50781, *P. bilaiae* NRRL 50782, *P. bilaiae* NRRL 50783, *P. bilaiae* NRRL 50784, *P. bilaiae* NRRL 50785, *P. bilaiae* NRRL 50786, *P. bilaiae* NRRL 50787, *P. bilaiae* NRRL 50788, *P. bilaiae* RSTB-SD1, *P. brevicompactum* AgRF18, *P. canescens* ATCC 10419, *P. expansum* ATCC 24692, *P. expansum* YT02, *P. fellatatum* ATCC 48694, *P. gaestrivorus* NRRL 50170, *P. glabrum* DAOM 239074, *P. glabrum* CBS 229.28, *P. janthinellum* ATCC 10455, *P. lanosocoeruleum* ATCC 48919, *P. radicum* ATCC 201836, *P. radicum* FRR 4717, *P. radicum* FRR 4719, *P. radicum* N93/47267 and/or *P. raistrickii* ATCC 10490), one or more *Pseudomonas* extracts (e.g., an extract of media comprising *P. jessenii* PS06), one or more acaricidal, insecticidal and/or nematocidal extracts (e.g., an extract of media comprising *Bacillus firmus* I-1582, *Bacillus mycoides* AQ726, NRRL B-21664; *Beauveria bassiana* ATCC-74040, *Beauveria bassiana* ATCC-74250, *Burkholderia* sp. A396 sp. nov. rinojensis, NRRL B-50319, *Chromobacterium subtsugae* NRRL B-30655, *Chromobacterium vaccinii* NRRL B-50880, *Flavobacterium* H492, NRRL B-50584, *Metarhizium anisopliae* F52 (also known as *Metarhizium anisopliae* strain 52, *Metarhizium anisopliae* strain 7, *Metarhizium anisopliae* strain 43 and *Metarhizium anisopliae* BIO-1020, TAE-001; deposited as DSM 3884, DSM 3885, ATCC 90448, SD 170 and ARSEF 7711) and/or *Paecilomyces fumosoroseus* FE991), and/or one or more fungicidal extracts (e.g., an extract of media comprising *Ampelomyces quisqualis* AQ 10® (Intrachem Bio GmbH & Co. KG, Germany), *Aspergillus flavus* AFLA-GUARD® (Syngenta Crop Protection, Inc., CH), *Aureobasidium pullulans* BOTEOTOR® (bio-ferm GmbH, Germany), *Bacillus pumilus* AQ717 (NRRL B-21662), *Bacillus pumilus* NRRL B-30087, *Bacillus* AQ175 (ATCC 55608), *Bacillus* AQ177 (ATCC 55609), *Bacillus subtilis* AQ713 (NRRL B-21661), *Bacillus subtilis* AQ743 (NRRL B-21665), *Bacillus amyloliquefaciens* FZB24, *Bacillus amyloliquefaciens* NRRL B-50349, *Bacillus amyloliquefaciens* TJ1000 (also known as 1BE, isolate ATCC BAA-390), *Bacillus thuringiensis* AQ52 (NRRL B-21619), *Candida oleophila* 1-82 (e.g., ASPIRE® from Ecogen Inc., USA), *Candida saitoana* BIO-

CURE® (in mixture with lysozyme; BASF, USA) and BIOCOAT® (ArystaLife Science, Ltd., Cary, NC), *Clonostachys rosea* f. *catenulata* (also referred to as *Gliocladium catenulatum*) J1446 (PRESTOP®, Verdera, Finland), *Coniothyrium minitans* CONTANS® (Prophyta, Germany), *Cryphonectria parasitica* (CNICM, France), *Cryptococcus albidus* YIELD PLUS® (Anchor Bio-Technologies, South Africa), *Fusarium oxysporum* BIOFOX® (from S.I.A.P.A., Italy) and FUSACLEAN® (Natural Plant Protection, France), *Metschnikowia fructicola* SHEMER® (Agrogreen, Israel), *Microdochium dimerum* ANTIBOT® (Agrauxine, France), *Muscodor albus* NRRL 30547, *Muscodor roseus* NRRL 30548, *Phlebiopsis gigantea* ROTSOP® (Verdera, Finland), *Pseudozyma flocculosa* SPORODEX® (Plant Products Co. Ltd., Canada), *Pythium oligandrum* DV74 (POLYVERSUM®, Remeslo SSRO, Biopreparaty, Czech Rep.), *Reynoutria sachalinensis* (e.g., REGALIA® from Marrone BioInnovations, USA), *Streptomyces* NRRL B-30145, *Streptomyces* M1064, *Streptomyces galbus* NRRL 30232, *Streptomyces lydicus* WYEC 108 (ATCC 55445), *Streptomyces violaceusniger* YCED 9 (ATCC 55660; DE-THATCH-9®, DECOMP-9® and THATCH CONTROL®, Idaho Research Foundation, USA), *Streptomyces* WYE 53 (ATCC 55750; DE-THATCH-9®, DECOMP-9® and THATCH CONTROL®, Idaho Research Foundation, USA), *Talaromyces flavus* V117b (PROTUS®, Prophyta, Germany), *Trichoderma asperellum* SKT-1 (ECO-HOPE®, Kumiai Chemical Industry Co., Ltd., Japan), *Trichoderma atroviride* LC52 (SENTINEL®, Agrimm Technologies Ltd, NZ), *Trichoderma harzianum* T-22 (PLANTSHIELD®, der Firma BioWorks Inc., USA), *Trichoderma harzianum* TH-35 (ROOT PRO®, from Mycontrol Ltd., Israel), *Trichoderma harzianum* T-39 (TRICHODEX®, Mycontrol Ltd., Israel; TRICHODERMA 2000®, Makhteshim Ltd., Israel), *Trichoderma harzianum* ICC012 and *Trichoderma viride* TRICHOPEL (Agrimm Technologies Ltd, NZ), *Trichoderma harzianum* ICC012 and *Trichoderma viride* ICC080 (REMEDIER® WP, Isagro Ricerca, Italy), *Trichoderma polysporum* and *Trichoderma harzianum* (BINAB®, BINAB Bio-Innovation AB, Sweden), *Trichoderma stromaticum* TRICOVAB® (C.E.P.L.A.C., Brazil), *Trichoderma virens* GL-21 (SOILGARD®, Certis LLC, USA), *Trichoderma virens* G1-3, ATCC 57678, *Trichoderma virens* G1-21 (Thermo Trilogy Corporation, Wasco, CA), *Trichoderma virens* G1-3 and *Bacillus amyloliquefaciens* FZB2, *Trichoderma virens* G1-3 and *Bacillus amyloliquefaciens* NRRL B-50349, *Trichoderma virens* G1-3 and *Bacillus amyloliquefaciens* TJ1000, *Trichoderma virens* G1-21 and *Bacillus amyloliquefaciens* FZB24, *Trichoderma virens* G1-21 and *Bacillus amyloliquefaciens* NRRL B-50349, *Trichoderma virens* G1-21 and *Bacillus amyloliquefaciens* TJ1000, *Trichoderma viride* TRIECO® (Ecosense Labs. (India) Pvt. Ltd., Indien, BIO-CURE® F from T. Stanes & Co. Ltd., Indien), *Trichoderma viride* TV1 (Agribiotec srl, Italy), *Trichoderma viride* ICC080, and/or *Ulocladium oudemansii* HRU3 (BOTRY-ZEN®, Botry-Zen Ltd, NZ)), and combinations thereof.

[0224] According to some embodiments, the compound and the composition of the invention may be combined with one or more lipo-chitooligosaccharides (LCOs), chitooligosaccharides (COs), and/or chitinous compounds. LCOs, sometimes referred to as symbiotic nodulation (Nod) signals (or Nod factors) or as Myc factors, consist of an oligosaccharide backbone of β -1,4-linked N-acetyl-D-glucosamine

("GlcNAc") residues with an N-linked fatty acyl chain condensed at the non-reducing end. As understood in the art, LCOs differ in the number of GlcNAc residues in the backbone, in the length and degree of saturation of the fatty acyl chain and in the substitutions of reducing and non-reducing sugar residues. See, e.g., Denarie et al., *Ann. Rev. Biochem.* 65:503 (1996); Diaz et al., *Mol. Plant-Microbe Interactions* 13:268 (2000); Hungria et al., *Soil Biol. Biochem.* 29:819 (1997); Hamel et al., *Planta* 232:787 (2010); and Prome et al., *Pure & Appl. Chem.* 70 (1): 55 (1998).

[0225] LCOs (and derivatives thereof) may be included or utilized in various forms of purity and can be used alone or in the form of a culture of LCO-producing bacteria or fungi. For example, OPTIMIZE® (commercially available from Bayer Company) contains a culture of *Bradyrhizobium japonicum* that produces LCO. Methods to provide substantially pure LCOs include removing the microbial cells from a mixture of LCOs and the microbe, or continuing to isolate and purify the LCO molecules through LCO solvent phase separation followed by HPLC chromatography as described, for example, in U.S. Pat. No. 5,549,718. Purification can be enhanced by repeated HPLC and the purified LCO molecules can be freeze-dried for long-term storage. Compositions and methods of the present disclosure may comprise analogues, derivatives, hydrates, isomers, salts and/or solvates of LCOs. LCOs may be incorporated into the composition according to the invention in any suitable amount(s)/concentration(s). For example, the composition according to the invention comprise about 1×10^{-20} M to about 1×10^{-1} M LCO(s). The amount/concentration of LCO may be an amount effective to impart a positive trait or benefit to a plant, such as to enhance the growth and/or yield of the plant to which the composition is applied. According to some embodiments, the LCO amount/concentration is not effective to enhance the yield of the plant without beneficial contributions from one or more other constituents of the composition, such as CO and/or one or more pesticides.

[0226] The compound and the composition of the invention may be combined with any suitable COs, perhaps in combination with one or more LCOs. COs differ from LCOs in that they lack the pendant fatty acid chain that is characteristic of LCOs. COs, sometimes referred to as N-acetylchitooligosaccharides, are also composed of GlcNAc residues but have side chain decorations that make them different from chitin molecules $[(C_8H_{13}NO_5)_n]$, CAS No. 1398-61-4] and chitosan molecules $[(C_5H_{11}NO_4)_n]$, CAS No. 9012-76-4]. See, e.g., D'Haese et al., *Glycobiol.* 12 (6): 79R (2002); Demont-Caulet et al., *Plant Physiol.* 120 (1): 83 (1999); Hanel et al., *Planta* 232:787 (2010); Muller et al., *Plant Physiol.* 124:733 (2000); Robina et al., *Tetrahedron* 58:521-530 (2002); Rouge et al., Docking of Chitin Oligomers and Nod Factors on Lectin Domains of the LysM-RLK Receptors in the Medicago-Rhizobium Symbiosis, in *The Molecular Immunology of Complex Carbohydrates-3* (Springer Science, 2011); Van der Holst et al., *Curr. Opin. Struc. Biol.* 11:608 (2001); and Wan et al., *Plant Cell* 21:1053 (2009). COs may be obtained from any suitable source. For example, the CO may be derived from an LCO. For example, in an aspect, the composition according to the invention comprise one or more COs derived from an LCO obtained (i.e., isolated and/or purified) from a strain of *Azorhizobium*, *Bradyrhizobium* (e.g., *B. japonicum*), *Mesorhizobium*, *Rhizobium* (e.g., *R. leguminosarum*),

Sinorhizobium (e.g., *S. meliloti*), or mycorrhizal fungi (e.g., *Glomus intraradicis*). Alternatively, the CO may be synthetic. Methods for the preparation of recombinant COs are known in the art. See, e.g., Cottaz et al., *Meth. Eng.* 7 (4): 311 (2005); Samain et al., *Carbohydrate Res.* 302:35 (1997.); and Samain et al., *J. Biotechnol.* 72:33 (1999), the contents and disclosures of which are incorporated herein by reference.

[0227] COs (and derivatives thereof) may be included or utilized in various forms of purity and can be used alone or in the form of a culture of CO-producing bacteria or fungi. It is to be understood that the compound and the composition of the invention may be combined with hydrates, isomers, salts and/or solvates of COs. COs may be used in any suitable amount(s)/concentration(s). For example, the composition according to the invention may comprise about 1×10^{-20} M to about 1×10^{-1} M COs. The amount/concentration of CO may be an amount effective to impart or confer a positive trait or benefit to a plant, such as to enhance the soil microbial environment, nutrient uptake, or increase the growth and/or yield of the plant to which the composition is applied. According to some embodiments, a CO amount/concentration may not be effective to enhance the growth of the plant without beneficial contributions from one or more other ingredients of the composition, such as LCO and/or one or more inoculants, biomolecules, nutrients, or pesticides.

[0228] The compound and the composition of the invention may be combined with one or more suitable chitinous compounds, such as, for example, chitin, chitosan, and isomers, salts and solvates thereof. Chitins and chitosans, which are major components of the cell walls of fungi and the exoskeletons of insects and crustaceans, are composed of GlcNAc residues. Chitins and chitosans may be obtained commercially or prepared from insects, crustacean shells, or fungal cell walls. Methods for the preparation of chitin and chitosan are known in the art. See, e.g., U.S. Pat. No. 4,536,207 (preparation from crustacean shells) and U.S. Pat. No. 5,965,545 (preparation from crab shells and hydrolysis of commercial chitosan); and Pochanavanich et al., *Lett. Appl. Microbiol.* 35:17 (2002) (preparation from fungal cell walls). Deacetylated chitins and chitosans may be obtained that range from less than 35% to greater than 90% deacetylation and cover a broad spectrum of molecular weights, e.g., low molecular weight chitosan oligomers of less than 15 kD and chitin oligomers of 0.5 to 2 kD; “practical grade” chitosan with a molecular weight of about 15 kD; and high molecular weight chitosan of up to 70 kD. Chitin and chitosan compositions formulated for seed treatment are commercially available. Commercial products include, for example, ELEXAR (Plant Defense Boosters, Inc.) and BEYOND™ (Agrihouse, Inc.).

[0229] The compound and the composition of the invention may be combined with one or more suitable flavonoids, including, but not limited to, anthocyanidins, anthoxanthins, chalcones, coumarins, flavanones, flavanols, flavans and isoflavonoids, as well as analogues, derivatives, hydrates, isomers, polymers, salts and solvates thereof. Flavonoids are phenolic compounds having the general structure of two aromatic rings connected by a three-carbon bridge. Classes of flavonoids are known in the art. See, e.g., Jain et al., *J. Plant Biochem. & Biotechnol.* 11:1 (2002); and Shaw et al., *Environ. Microbiol.* 11:1867 (2006), the contents and disclosures of which are incorporated herein by reference.

Several flavonoid compounds are commercially available. Flavonoid compounds may be isolated from plants or seeds, e.g., as described in U.S. Pat. Nos. 5,702,752; 5,990,291; and 6,146,668. Flavonoid compounds may also be produced by genetically engineered organisms, such as yeast. See, e.g., Ralston et al., *Plant Physiol.* 137:1375 (2005).

[0230] According to some embodiments, the compound and the composition of the invention may be combined with one or more flavanones, such as one or more of butin, eriodictyol, hesperetin, hesperidin, homoeriodictyol, isosakuranetin, naringenin, naringin, pinocembrin, poncirin, sakuranetin, sakuranin, and/or sterubin, one or more flavanols, such as dihydrokaempferol and/or taxifolin, one or more flavans, such as one or more flavan-3-ols (e.g., catechin (C), catechin 3-gallate (Cg), epicatechins (EC), epigallocatechin (EGC) epicatechin 3-gallate (ECg), epigallocatechin 3-gallate (EGCg), epiafzelechin, fisetinidol, gallocatechin (GC), gallocatechin 3-gallate (GCg), glibourtinidol, mesquitol, robinetinidol, theaflavin-3-gallate, theaflavin-3'-gallate, theflavin-3,3'-digallate, thearubigin), flavan-4-ols (e.g., apiforol and/or luteoforol) and/or flavan-3, 4-diols (e.g., leucocyanidin, leucodelphinidin, leucofisetinidin, leucomalvidin, leucopelargonidin, leucoponidin, leucorobinetinidin, melacacidin and/or teracacidin) and/or dimers, trimers, oligomers and/or polymers thereof (e.g., one or more proanthocyanidins), one or more isoflavonoids, such as one or more isoflavones or flavonoid derivatives (e.g., biochanin A, daidzein, formononetin, genistein and/or glycitein), isoflavanes (e.g., equol, ionchocarpene and/or laxifloerane), isoflavandiols, isoflavones (e.g., glabrene, hagin D and/or 2-methoxyjudaicin), coumestans (e.g., coumestrol, plicadin and/or wedelolactone), pterocarpanes, rotonoids, neoflavonoids (e.g., calophyllolide, coustareagenin, dalbergichromene, dalbergin, nivetin), and/or pterocarpanes (e.g., bitucarpin A, bitucarpin B, erybraedin A, erybraedin B, erythrabysins II, erythrabysin-1, erycristagallin, glycinol, glyceollidins, glyceollins, glycyrrhizol, maackiain, medicarpin, morisianine, orientanol, phaseolin, pisatin, striatine, trifolirhizin), and combinations thereof. Flavonoids and their derivatives may be included in the present composition in any suitable form, including, but not limited to, polymorphic and crystalline forms. Flavonoids may be included in the composition according to the invention in any suitable amount(s) or concentration(s). The amount/concentration of a flavonoid (s) may be an amount effective to impart a benefit to a plant, which may be indirectly through activity on soil microorganisms or other means, such as to enhance plant nutrition and/or yield. According to some embodiments, a flavonoid amount/concentration may not be effective to enhance the nutrition or yield of the plant without the beneficial contributions from one or more other ingredients of the composition, such as LCO, CO, and/or one or more pesticides.

[0231] The compound and the composition of the invention may be combined with one or more suitable non-flavonoid nod-gene inducer(s), including, but not limited to, jasmonic acid ([1R-[1a,2B (Z)]]-3-oxo-2-(pentenyl)cyclopentaneacetic acid; JA), linoleic acid ((Z,Z)-9,12-Octadecadienoic acid) and/or linolenic acid ((Z,Z,Z)-9,12,15-octadecatrienoic acid), and analogues, derivatives, hydrates, isomers, polymers, salts and solvates thereof. Jasmonic acid and its methyl ester, methyl jasmonate (MeJA), collectively known as jasmonates, are octadecanoid-based compounds that occur naturally in some plants (e.g., wheat), fungi (e.g.,

Botryodiplodia theobromae, *Gibberella fujikuroi*), yeast (e.g., *Saccharomyces cerevisiae*) and bacteria (e.g., *Escherichia coli*). Linoleic acid and linolenic acid may be produced in the course of the biosynthesis of jasmonic acid. Jasmonates, linoleic acid and linolenic acid (and their derivatives) are reported to be inducers of nod gene expression or LCO production by rhizobacteria. See, e.g., Mabood et al., *PLANT PHYSIOL. BIOCHEM.* 44 (11): 759 (2006); Mabood et al., *AGR. J.* 98 (2): 289 (2006); Mabood et al., *FIELD CROPS RES.* 95 (2-3): 412 (2006); and Mabood & Smith, *Linoleic and linolenic acid induce the expression of nod genes in Bradyrhizobium japonicum* USDA 3, *PLANT BIOL.* (2001).

[0232] Derivatives of jasmonic acid, linoleic acid, and linolenic acid that may be included or used in combination with the compound and the composition according to the invention include esters, amides, glycosides and salts thereof. Representative esters are compounds in which the carboxyl group of linoleic acid, linolenic acid, or jasmonic acid has been replaced with a —COR group, where R is an —OR¹ group, in which R¹ is: an alkyl group, such as a C₁–C₈ unbranched or branched alkyl group, e.g., a methyl, ethyl or propyl group; an alkenyl group, such as a C₂–C₈ unbranched or branched alkenyl group; an alkynyl group, such as a C₂–C₈ unbranched or branched alkynyl group; an aryl group having, for example, 6 to 10 carbon atoms; or a heteroaryl group having, for example, 4 to 9 carbon atoms, wherein the heteroatoms in the heteroaryl group can be, for example, N, O, P, or S. Representative amides are compounds in which the carboxyl group of linoleic acid, linolenic acid, or jasmonic acid has been replaced with a —COR group, where R is an NR²R³ group, in which R² and R³ are each independently: a hydrogen; an alkyl group, such as a C₁–C₈ unbranched or branched alkyl group, e.g., a methyl, ethyl or propyl group; an alkenyl group, such as a C₂–C₈ unbranched or branched alkenyl group; an alkynyl group, such as a C₂–C₈ unbranched or branched alkynyl group; an aryl group having, for example, 6 to 10 carbon atoms; or a heteroaryl group having, for example, 4 to 9 carbon atoms, wherein the heteroatoms in the heteroaryl group can be, for example, N, O, P, or S. Esters may be prepared by known methods, such as acid-catalyzed nucleophilic addition, wherein the carboxylic acid is reacted with an alcohol in the presence of a catalytic amount of a mineral acid. Amides may also be prepared by known methods, such as by reacting the carboxylic acid with the appropriate amine in the presence of a coupling agent, such as dicyclohexyl carbodiimide (DCC), under neutral conditions. Suitable salts of linoleic acid, linolenic acid and jasmonic acid include, for example, base addition salts. The bases that may be used as reagents to prepare metabolically acceptable base salts of these compounds include those derived from cations such as alkali metal cations (e.g., potassium and sodium) and alkaline earth metal cations (e.g., calcium and magnesium). These salts may be readily prepared by mixing a solution of linoleic acid, linolenic acid, or jasmonic acid with a solution of the base. The salts may be precipitated from solution and collected by filtration, or may be recovered by other means such as by evaporation of the solvent.

[0233] Non-flavonoid nod-gene inducers may be used in combination with the compound and the composition according to the invention in any suitable amount(s)/concentration(s). For example, the amount/concentration of non-flavonoid nod-gene inducers may be an amount effective to impart or confer a positive trait or benefit to a plant,

such as to enhance the growth and/or yield of the plant to which the composition is applied. According to some embodiments, the amount/concentration of non-flavonoid nod-gene inducers may not be effective to enhance the growth and/or yield of the plant without beneficial contributions from one or more other ingredients of the composition, such as a LCO, CO and/or one or more pesticides.

[0234] The compound and the composition of the invention may be combined with karrakins, including but not limited to 2H-furo[2,3-c]pyran-2-ones, as well as analogues, derivatives, hydrates, isomers, polymers, salts and solvates thereof. Examples of biologically acceptable salts of karrakins include acid addition salts formed with biologically acceptable acids, examples of which include hydrochloride, hydrobromide, sulphate or bisulphate, phosphate or hydrogen phosphate, acetate, benzoate, succinate, fumarate, maleate, lactate, citrate, tartrate, gluconate; methanesulphonate, benzenesulphonate and p-toluenesulphonic acid. Additional biologically acceptable metal salts may include alkali metal salts, with bases, examples of which include the sodium and potassium salts. Karrakins may be incorporated into the composition according to the invention in any suitable amount(s) or concentration(s). For example, the amount/concentration of a karrakin may be an amount or concentration effective to impart or confer a positive trait or benefit to a plant, such as to enhance the growth and/or yield of the plant to which the composition is applied. In an aspect, a karrakin amount/concentration may not be effective to enhance the growth and/or yield of the plant without beneficial contributions from one or more other ingredients of the composition, such as a LCO, CO and/or one or more pesticides.

[0235] The compound and the composition of the invention may be combined with one or more anthocyanidins and/or anthoxanthins, such as one or more of cyanidin, delphinidin, malvidin, pelargonidin, peonidin, petunidin, flavones (e.g., apigenin, baicalein, chrysin, 7,8-dihydroxyflavone, diosmin, flavoxate, 6-hydroxyflavone, luteolin, scutellarein, tangeritin and/or wogonin) and/or flavonols (e.g., amurensin, astragalin, azaleatin, azalein, fisetin, furanoflavonols galangin, gossypetin, 3-hydroxyflavone, hyperoside, icariin, isoquercetin, kaempferide, kaempferitrin, kaempferol, isorhamnetin, morin, myricetin, myricitrin, natsudaaidain, pachypodol, pyranoflavonols quercetin, quercitrin, rhamnazin, rhamnetin, robinin, rutin, spiraeoside, troxerutin and/or zanthorhamnin), and combinations thereof.

[0236] The compound and the composition of the invention may be combined with gluconolactone and/or an analogue, derivative, hydrate, isomer, polymer, salt and/or solvate thereof. Gluconolactone may be incorporated into the composition according to the invention in any suitable amount(s)/concentration(s). For example, the amount/concentration of a gluconolactone amount/concentration may be an amount effective to impart or confer a positive trait or benefit to a plant, such as to enhance the growth and/or yield of the plant to which the composition is applied. In an aspect, the gluconolactone amount/concentration may not be effective to enhance the growth and/or yield of the plant without beneficial contributions from one or more other ingredients of the composition, such as a LCO, CO and/or one or more pesticides.

[0237] The compound and the composition of the invention may be combined with one or more suitable nutrient(s) and/or fertilizer(s), such as organic acids (e.g., acetic acid,

citric acid, lactic acid, malic acid, taurine, etc.), macromineral (e.g., phosphorous, calcium, magnesium, potassium, sodium, iron, etc.), trace minerals (e.g., boron, cobalt, chloride, chromium, copper, fluoride, iodine, iron, manganese, molybdenum, selenium, zinc, etc.), vitamins, (e.g., vitamin A, vitamin B complex (i.e., vitamin B₁, vitamin B₂, vitamin B₃, vitamin B₅, vitamin B₆, vitamin B₇, vitamin B₈, vitamin B₉, vitamin B₁₂, choline) vitamin C, vitamin D, vitamin E, vitamin K.), and/or carotenoids (α -carotene, β -carotene, cryptoxanthin, lutein, lycopene, zeaxanthin, etc.), and combinations thereof. In an aspect, the compound and the composition of the invention may be combined with macro- and micronutrients of plants or microbes, including phosphorous, boron, chlorine, copper, iron, manganese, molybdenum and/or zinc. According to some embodiments, the compound and the composition of the invention may be combined with one or more beneficial micronutrients. Non-limiting examples of micronutrients for use in compositions described herein may include vitamins, (e.g., vitamin A, vitamin B complex (i.e., vitamin B₁, vitamin B₂, vitamin B₃, vitamin B₅, vitamin B₆, vitamin B₇, vitamin B₈, vitamin B₉, vitamin B₁₂, choline) vitamin C, vitamin D, vitamin E, vitamin K, carotenoids (α -carotene, β -carotene, cryptoxanthin, lutein, lycopene, zeaxanthin, etc.), macromineral (e.g., phosphorous, calcium, magnesium, potassium, sodium, iron, etc.), trace minerals (e.g., boron, cobalt, chloride, chromium, copper, fluoride, iodine, iron, manganese, molybdenum, selenium, zinc, etc.), organic acids (e.g., acetic acid, citric acid, lactic acid, malic acid, taurine, etc.), and combinations thereof (BAYFOLAN secure, BAYFOLAN complete, BAYFOLAN energy, BAYFOLAN power, Aglukon GmbH, Germany). In a particular aspect, compositions may comprise phosphorous, boron, chlorine, copper, iron, manganese, molybdenum, and/or zinc, and combinations thereof. For compositions comprising phosphorous, it is envisioned that any suitable source of phosphorous may be used. For example, phosphorus may be derived from a rock phosphate source, such as monoammonium phosphate, diammonium phosphate, monocalcium phosphate, super phosphate, triple super phosphate, and/or ammonium polyphosphate, an organic phosphorous source, or a phosphorous source capable of solubilization by one or more microorganisms (e.g., *Penicillium bilaiae*).

Compositions/Formulations

[0238] The present invention further relates to compositions, in particular compositions for controlling unwanted microorganisms. The composition may be applied to the microorganisms and/or in their habitat.

[0239] The composition comprises at least one compound of the invention and at least one agriculturally suitable auxiliary, e.g. carrier(s) and/or surfactant(s).

[0240] A carrier is a solid or liquid, natural or synthetic, organic or inorganic substance that is generally inert. The carrier generally improves the application of the compounds, for instance, to plants, plant parts or seeds. Examples of suitable solid carriers include, but are not limited to, ammonium salts, in particular ammonium sulfates, ammonium phosphates and ammonium nitrates, natural rock flours, such as kaolins, clays, talc, chalk, quartz, attapulgite, montmorillonite and diatomaceous earth, silica gel and synthetic rock flours, such as finely divided silica, alumina and silicates. Examples of typically useful solid carriers for preparing granules include, but are not limited to

crushed and fractionated natural rocks such as calcite, marble, pumice, sepiolite and dolomite, synthetic granules of inorganic and organic flours and granules of organic material such as paper, sawdust, coconut shells, maize cobs and tobacco stalks. Examples of suitable liquid carriers include, but are not limited to, water, organic solvents and combinations thereof. Examples of suitable solvents include polar and nonpolar organic chemical liquids, for example from the classes of aromatic and nonaromatic hydrocarbons (such as cyclohexane, paraffins, alkylbenzenes, xylene, toluene, tetrahydronaphthalene, alkyl naphthalenes, chlorinated aromatics or chlorinated aliphatic hydrocarbons such as chlorobenzenes, chloroethylenes or methylene chloride), alcohols and polyols (which may optionally also be substituted, etherified and/or esterified, such as ethanol, propanol, butanol, benzylalcohol, cyclohexanol or glycol), ketones (such as acetone, methyl ethyl ketone, methyl isobutyl ketone or cyclohexanone), esters (including fats and oils) and (poly) ethers, unsubstituted and substituted amines, amides (such as dimethylformamide or fatty acid amides) and esters thereof, lactams (such as N-alkylpyrrolidones, in particular N-methylpyrrolidone) and lactones, sulfones and sulfoxides (such as dimethyl sulfoxide), oils of vegetable or animal origin. The carrier may also be a liquefied gaseous extender, i.e. liquid which is gaseous at standard temperature and under standard pressure, for example aerosol propellants such as halohydrocarbons, butane, propane, nitrogen and carbon dioxide.

[0241] Preferred solid carriers are selected from clays, talc and silica.

[0242] Preferred liquid carriers are selected from water, fatty acid amides and esters thereof, aromatic and nonaromatic hydrocarbons, lactams and carbonic acid esters.

[0243] The amount of carrier typically ranges from 1 to 99.99%, preferably from 5 to 99.9%, more preferably from 10 to 99.5%, and most preferably from 20 to 99% by weight of the composition.

[0244] Liquid carriers are typically present in a range of from 20 to 90%, for example 30 to 80% by weight of the composition.

[0245] Solid carriers are typically present in a range of from 0 to 50%, preferably 5 to 45%, for example 10 to 30% by weight of the composition.

[0246] If the composition comprises two or more carriers, the outlined ranges refer to the total amount of carriers.

[0247] The surfactant can be an ionic (cationic or anionic), amphoteric or non-ionic surfactant, such as ionic or non-ionic emulsifier(s), foam former(s), dispersant(s), wetting agent(s), penetration enhancer(s) and any mixtures thereof. Examples of suitable surfactants include, but are not limited to, salts of polyacrylic acid, salts of lignosulfonic acid (such as sodium lignosulfonate), salts of phenolsulfonic acid or naphthalenesulfonic acid, polycondensates of ethylene oxide and/or propylene oxide with fatty alcohols, fatty acids or fatty amines (for example, polyoxyethylene fatty acid esters such as castor oil ethoxylate, polyoxyethylene fatty alcohol ethers, for example alkylaryl polyglycol ethers), substituted phenols (preferably alkylphenols or arylphenols) and ethoxylates thereof (such as tristyrilphenol ethoxylate), salts of sulfosuccinic esters, taurine derivatives (preferably alkyl taurates), phosphoric esters of polyethoxylated alcohols or phenols, fatty esters of polyols (such as fatty acid esters of glycerol, sorbitol or sucrose), sulfates (such as alkyl sulfates and alkyl ether sulfates), sulfonates (for example,

alkylsulfonates, arylsulfonates and alkylbenzene sulfonates), phosphate esters, protein hydrolysates, ligno-sulfite waste liquors and methylcellulose. Any reference to salts in this paragraph refers preferably to the respective alkali, alkaline earth and ammonium salts.

[0248] Preferred surfactants are selected from polyoxyethylene fatty alcohol ethers, polyoxyethylene fatty acid esters, alkylbenzene sulfonates, such as calcium dodecylbenzenesulfonate, castor oil ethoxylate, sodium lignosulfonate and arylphenol ethoxylates, such as tristyrylphenol ethoxylate.

[0249] The amount of surfactants typically ranges from 5 to 40%, for example 10 to 20%, by weight of the composition.

[0250] Further examples of suitable auxiliaries include water repellents, siccatives, binders (adhesive, tackifier, fixing agent, such as carboxymethylcellulose, natural and synthetic polymers in the form of powders, granules or latices, such as gum arabic, polyvinyl alcohol and polyvinyl acetate, natural phospholipids such as cephalins and lecithins and synthetic phospholipids, polyvinylpyrrolidone and tylose), thickeners and secondary thickeners (such as cellulose ethers, acrylic acid derivatives, xanthan gum, modified clays, e.g. the products available under the name Bentone, and finely divided silica), stabilizers (e.g. cold stabilizers, preservatives (e.g. dichlorophene and benzyl alcohol hemiformal), antioxidants, light stabilizers, in particular UV stabilizers, or other agents which improve chemical and/or physical stability), dyes or pigments (such as inorganic pigments, e.g. iron oxide, titanium oxide and Prussian Blue; organic dyes, e.g. alizarin, azo and metal phthalocyanine dyes), antifoams (e.g. silicone antifoams and magnesium stearate), antifreezes, stickers, gibberellins and processing auxiliaries, mineral and vegetable oils, perfumes, waxes, nutrients (including trace nutrients, such as salts of iron, manganese, boron, copper, cobalt, molybdenum and zinc), protective colloids, thixotropic substances, penetrants, sequestering agents and complex formers.

[0251] The choice of the auxiliaries depends on the intended mode of application of the compound of the invention and/or on the physical properties of the compound (s). Furthermore, the auxiliaries may be chosen to impart particular properties (technical, physical and/or biological properties) to the compositions or use forms prepared therefrom. The choice of auxiliaries may allow customizing the compositions to specific needs.

[0252] The composition of the invention may be provided to the end user as ready-for-use formulation, i.e. the compositions may be directly applied to the plants or seeds by a suitable device, such as a spraying or dusting device. Alternatively, the compositions may be provided to the end user in the form of concentrates which have to be diluted, preferably with water, prior to use.

[0253] The composition of the invention can be prepared in conventional manners, for example by mixing the compound of the invention with one or more suitable auxiliaries, such as disclosed herein above.

[0254] The composition comprises a fungicidally effective amount of the compound(s) of the invention. The term "effective amount" denotes an amount, which is sufficient for controlling harmful fungi on cultivated plants or in the protection of materials and which does not result in a substantial damage to the treated plants. Such an amount can vary in a broad range and is dependent on various factors, such as the fungal species to be controlled, the treated

cultivated plant or material, the climatic conditions and the specific compound of the invention used. Usually, the composition according to the invention contains from 0.01 to 99% by weight, preferably from 0.05 to 98% by weight, more preferred from 0.1 to 95% by weight, even more preferred from 0.5 to 90% by weight, most preferably from 1 to 80% by weight of the compound of the invention. It is possible that a composition comprises two or more compounds of the invention. In such case the outlined ranges refer to the total amount of compounds of the present invention.

[0255] The composition of the invention may be in any customary composition type, such as solutions (e.g. aqueous solutions), emulsions, water- and oil-based suspensions, powders (e.g. wettable powders, soluble powders), dusts, pastes, granules (e.g. soluble granules, granules for broadcasting), suspoemulsion concentrates, natural or synthetic products impregnated with the compound of the invention, fertilizers and also microencapsulations in polymeric substances. The compound of the invention may be present in a suspended, emulsified or dissolved form. Examples of particular suitable composition types are solutions, water-soluble concentrates (e.g. SL, LS), dispersible concentrates (DC), suspensions and suspension concentrates (e.g. SC, OD, OF, FS), emulsifiable concentrates (e.g. EC), emulsions (e.g. EW, EO, ES, ME, SE), capsules (e.g. CS, ZC), pastes, pastilles, wettable powders or dusts (e.g. WP, SP, WS, DP, DS), pressings (e.g. BR, TB, DT), granules (e.g. WG, SG, GR, FG, GG, MG), insecticidal articles (e.g. LN), as well as gel formulations for the treatment of plant propagation materials such as seeds (e.g. GW, GF). These and further compositions types are defined by the Food and Agriculture Organization of the United Nations (FAO). An overview is given in the "Catalogue of pesticide formulation types and international coding system", Technical Monograph No. 2, 6th Ed. May 2008, Croplife International.

[0256] Preferably, the composition of the invention is in form of one of the following types: EC, SC, FS, SE, OD and WG, more preferred EC, SC, OD and WG.

[0257] Further details about examples of composition types and their preparation are given below. If two or more compounds of the invention are present, the outlined amount of compound of the invention refers to the total amount of compounds of the present invention. This applies mutatis mutandis for any further component of the composition, if two or more representatives of such component, e.g. wetting agent, binder, are present.

i) Water-Soluble Concentrates (SL, LS)

[0258] 10-60% by weight of at least one compound of the invention and 5-15% by weight surfactant (e.g. polyoxyethylene fatty alcohol ether) are dissolved in such amount of water and/or water-soluble solvent (e.g. alcohols such as propylene glycol or carbonates such as propylene carbonate) to result in a total amount of 100% by weight. Before application the concentrate is diluted with water.

ii) Dispersible Concentrates (DC)

[0259] 5-25% by weight of at least one compound of the invention and 1-10% by weight surfactant and/or binder (e.g. polyvinylpyrrolidone) are dissolved in such amount of

organic solvent (e.g. cyclohexanone) to result in a total amount of 100% by weight. Dilution with water gives a dispersion.

iii) Emulsifiable Concentrates (EC)

[0260] 15-70% by weight of at least one compound of the invention and 5-10% by weight surfactant (e.g. a mixture of calcium dodecylbenzenesulfonate and castor oil ethoxylate) are dissolved in such amount of water-insoluble organic solvent (e.g. aromatic hydrocarbon or fatty acid amide) and if needed additional water-soluble solvent to result in a total amount of 100% by weight. Dilution with water gives an emulsion.

iv) Emulsions (EW, EO, ES)

[0261] 5-40% by weight of at least one compound of the invention and 1-10% by weight surfactant (e.g. a mixture of calcium dodecylbenzenesulfonate and castor oil ethoxylate) are dissolved in 20-40% by weight water-insoluble organic solvent (e.g. aromatic hydrocarbon). This mixture is added to such amount of water by means of an emulsifying machine to result in a total amount of 100% by weight. The resulting composition is a homogeneous emulsion. Before application the emulsion may be further diluted with water.

v) Suspensions and Suspension Concentrates

v-1) Water-Based (SC, FS)

[0262] In a suitable grinding equipment, e.g. an agitated ball mill, 20-60% by weight of at least one compound of the invention are comminuted with addition of 2-10% by weight surfactant (e.g. sodium lignosulfonate and polyoxyethylene fatty alcohol ether), 0.1-2% by weight thickener (e.g. xanthan gum) and water to give a fine active substance suspension. The water is added in such amount to result in a total amount of 100% by weight. Dilution with water gives a stable suspension of the active substance. For FS type compositions up to 40% by weight binder (e.g. polyvinyl-alcohol) is added.

v-2) Oil-Based (OD, OF)

[0263] In a suitable grinding equipment, e.g. an agitated ball mill, 20-60% by weight of at least one compound of the invention are comminuted with addition of 2-10% by weight surfactant (e.g. sodium lignosulfonate and polyoxyethylene fatty alcohol ether), 0.1-2% by weight thickener (e.g. modified clay, in particular Bentone, or silica) and an organic carrier to give a fine active substance oil suspension. The organic carrier is added in such amount to result in a total amount of 100% by weight. Dilution with water gives a stable dispersion of the active substance.

vi) Water-Dispersible Granules and Water-Soluble Granules (WG, SG)

[0264] 50-80% by weight of at least one compound of the invention are ground finely with addition of surfactant (e.g. sodium lignosulfonate and polyoxyethylene fatty alcohol ether) and converted to water-dispersible or water-soluble granules by means of technical appliances (e.g. extrusion, spray tower, fluidized bed). The surfactant is used in such amount to result in a total amount of 100% by weight. Dilution with water gives a stable dispersion or solution of the active substance.

vii) Water-Dispersible Powders and Water-Soluble Powders (WP, SP, WS)

[0265] 50-80% by weight of at least one compound of the invention are ground in a rotor-stator mill with addition of 1-8% by weight surfactant (e.g. sodium lignosulfonate, polyoxyethylene fatty alcohol ether) and such amount of solid carrier, e.g. silica gel, to result in a total amount of 100% by weight. Dilution with water gives a stable dispersion or solution of the active substance.

viii) Gel (GW, GF)

[0266] In an agitated ball mill, 5-25% by weight of at least one compound of the invention are comminuted with addition of 3-10% by weight surfactant (e.g. sodium lignosulfonate), 1-5% by weight binder (e.g. carboxymethylcellulose) and such amount of water to result in a total amount of 100% by weight. This results in a fine suspension of the active substance. Dilution with water gives a stable suspension of the active substance.

ix) Microemulsion (ME)

[0267] 5-20% by weight of at least one compound of the invention are added to 5-30% by weight organic solvent blend (e.g. fatty acid dimethylamide and cyclohexanone), 10-25% by weight surfactant blend (e.g. polyoxyethylene fatty alcohol ether and arylphenol ethoxylate), and such amount of water to result in a total amount of 100% by weight. This mixture is stirred for 1 h to produce spontaneously a thermodynamically stable microemulsion.

x) Microcapsules (CS)

[0268] An oil phase comprising 5-50% by weight of at least one compound of the invention, 0-40% by weight water-insoluble organic solvent (e.g. aromatic hydrocarbon), 2-15% by weight acrylic monomers (e.g. methylmethacrylate, methacrylic acid and a di- or triacrylate) are dispersed into an aqueous solution of a protective colloid (e.g. polyvinyl alcohol). Radical polymerization initiated by a radical initiator results in the formation of poly(meth)acrylate microcapsules. Alternatively, an oil phase comprising 5-50% by weight of at least one compound of the invention, 0-40% by weight water-insoluble organic solvent (e.g. aromatic hydrocarbon), and an isocyanate monomer (e.g. diphenylmethane-4,4'-diisocyanate) are dispersed into an aqueous solution of a protective colloid (e.g. polyvinyl alcohol). The addition of a polyamine (e.g. hexamethylenediamine) results in the formation of polyurea microcapsules. The monomers amount to 1-10% by weight of the total CS composition.

xi) Dustable Powders (DP, DS)

[0269] 1-10% by weight of at least one compound of the invention are ground finely and mixed intimately with such amount of solid carrier, e.g. finely divided kaolin, to result in a total amount of 100% by weight.

xii) Granules (GR, FG)

[0270] 0.5-30% by weight of at least one compound of the invention are ground finely and associated with such amount of solid carrier (e.g. silicate) to result in a total amount of 100% by weight. Granulation is achieved by extrusion, spray-drying or the fluidized bed.

xiii) Ultra-Low Volume Liquids (UL)

[0271] 1-50% by weight of at least one compound of the invention are dissolved in such amount of organic solvent, e.g. aromatic hydrocarbon, to result in a total amount of 100% by weight.

[0272] The compositions types i) to xiii) may optionally comprise further auxiliaries, such as 0.1-1% by weight preservatives, 0.1-1% by weight antifoams, 0.1-1% by weight dyes and/or pigments, and 5-10% by weight anti-freezes.

[0273] The invention is illustrated by the examples below. However, the invention is not limited to the examples.

EXAMPLES

a-1. In Vivo Preventive Test on *Alternaria brassicae* (Leaf Spot on Radish or Cabbage)

Solvent:	5% by volume of Dimethyl sulfoxide 10% by volume of Acetone
Emulsifier:	1 µl of Tween® 80 per mg of active ingredient

[0274] The active ingredients were made soluble and homogenized in a mixture of Dimethyl sulfoxide/Acetone// Tween® 80 and then diluted in water to the desired concentration.

[0275] The young plants of radish or cabbage were treated by spraying the active ingredient prepared as described above. Control plants were treated only with an aqueous solution of Acetone/Dimethyl sulfoxide/Tween® 80.

[0276] After 24 hours, the plants were contaminated by spraying the leaves with an aqueous suspension of *Alternaria brassicae* spores. The contaminated radish or cabbage plants were incubated for 3 to 4 days at 20° C. and at 100% relative humidity.

[0277] The test was evaluated 3 to 4 days after the inoculation. 0% means an efficacy which corresponds to that of the control plants while an efficacy of 100% means that no disease was observed.

[0278] In this test, compound (I) showed efficacy of at least 70% at a concentration of 500 ppm of active ingredient.

a-2. In Vivo Preventive Test on *Botrytis cinerea* (Grey Mould)

Solvent:	5% by volume of Dimethyl sulfoxide 10% by volume of Acetone
Emulsifier:	1 µl of Tween® 80 per mg of active ingredient

[0279] The active ingredients were made soluble and homogenized in a mixture of Dimethyl sulfoxide/Acetone// Tween® 80 and then diluted in water to the desired concentration.

[0280] The young plants of gherkin or cabbage were treated by spraying the active ingredient prepared as described above. Control plants were treated only with an aqueous solution of Acetone/Dimethyl sulfoxide/Tween® 80.

[0281] After 24 hours, the plants were contaminated by spraying the leaves with an aqueous suspension of *Botrytis cinerea* spores. The contaminated gherkin plants were incubated for 4 to 5 days at 17° C. and at 90% relative humidity. The contaminated cabbage plants were incubated for 4 to 5 days at 20° C. and at 100% relative humidity.

[0282] The test was evaluated 4 to 5 days after the inoculation. 0% means an efficacy which corresponds to that of the control plants while an efficacy of 100% means that no disease was observed.

[0283] In this test, compound (I) showed efficacy of at least 70% at a concentration of 500 ppm of active ingredient.

a-3. In Vivo Preventive Test on *Pyrenophora teres* (Net Blotch on Barley)

Solvent:	5% by volume of Dimethyl sulfoxide 10% by volume of Acetone
Emulsifier:	1 µl of Tween® 80 per mg of active ingredient

[0284] The active ingredients were made soluble and homogenized in a mixture of Dimethyl sulfoxide/Acetone// Tween® 80 and then diluted in water to the desired concentration.

[0285] The young plants of barley were treated by spraying the active ingredient prepared as described above. Control plants were treated only with an aqueous solution of Acetone/Dimethyl sulfoxide/Tween® 80.

[0286] After 24 hours, the plants were contaminated by spraying the leaves with an aqueous suspension of *Pyrenophora teres* spores. The contaminated barley plants were incubated for 48 hours at 20° C. and at 100% relative humidity and then for 8 days at 20° C. and at 70-80% relative humidity.

[0287] The test was evaluated 10 days after the inoculation. 0% means an efficacy which corresponds to that of the control plants while an efficacy of 100% means that no disease was observed.

[0288] In this test, compound (I) showed efficacy of at least 70% at a concentration of 500 ppm of active ingredient.

a-4. In Vivo Preventive Test on *Septoria tritici* (Leaf Spot on Wheat)

Solvent:	5% by volume of Dimethyl sulfoxide 10% by volume of Acetone
Emulsifier:	1 µl of Tween® 80 per mg of active ingredient

[0289] The active ingredients were made soluble and homogenized in a mixture of Dimethyl sulfoxide/Acetone// Tween® 80 and then diluted in water to the desired concentration.

[0290] The young plants of wheat were treated by spraying the active ingredient prepared as described above. Control plants were treated only with an aqueous solution of Acetone/Dimethyl sulfoxide/Tween® 80.

[0291] After 24 hours, the plants were contaminated by spraying the leaves with an aqueous suspension of *Septoria tritici* spores. The contaminated wheat plants were incubated for 72 hours at 17° C. and at 100% relative humidity and then for 14 to 16 days at 20° C. and at 90% relative humidity.

[0292] The test was evaluated 17 to 19 days after the inoculation. 0% means an efficacy which corresponds to that of the control plants while an efficacy of 100% means that no disease was observed.

[0293] In this test, compound (I) showed efficacy of at least 70% at a concentration of 500 ppm of active ingredient.

a-5. In Vivo Preventive Test on *Sphaerotheca fuliginea* (Powdery Mildew on Cucurbits)

Solvent:	5% by volume of Dimethyl sulfoxide 10% by volume of Acetone
Emulsifier:	1 µl of Tween ® 80 per mg of active ingredient

[0294] The active ingredients were made soluble and homogenized in a mixture of Dimethyl sulfoxide/Acetone// Tween® 80 and then diluted in water to the desired concentration.

[0295] The young plants of gherkin were treated by spraying the active ingredient prepared as described above. Control plants were treated only with an aqueous solution of Acetone/Dimethyl sulfoxide/Tween® 80.

[0296] After 24 hours, the plants were contaminated by spraying the leaves with an aqueous suspension of *Sphaerotheca fuliginea* spores. The contaminated gherkin plants were incubated for 8 days at 20° C. and at 70-80% relative humidity.

[0297] The test was evaluated 8 days after the inoculation. 0% means an efficacy which corresponds to that of the control plants while an efficacy of 100% means that no disease was observed.

[0298] In this test, compound (I) showed efficacy of at least 70% at a concentration of 500 ppm of active ingredient.

a-6. In Vivo Preventive Test on *Colletotrichum lindemuthianum* (Leaf Spot on Bean)

Solvent:	5% by volume of Dimethyl sulfoxide 10% by volume of Acetone
Emulsifier:	1 µl of Tween ® 80 per mg of active ingredient

[0299] The active ingredients were made soluble and homogenized in a mixture of Dimethyl sulfoxide/Acetone// Tween® 80 and then diluted in water to the desired concentration.

[0300] The young plants of bean were treated by spraying the active ingredient prepared as described above. Control plants were treated only with an aqueous solution of Acetone/Dimethyl sulfoxide/Tween® 80.

[0301] After 24 hours, the plants were contaminated by spraying the leaves with an aqueous suspension of *Colletotrichum lindemuthianum* spores. The contaminated bean plants were incubated for 24 hours at 20° C. and at 100% relative humidity and then for 6 days at 20° C. and at 90% relative humidity.

[0302] The test was evaluated 7 days after the inoculation. 0% means an efficacy which corresponds to that of the control plants while an efficacy of 100% means that no disease was observed.

[0303] In this test, compound (I) showed efficacy of at least 70% at a concentration of 500 ppm of active ingredient.

a-7. *Alternaria alternata* In Vitro Cell Test

Solvent:	DMSO
Culture medium:	14.6 g anhydrous D-glucose (VWR), 7.1 g Mycological Peptone (Oxoid), 1.4 g granulated Yeast Extract (Merck), QSP 1 liter
Inoculum:	spores suspension

[0304] Fungicides were solubilized in DMSO and the solution used to prepare the required range of concentrations. The final concentration of DMSO used in the assay was ≤1%.

[0305] A spore suspension of *A. alternata* was prepared and diluted to the desired spore density.

[0306] Fungicides were evaluated for their ability to inhibit spore germination and mycelium growth in liquid culture assay. The compounds were added in the desired concentration to the culture medium with spores. After 5 days incubation, fungi-toxicity of compounds was determined by spectrometric measurement of mycelium growth. Inhibition of fungal growth was determined by comparing the absorbance values in wells containing the fungicides with the absorbance in control wells without fungicides.

[0307] In this test, compound (I) showed efficacy of at least 70% at a concentration of 20 ppm of active ingredient.

a-8. *Fusarium culmorum* In Vitro Cell Test

Solvent:	DMSO
Culture medium:	14.6 g anhydrous D-glucose (VWR), 7.1 g Mycological Peptone (Oxoid), 1.4 g granulated Yeast Extract (Merck), QSP 1 liter
Inoculum:	spore suspension

[0308] Fungicides were solubilized in DMSO and the solution used to prepare the required range of concentrations. The final concentration of DMSO used in the assay was ≤1%.

[0309] A spore suspension of *F. culmorum* was prepared and diluted to the desired spore density.

[0310] Fungicides were evaluated for their ability to inhibit spore germination and mycelium growth in liquid culture assay. The compounds were added in the desired concentration to the culture medium with spores. After 4 days incubation, fungi-toxicity of compounds was determined by spectrometric measurement of mycelium growth. Inhibition of fungal growth was determined by comparing the absorbance values in wells containing the fungicides with the absorbance in control wells without fungicides.

[0311] In this test, compound (I) showed efficacy of at least 70% at a concentration of 20 ppm of active ingredient.

a-9. *Pyricularia Oryzae* In Vitro Cell Test

Solvent:	DMSO
Culture medium:	14.6 g anhydrous D-glucose (VWR), 7.1 g Mycological Peptone (Oxoid), 1.4 g granulated Yeast Extract (Merck), QSP 1 liter
Inoculum:	spore suspension

[0312] Fungicides were solubilized in DMSO and the solution used to prepare the required range of concentrations. The final concentration of DMSO used in the assay was ≤1%.

[0313] A spore suspension of *P. oryzae* was prepared and diluted to the desired spore density.

[0314] Fungicides were evaluated for their ability to inhibit spore germination and mycelium growth in liquid culture assay. The compounds were added in the desired concentration to the culture medium with spores. After 5 days incubation, fungi-toxicity of compounds was determined by spectrometric measurement of mycelium growth.

Inhibition of fungal growth was determined by comparing the absorbance values in wells containing the fungicides with the absorbance in control wells without fungicides.

[0315] In this test, compound (I) showed efficacy of at least 70% at a concentration of 20 ppm of active ingredient.

a-10. *Ustilago avenae* In Vitro Cell Test

Solvent:	DMSO
Culture medium:	14.6 g anhydrous D-glucose (VWR), 7.1 g Mycological Peptone (Oxoid), 1.4 g granulated Yeast Extract (Merck), QSP 1 liter
Inoculum:	spore suspension

[0316] Fungicides were solubilized in DMSO and the solution used to prepare the required range of concentrations. The final concentration of DMSO used in the assay was $\leq 1\%$.

[0317] Inoculum was prepared from a pre-culture of *U. avenae* grown in liquid medium and diluted to the desired optical density (OD).

[0318] Fungicides were evaluated for their ability to inhibit mycelium growth in liquid culture assay. The compounds were added in the desired concentrations to culture medium containing the spore suspension. After 4 days of incubation, the fungicidal efficacy of compounds was determined by spectrometric measurement of mycelium growth. Inhibition of fungal growth was determined by comparing the absorbance values in wells containing the fungicides with the absorbance in control wells without fungicides.

[0319] In this test, compound (I) showed efficacy of at least 70% at a concentration of 20 ppm of active ingredient.

a-11. *Colletotrichum lindemuthianum* In Vitro Cell Test

Solvent:	DMSO
Culture medium:	14.6 g anhydrous D-glucose (VWR), 7.1 g Mycological Peptone (Oxoid), 1.4 g granulated Yeast Extract (Merck), QSP 1 liter
Inoculum:	spores suspension

[0320] Fungicides were solubilized in DMSO and the solution used to prepare the required range of concentrations. The final concentration of DMSO used in the assay was $\leq 1\%$.

[0321] A spore suspension of *C. lindemuthianum* was prepared and diluted to the desired spore density.

[0322] Fungicides were evaluated for their ability to inhibit spores germination and mycelium growth in liquid culture assay. The compounds were added in the desired concentration to the culture medium with spores. After 6 days incubation, fungi-toxicity of compounds was determined by spectrometric measurement of mycelium growth. Inhibition of fungal growth was determined by comparing the absorbance values in wells containing the fungicides with the absorbance in control wells without fungicides.

[0323] In this test, compound (I) showed efficacy of at least 70% at a concentration of 20 ppm of active ingredient.

a-12. *Pyrenophora teres* In Vitro Cell Test

Solvent:	DMSO
Culture medium:	14.6 g anhydrous D-glucose (VWR), 7.1 g Mycological Peptone (Oxoid), 1.4 g granulated Yeast Extract (Merck), QSP 1 liter
Inoculum:	spore suspension

[0324] Fungicides were solubilized in DMSO and the solution used to prepare the required range of concentrations. The final concentration of DMSO used in the assay was $\leq 1\%$.

[0325] A spore suspension of *P. teres* was prepared and diluted to the desired spore density.

[0326] Fungicides were evaluated for their ability to inhibit spore germination and mycelium growth in liquid culture assay. The compounds were added in the desired concentration to the culture medium with spores. After 6 days incubation, fungi-toxicity of compounds was determined by spectrometric measurement of mycelium growth. Inhibition of fungal growth was determined by comparing the absorbance values in wells containing the fungicides with the absorbance in control wells without fungicides.

[0327] In this test, compound (I) showed efficacy of at least 70% at a concentration of 20 ppm of active ingredient.

a-13. *Fusarium virguliforme* In Vitro Cell Test

Solvent:	DMSO
Culture medium:	14.6 g anhydrous D-glucose (VWR), 7.1 g Mycological Peptone (Oxoid), 1.4 g granulated Yeast Extract (Merck), QSP 1 liter
Inoculum:	spore suspension

[0328] Fungicides were solubilized in DMSO and the solution used to prepare the required range of concentrations. The final concentration of DMSO used in the assay was $\leq 1\%$.

[0329] A spore suspension of *F. virguliforme* was prepared and diluted to the desired spore density.

[0330] Fungicides were evaluated for their ability to inhibit spore germination and mycelium growth in liquid culture assay. The compounds were added in the desired concentration to the culture medium with spores. After 6 days incubation, fungi-toxicity of compounds was determined by spectrometric measurement of mycelium growth. Inhibition of fungal growth was determined by comparing the absorbance values in wells containing the fungicides with the absorbance in control wells without fungicides.

[0331] In this test, compound (I) showed efficacy of at least 70% at a concentration of 20 ppm of active ingredient.

a-14. *Botrytis cinerea* In Vitro Cell Test

Solvent:	DMSO
Culture medium:	1 g KH ₂ PO ₄ (VWR), 1 g K ₂ HPO ₄ (VWR), 0.5 g Urea (VWR), 3 g KNO ₃ (Prolabo), 10 g saccharose (VWR), 0.5 g MgSO ₄ · 7H ₂ O (Sigma), 0.07 g CaCl ₂ · 2H ₂ O (Prolabo), 0.2 mg MnSO ₄ · H ₂ O (Sigma), 0.6 mg CuSO ₄ · 5H ₂ O (Sigma), 7.9 mg ZnSO ₄ · 7H ₂ O

-continued

(Sigma), 0.1 mg H₃BO₃ (Merck), 0.14 mg NaMoO₄·2H₂O (Sigma), 2 mg thiamine (Sigma), 0.1 mg biotine (VWR), 4 mg FeSO₄·7H₂O (Sigma), QSP 1 liter

Inoculum: spore suspension

[0332] Fungicides were solubilized in DMSO and the solution used to prepare the required range of concentrations. The final concentration of DMSO used in the assay was ≤1%.

[0333] A spore suspension of *B. cinerea* was prepared and diluted to the desired spore density.

[0334] Fungicides were evaluated for their ability to inhibit spore germination and mycelium growth in liquid culture assay. The compounds were added in the desired concentration to the culture medium with spores. After 6 days incubation, fungi-toxicity of compounds was determined by spectrometric measurement of mycelium growth. Inhibition of fungal growth was determined by comparing the absorbance values in wells containing the fungicides with the absorbance in control wells without fungicides.

[0335] In this test, compound (I) showed efficacy of at least 70% at a concentration of 20 ppm of active ingredient.

a-15. *Septoria tritici* In Vitro Cell Test

Solvent: DMSO

Culture medium: 1 g KH₂PO₄ (VWR), 1 g K₂HPO₄ (VWR), 0.5 g Urea (VWR), 3 g KNO₃ (Prolabo), 10 g saccharose (VWR), 0.5 g MgSO₄·7H₂O (Sigma), 0.07 g CaCl₂·2H₂O (Prolabo), 0.2 mg MnSO₄·H₂O (Sigma), 0.6 mg CuSO₄·5H₂O (Sigma), 7.9 mg ZnSO₄·7H₂O (Sigma), 0.1 mg H₃BO₃ (Merck), 0.14 mg NaMoO₄·2H₂O (Sigma), 2 mg thiamine (Sigma), 0.1 mg biotine (VWR), 4 mg FeSO₄·7H₂O (Sigma), QSP 1 liter

Inoculum: spore suspension

[0336] Fungicides were solubilized in DMSO and the solution used to prepare the required range of concentrations. The final concentration of DMSO used in the assay was ≤1%.

[0337] A spore suspension of *S. tritici* was prepared and diluted to the desired spore density.

[0338] Fungicides were evaluated for their ability to inhibit spore germination and mycelium growth in liquid culture assay. The compounds were added in the desired concentration to the culture medium with spores. After 7 days incubation, fungi-toxicity of compounds was determined by spectrometric measurement of mycelium growth. Inhibition of fungal growth was determined by comparing the absorbance values in wells containing the fungicides with the absorbance in control wells without fungicides.

[0339] In this test, compound (I) showed efficacy of at least 70% at a concentration of 20 ppm of active ingredient.

a-16. In Vivo Preventive *Fusarium graminearum* Test (Wheat)

Solvent: 24.5 parts by weight of acetone
24.5 parts by weight of dimethyl sulfoxide

Emulsifier: 1 part by weight of polyoxyethylene sorbitan monooleate

[0340] To produce a suitable preparation of active compound, 1 part by weight of active compound or active compound combination was mixed with the stated amounts

of solvent and emulsifier, and the concentrate was diluted with water to the desired concentration.

[0341] To test for preventive activity, young plants were sprayed with the preparation of active compound or active compound combination at the stated rate of application.

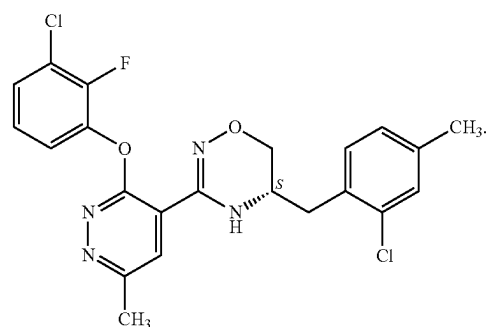
[0342] The following day the plants were slightly injured by using a sandblast and afterwards they were sprayed with a conidia suspension of *Fusarium graminearum*.

[0343] The plants were placed in the greenhouse under a translucent incubation cabinet at a temperature of approximately 25° C. and a relative atmospheric humidity of approximately 100%.

[0344] The test was evaluated 5 days after the inoculation. 0% means an efficacy which corresponds to that of the untreated control, while an efficacy of 100% means that no disease is observed.

[0345] In this test, compound (I) showed efficacy of at least 70% at a concentration of 500 ppm of active ingredient.

1. A use of (5S)-3-[3-(3-chloro-2-fluorophenoxy)-6-methylpyridazin-4-yl]-5-(2-chloro-4-methylbenzyl)-5,6-dihydro-4H-1,2,4-oxadiazine according to formula (I)



(I)

for controlling unwanted microorganisms causing diseases on plants.

2. The use according to claim 1, wherein the unwanted microorganisms are selected from the group consisting of phytopathogenic fungi and phytopathogenic viruses.

3. The use according to claim 1, wherein the diseases are selected from the group consisting of diseases caused by powdery mildew pathogens, diseases caused by rust disease pathogens, leaf blotch diseases and leaf wilt diseases, root and stem diseases, ear and panicle diseases, diseases caused

by smut fungi, fruit rot, seed- and soil-borne rot and wilt diseases, cancers, galls and witches' broom, wilt diseases, deformations of leaves, flowers and fruits, degenerative diseases in woody plants, diseases of soya beans, fungal diseases on leaves, stems, pods and seeds and fungal diseases on roots and the stem base.

4. The use according to claim 1, wherein the diseases are selected from the group consisting of diseases caused by powdery mildew pathogens, leaf blotch diseases and leaf wilt diseases, root and stem diseases, ear and panicle diseases, fruit rot, seed- and soil-borne rot and wilt diseases, cancers, galls, witches' broom and wilt diseases.

5. The use according to claim 1, wherein the plants are selected from the group consisting of cotton, flax, grapevine, fruit, vegetables the vegetables selected from the group consisting of *Rosaceae* sp., *Ribesioideae* sp., *Juglandaceae* sp., *Betulaceae* sp., *Anacardiaceae* sp., *Fagaceae* sp., *Moraceae* sp., *Oleaceae* sp., *Actinidaceae* sp., *Lauraceae* sp., *Musaceae* sp., *Rubiaceae* sp., *Theaceae* sp., *Sterculiaceae* sp., *Rutaceae* sp., *Solanaceae* sp., *Liliaceae* sp., *Asteraceae* sp., *Umbelliferae* sp., *Cruciferae* sp., *Chenopodiaceae* sp., *Cucurbitaceae* sp., *Alliaceae* sp., and *Papilionaceae* sp.; major crop plants, the major crop plants selected from the group consisting of *Gramineae* sp., *Asteraceae* sp., *Brassicaceae* sp., *Fabaceae* sp., *Papilionaceae* sp., and *Chenopodiaceae* sp.; useful plants and ornamental plants for gardens and wooded areas; and genetically modified varieties thereof.

6. The use according to claim 1, wherein the unwanted microorganisms are selected from the group consisting of *Blumeria* species; *Leveillula* species; *Podosphaera* species; *Sphaerotheca* species; *Erysiphe* species; *Gymnosporangium* species; *Hemileia* species; *Phakopsora* species; *Puccinia* species; *Uromyces* species; *Alternaria* species; *Cercospora* species; *Cladosporium* species; *Cochliobolus* species; *Colletotrichum* species; *Corynespora* species; *Cycloconium* species; *Diaporthe* species; *Elsinoe* species; *Gloeosporium* species; *Glomerella* species; *Guignardia* species; *Leptosphaeria* species; *Magnaporthe* species; *Marsonina* species; *Microdochium* species; *Monilinia* species; *Mycosphaerella* species; *Phaeosphaeria* species; *Phyllachora* species; *Pyrenophora* species; *Ramularia* species; *Rhynchosporium* species; *Septoria* species; *Setosphaeria* species; *Sclerotinia* species; *Stagonospora* species; *Stemphylium* species; *Typhula* species; *Venturia* species; *Corticium* species; *Fusarium* species; *Gaeumannomyces* species; *Marsonina* species; *Rhizoctonia* species; *Sarocladium* species; *Sclerotium* species; *Tapesia* species; *Thielaviopsis* species; *Alternaria* species; *Aspergillus* species; *Cladosporium* species; *Claviceps* species; *Fusarium* species; *Gibberella* species; *Monographella* species; *Stagnospora* species; *Sphacelotheca* species; *Tilletia* species; *Urocystis* species; *Ustilago* species; *Aspergillus* species; *Botrytis* species; *Monilinia* species; *Penicillium* species; *Phomopsis* species; *Rhizopus* species; *Sclerotinia* species; *Verticillium* species; *Alternaria* species; *Aphanomyces* species; *Ascochyta* species; *Aspergillus* species; *Cladosporium* species; *Cochliobolus* species; *Colletotrichum* species; *Fusarium* species; *Gibberella* species; *Macrophomina* species; *Microdochium* species; *Monographella* species; *Penicillium* species; *Phoma* species; *Phomopsis* species; *Pyrenophora* species; *Pyricularia* species; *Rhizoctonia* species; *Rhizopus* species; *Sclerotium* species; *Septoria* species; *Typhula* species; *Verticillium* species; *Nectria* species; *Verticillium* species; *Fusarium* species;

Exobasidium species; *Taphrina* species; *Esca* species; *Ganoderma* species; *Rhizoctonia* species; *Helminthosporium* species; *Alternaria* spec. *atrans tenuissima*, *Colletotrichum gloeosporoides dematium* var. *truncatum*, *Septoria glycines*, *Cercospora kikuchii*, *Choanephora infundibulifera trispora* (Syn.), *Cordana musae*, *Dactuliophora glycines*, *Peronospora manshurica*, *Drechslera glycini*, *Cercospora sojae*, *Leptosphaerulina trifolii*, *Leveillula Taurica*, *Phyllosticta sojaecola*, *Phomopsis sojae*, *Microsphaera diffusa*, *Pyrenochaeta glycines*, *Rhizoctonia solani*, *Phakopsora pachyrhizi*, *Phakopsora meibomia*, *Sphaceloma glycines*, *Stemphylium botryosum*, *Fusarium virguliforme*, *Corynespora cassicola*; *Calonectria crotalariae*, *Macrophomina phaseolina*, *Fusarium oxysporum*, *Fusarium orthoceras*, *Fusarium semitectum*, *Fusarium equiseti*, *Mycocleptodiscus terrestris*, *Neocosmospora vasinfecta*, *Diaporthe phaseolorum*, *Diaporthe phaseolorum* var. *caulivora*, *Phialophora gregata*, *Rhizoctonia solani*, *Sclerotinia sclerotiorum*, *Sclerotinia rolfsii*, *Thielaviopsis basicola*, tobacco mosaic virus (TMV), tobacco rattle virus, tobacco stunt virus (TStuV), tobacco leaf curl virus (VLCV), tobacco nervilia mosaic virus (TVBMV), tobacco necrotic dwarf virus (TNDV), tobacco streak virus (TSV), potato virus X (PVX), potato viruses Y, S, M, and A, potato acuba mosaic virus (PAMV), potato mop-top virus (PMTV), potato leaf-roll virus (PLRV), alfalfa mosaic virus (AMV), cucumber mosaic virus (CMV), cucumber green mottle mosaic virus (CGMMV), cucumber yellows virus (CuYV), watermelon mosaic virus (WMV), tomato spotted wilt virus (TSWV), tomato ringspot virus (TomRSV), sugarcane mosaic virus (SCMV), rice dwarf virus, rice stripe virus, rice black-streaked dwarf virus, strawberry mottle virus (SMoV), strawberry vein banding virus (SVBV), strawberry mild yellow edge virus (SMYEV), strawberry crinkle virus (SCrV), broad beanwilt virus (BBWV), and melon necrotic spot virus (MNSV).

7. The use according to claim 1, wherein the unwanted microorganisms are selected from the group consisting of *Sphaerotheca* species *Alternaria* species, *Cercospora* species, *Colletotrichum* species; *Diaporthe* species; *Phaeosphaeria* species; *Pyrenophora* species; *Septoria* species; *Fusarium* species; *Alternaria* species; *Fusarium* species; *Botrytis* species; *Alternaria* species; *Colletotrichum* species; *Fusarium* species; *Pyrenophora* species; *Pyricularia* species; and *Fusarium* species.

8. A composition for controlling phytopathogenic fungi and/or phatopathogenic viruses, comprising compound of formula (I) and at least one carrier or surfactant.

9. The composition according to claim 8, which further comprises one or more additional active substance(s) selected from the group consisting of herbicides, insecticides, acaricides, fungicides, safeners and plant growth regulator.

10. The method for controlling unwanted microorganisms and/or fungi comprising application of the compound of formula (I) or a composition according to claim 8 to the plant or parts thereof.

11. The method according to claim 6, wherein the plant is selected from the group consisting of cotton, flax, grapevine, fruit, vegetables, wherein the vegetables are selected from the group consisting of *Rosaceae* sp., *Ribesioideae* sp., *Juglandaceae* sp., *Betulaceae* sp., *Anacardiaceae* sp., *Fagaceae* sp., *Moraceae* sp., *Oleaceae* sp., *Actinidaceae* sp., *Lauraceae* sp., *Musaceae* sp., *Rubiaceae* sp., *Theaceae* sp.,

Sterculiaceae sp., *Rutaceae* sp.; *Solanaceae* sp., *Liliaceae* sp., *Asteraceae* sp., *Umbelliferae* sp., *Cruciferae* sp., *Chenopodiaceae* sp., *Cucurbitaceae* sp., *Alliaceae* sp., and *Papilionaceae* sp.; major crop plants, wherein the major crop plants are selected from the group consisting of *Gramineae* sp., *Asteraceae* sp., *Brassicaceae* sp. *Fabaceae* sp., *Papilionaceae* sp., *Chenopodiaceae* sp.; useful plants and ornamental plants for gardens and wooded areas; and genetically modified varieties thereof.

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